

# Chapter 23

## Three Dimensional Geometry

### Solutions

#### SECTION - A

##### Objective Type Questions (One Option is correct)

1. The perpendicular distance of the point  $P(6, 8, 9)$  from  $XY$ -plane is  
(1) 6 (2) 9 (3) 7 (4) 8

**Sol.** Answer (2)

$z$ -coordinate is the distance of a point  $P(6, 8, 9)$  from  $XY$ -plane.

2. Foot of perpendicular drawn from a point  $P(-2, 3, 5)$  on the  $YZ$ -plane is  
(1)  $(1, 3, 5)$  (2)  $(-2, 3, 0)$  (3)  $(0, 3, 5)$  (4)  $(0, 3, 0)$

**Sol.** Answer (3)

$x$ -coordinate is the distance of a point  $P(-2, 3, 5)$  from  $YZ$ -plane.

3. The locus of the point for which  $x = 0, y = 0$  is  
(1) Equation of  $x$ -axis (2) Equation of  $y$ -axis  
(3) Equation of  $z$ -axis (4) Equation of  $XY$ -plane

**Sol.** Answer (3)

Equation of  $z$ -axis is  $x = 0, y = 0$ , because  $x$  and  $y$ -coordinates on the  $z$ -axis is zero.

4. The length of the foot of perpendicular drawn from the point  $P(3, 6, 7)$  on  $y$ -axis is  
(1)  $\sqrt{48}$  (2)  $\sqrt{45}$  (3) 3 (4)  $\sqrt{58}$

**Sol.** Answer (4)

Length of perpendicular from the point  $P(3, 6, 7)$  on  $y$ -axis is

$$\sqrt{(x\text{-coordinate})^2 + (z\text{-coordinate})^2} = \sqrt{3^2 + 7^2} = \sqrt{58} \text{ units}$$

5. The point  $(-2, 3, 4)$  lies in  
(1) I octant (2) VI octant (3) II octant (4) V octant

**Sol.** Answer (3)

$(-2, 3, 4)$  lies in II<sup>nd</sup> octant.

6. Equation of XZ-plane is

- (1)  $x = 0$  (2)  $x = 0, y = 0$  (3)  $y = 0$  (4)  $y = 0, z = 0$

**Sol.** Answer (3)

y-coordinate on XZ-plane is zero.

$\therefore$  Equation of XZ-plane is  $y = 0$ .

7. A point on y-axis at a distance  $\sqrt{20}$  units from the point (2, 2, 5) is

- (1) (1, 1, 2) (2) (0, 4, 0) (3) (4, -4, 0) (4) None of these

**Sol.** Answer (4)

Any point on y-axis can be taken as (0, y, 0)

$$\text{Now, } \sqrt{(0-2)^2 + (y-2)^2 + (0-5)^2} = \sqrt{20}$$

$$\Rightarrow \sqrt{4 + (y-2)^2 + 25} = \sqrt{20}$$

$$\Rightarrow (y-2)^2 = -9$$

which is not true.

8. If a parallelopiped is formed by planes drawn through the points (2, 5, 3) and (6, 7, 9) parallel to the coordinate planes, then the length of its diagonal is

- (1)  $\sqrt{48}$  units (2)  $2\sqrt{14}$  units (3)  $3\sqrt{14}$  units (4)  $\sqrt{14}$  units

**Sol.** Answer (2)

Length of diagonal

$$= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

$$= \sqrt{(6-2)^2 + (7-5)^2 + (9-3)^2}$$

$$= \sqrt{16 + 4 + 36}$$

$$= 2\sqrt{14} \text{ units}$$

9. The points (0, 7, 10), (-1, 6, 6) and (-4, 9, 6) form a/an

- (1) Right triangle (2) Isosceles triangle  
(3) Isosceles right triangle (4) Equilateral triangle

**Sol.** Answer (3)

Let A(0, 7, 10), B(-1, 6, 6) and C(-4, 9, 6) be given points

$$\text{Now, } AB = \sqrt{(-1-0)^2 + (6-7)^2 + (6-10)^2} = 3\sqrt{2} \text{ units}$$

$$BC = \sqrt{(-4+1)^2 + (9-6)^2 + (6-6)^2} = 3\sqrt{2} \text{ units}$$

$$AC = \sqrt{(-4-0)^2 + (9-7)^2 + (6-10)^2} = 6 \text{ units}$$

$$\text{Since, } AB = BC \text{ and } AB^2 + BC^2 = AC^2$$

10. In which ratio the line segment joining the points (2, 4, 5) and (3, -4, -5) is divided by the XY-plane?

(1) 1 : 2

(2) 2 : 1

(3) 1 : 1

(4) 3 : 2

**Sol.** Answer (3)

Let the required ratio be  $k : 1$

$$\Rightarrow \left( \frac{3k+2}{k+1}, \frac{-4k+4}{k+1}, \frac{-5k+5}{k+1} \right) = (x, y, 0)$$

$$\Rightarrow \frac{-5k+5}{k+1} = 0$$

$$\Rightarrow k = 1$$

Hence, XY-plane divides the join of given points in 1 : 1.

11. In which ratio the line segment joining the points (3, 4, 10) and (-3, 2, 5) is divided by x-axis?

(1) 2 : 1 internally

(2) 2 : 1 externally

(3) 1 : 2 internally

(4) 1 : 2 externally

**Sol.** Answer (2)

Let x-axis divides the join of line segment in  $k : 1$ . Coordinates of the point are

$$\Rightarrow \left( \frac{-3k+3}{k+1}, \frac{2k+4}{k+1}, \frac{5k+10}{k+1} \right)$$

Since, y and z-coordinates on x-axis is zero

$$\Rightarrow \frac{2k+4}{k+1} = 0, \frac{5k+10}{k+1} = 0$$

On solving the equation, we get

$$k = -2$$

Hence, the required ratio is 2 : 1 externally.

12. In which ratio the line segment joining the points (3, 0, 5) and (-2, 3, 2) is divided by YZ-plane?

(1) 4 : 5

(2) 2 : 3

(3) 5 : 3

(4) 3 : 2

**Sol.** Answer (4)

Let YZ-plane divide the join of line segment (3, 0, 5) and (-2, 3, 2) in  $k : 1$ .

Therefore, coordinates of the point are

$$\left( \frac{-2k+3}{k+1}, \frac{3k}{k+1}, \frac{2k+5}{k+1} \right)$$

Since, x-coordinate on YZ-plane is zero.

$$\Rightarrow \frac{-2k+3}{k+1} = 0$$

$$\Rightarrow k = \frac{3}{2}$$

Hence, the required ratio is 3 : 2.

13. The length of the foot of perpendicular drawn from point (3, 4, 5) on z-axis is

- (1) 5 units (2)  $\sqrt{34}$  units (3)  $\sqrt{41}$  units (4)  $2\sqrt{5}$  units

**Sol.** Answer (1)

Length of perpendicular from the point (3, 4, 5) on z-axis is

$$\sqrt{(x\text{-coordinate})^2 + (y\text{-coordinate})^2} = \sqrt{3^2 + 4^2} = 5 \text{ units}$$

14. What is the perpendicular distance of the point (6, 7, 8) from YZ-plane?

- (1) 8 units (2) 6 units (3) 7 units (4) 21 units

**Sol.** Answer (2)

x-coordinate is the distance of a point from YZ-plane.

15. If L is the foot of the perpendicular from Q(-3, 6, 7) on the XY-plane, then the coordinates of the point L is

- (1) (3, 6, 7) (2) (-3, 6, 0) (3) (6, 7, 0) (4) (-3, 0, 7)

**Sol.** Answer (2)

Since, z-coordinate in XY-plane is zero.

Therefore, coordinates of the foot of perpendiculars of the point L is (-3, 6).

16. The coordinates of the foot of perpendicular from the point (3, -4, 5) on YZ-plane is

- (1) (0, -4, 5) (2) (3, -4, 5) (3) (3, 4, 5) (4) (-3, -4, -5)

**Sol.** Answer (1)

Since, x-coordinate in YZ-plane is zero.

Therefore, coordinates of the foot of perpendiculars of the point is (0, -4, 5).

17. The image of the point (-2, 3, 5) in XY-plane is

- (1) (-2, -3, -5) (2) (-2, 3, -5) (3) (2, 3, 5) (4) (-2, 3, 5)

**Sol.** Answer (2)

Image of the point in XY-plane is (-2, 3, -5)

18. If the image of the point (-4, 3, 5) in XZ-plane is Q, then the coordinates of Q is

- (1) (-4, -3, -5) (2) (-3, -4, 5) (3) (4, 3, 5) (4) (-4, -3, 5)

**Sol.** Answer (4)

Coordinates of Q is (-4, -3, 5).

19. If the extremities of the diagonal of a square are (1, -2, 3) and (3, -4, 3), then the length of the side is

- (1)  $2\sqrt{2}$  units (2) 2 units (3)  $\sqrt{2}$  units (4) 4 units

**Sol.** Answer (2)

Length of diagonal

$$= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

$$= \sqrt{(3 - 1)^2 + (-4 + 2)^2 + (3 - 3)^2}$$

$$= 2\sqrt{2} \text{ units}$$

Now, diagonal of square =  $\sqrt{2} \times \text{side}$

$\therefore$  Side = 2 units



20. If  $P(0, 1, 2)$ ,  $Q(4, -2, 1)$  and  $R(0, 0, 0)$  are three points, then  $\angle PRQ$  is

(1)  $\frac{\pi}{4}$

(2)  $\frac{\pi}{2}$

(3)  $\frac{\pi}{3}$

(4)  $\frac{\pi}{6}$

**Sol.** Answer (2)

$$PQ = \sqrt{(4-0)^2 + (-2-1)^2 + (1-2)^2} = \sqrt{26} \text{ units}$$

$$QR = \sqrt{(0-4)^2 + (0+2)^2 + (0-1)^2} = \sqrt{21} \text{ units}$$

$$PR = \sqrt{(0-0)^2 + (0-1)^2 + (0-2)^2} = \sqrt{5} \text{ units}$$

$$\text{Now, } QR^2 + PR^2 = PQ^2$$

It means  $PQR$  is a right angled triangle

$$\therefore \angle PRQ = \frac{\pi}{2}$$

21. If origin is the centroid of the triangle with vertices  $P(3a, 3, 6)$ ,  $Q(-4, 2b, -8)$  and  $R(8, 12, 2c)$ , then the value of  $a$ ,  $b$  and  $c$  are

(1)  $\frac{4}{3}, 1, 2$

(2)  $-\frac{4}{3}, -\frac{15}{2}, 1$

(3)  $3, 2, -\frac{4}{5}$

(4)  $\frac{4}{3}, \frac{15}{2}, 1$

**Sol.** Answer (2)

The coordinates of centroid are

$$\left( \frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3} \right)$$

$$= \left( \frac{3a - 4 + 8}{3}, \frac{3 + 2b + 12}{3}, \frac{6 - 8 + 2c}{3} \right)$$

$$= \left( \frac{3a + 4}{3}, \frac{2b + 15}{3}, \frac{2c - 2}{3} \right)$$

But it is given that, origin is centroid

$$\Rightarrow \frac{3a + 4}{3} = 0, \frac{2b + 15}{3} = 0 \text{ and } \frac{2c - 2}{3} = 0$$

On solving these equations, we get

$$a = -\frac{4}{3}, b = -\frac{15}{2} \text{ and } c = 1$$

22. If the three vertices of a parallelogram  $ABCD$  are  $A(3, -1, 5)$ ,  $B(1, -2, -4)$  and  $C(0, 3, 0)$ , then the coordinates of fourth vertex is
- (1)  $(2, 4, 9)$                       (2)  $(2, -4, -9)$                       (3)  $(0, 4, 9)$                       (4)  $(0, -4, -9)$

**Sol.** Answer (1)

Let the fourth vertex be  $(x, y, z)$ , then mid-point of  $AC$  = mid-point of  $BD$

$$\Rightarrow \left( \frac{0+3}{2}, \frac{-1+3}{2}, \frac{5+0}{2} \right) = \left( \frac{x+1}{2}, \frac{y-2}{2}, \frac{z-4}{2} \right)$$

$$\Rightarrow \left( \frac{3}{2}, 1, \frac{5}{2} \right) = \left( \frac{x+1}{2}, \frac{y-2}{2}, \frac{z-4}{2} \right)$$

$$\Rightarrow \frac{3}{2} = \frac{x+1}{2}, 1 = \frac{y-2}{2}, \frac{5}{2} = \frac{z-4}{2}$$

$$\Rightarrow x = 2, y = 4, z = 9$$

Hence, the fourth coordinates are  $(2, 4, 9)$ .

23. A point with  $y$ -coordinate 6 lies on the line segment joining the points  $(-2, 3, 4)$  and  $(6, 10, 18)$ . Coordinates of the point are

(1)  $\left( \frac{10}{7}, -6, 10 \right)$

(2)  $\left( -\frac{10}{7}, 6, -10 \right)$

(3)  $\left( \frac{10}{7}, 6, 10 \right)$

(4)  $(0, 0, 0)$

**Sol.** Answer (3)

Let  $R(x, 6, z)$  divides the join of the given points  $(-2, 3, 4)$  and  $(6, 10, 18)$  in  $k : 1$ .

The coordinates of point are

$$\left( \frac{6k-2}{k+1}, \frac{10k+3}{k+1}, \frac{18k+4}{k+1} \right) \quad \dots(i)$$

But  $y$ -coordinate is given as 6

$$\Rightarrow \frac{10k+3}{k+1} = 6$$

$$\Rightarrow 10k+3 = 6k+6$$

$$\Rightarrow 4k = 3$$

$$\Rightarrow k = \frac{3}{4}$$

On putting  $k = \frac{3}{4}$  in (i), we get the coordinates of the point as

$$\left( \frac{10}{7}, 6, 10 \right)$$

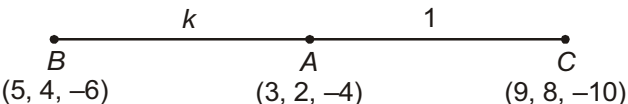
24. If the given points  $A(3, 2, -4)$ ,  $B(5, 4, -6)$  and  $C(9, 8, -10)$  are collinear, then the ratio in which  $A$  divides  $BC$  is

- (1)  $1 : 3$  internally      (2)  $1 : 3$  externally      (3)  $3 : 1$  internally      (4)  $3 : 1$  externally

**Sol.** Answer (2)

Since points are collinear, it means  $A$  lies on  $BC$ . Let  $A$  divides  $BC$  in  $k : 1$ .

Therefore,  $\left( \frac{9k+5}{k+1}, \frac{8k+4}{k+1}, \frac{-10k-6}{k+1} \right) = (3, 2, -4)$



$$\Rightarrow \frac{9k+5}{k+1} = 3, \frac{8k+4}{k+1} = 2 \text{ and } \frac{-10k-6}{k+1} = -4$$

On solving these equations, we get

$$k = \frac{-1}{3}$$

Hence, the required ratio is  $1 : 3$  externally.

25. The ratio in which the join of  $A(2, 1, 5)$  and  $B(3, 4, 3)$  is divided by the plane  $x + 2y - z = 0$  is

- (1)  $2 : 3$       (2)  $8 : 1$       (3)  $1 : 8$       (4)  $4 : 3$

**Sol.** Answer (3)

Let the join of  $A(2, 1, 5)$  and  $B(3, 4, 3)$  is divided by the plane  $x + 2y - z = 0$  in  $k : 1$ .

Therefore, the coordinates of the point are

$$\left( \frac{3k+2}{k+1}, \frac{4k+1}{k+1}, \frac{3k+5}{k+1} \right)$$

Since, the point lies on  $x + 2y - z = 0$

$$\Rightarrow \frac{3k+2}{k+1} + \frac{2(4k+1)}{k+1} - \frac{3k+5}{k+1} = 0$$

$$\Rightarrow 3k + 2 + 8k + 2 - 3k - 5 = 0$$

$$\Rightarrow 8k - 1 = 0$$

$$\therefore k = \frac{1}{8}$$

Hence, the required ratio is  $1 : 8$ .

26. The ratio in which the join of  $A(1, 2, 3)$  and  $B(3, 4, 6)$  is divided by  $XY$ -plane externally is

- (1)  $2 : 1$       (2)  $1 : 2$       (3)  $2 : 3$       (4)  $3 : 2$

**Sol.** Answer (2)

Let the join of  $A(1, 2, 3)$  and  $B(3, 4, 6)$  is divided by  $XY$ -plane in  $k : 1$ .

Therefore, the coordinates are

$$\left( \frac{3k+1}{k+1}, \frac{4k+2}{k+1}, \frac{6k+3}{k+1} \right)$$

Now,  $z$ -coordinate on  $XY$ -plane is zero

$$\Rightarrow \frac{6k+3}{k+1} = 0$$

$$\Rightarrow k = -\frac{1}{2}$$

Hence, the required ratio is 1 : 2.

27. The points  $(5, -4, 2)$ ,  $(4, -3, 1)$ ,  $(7, -6, 4)$  and  $(8, -7, 5)$  are the vertices of a

- (1) Parallelogram                      (2) Rectangle                      (3) Square                      (4) Rhombus

**Sol.** Answer (1)

Let  $A(5, -4, 2)$ ,  $B(4, -3, 1)$ ,  $C(7, -6, 4)$  and  $D(8, -7, 5)$  be given points

$$\begin{aligned} \text{Mid-point of } AC &= \left( \frac{7+5}{2}, \frac{-6-4}{2}, \frac{4+2}{2} \right) \\ &= (6, -5, 3) \end{aligned}$$

$$\begin{aligned} \text{Mid-point of } BD &= \left( \frac{8+4}{2}, \frac{-7-3}{2}, \frac{5+1}{2} \right) \\ &= (6, -5, 3) \end{aligned}$$

Since, mid-points are same.

Therefore,  $ABCD$  may be a parallelogram.

$$\text{Now, } AC = \sqrt{(7-5)^2 + (-6+4)^2 + (4-2)^2} = 3\sqrt{2} \text{ units}$$

$$BD = \sqrt{(8-4)^2 + (-7+3)^2 + (5-1)^2} = 4\sqrt{3} \text{ units}$$

$$\Rightarrow AC \neq BD$$

$\therefore ABCD$  is a parallelogram.

28. A line is parallel to  $YZ$ -plane, if all the points on the line have equal

- (1)  $x$ -coordinates                      (2)  $y$ -coordinates                      (3)  $z$ -coordinates                      (4) None of these

**Sol.** Answer (1)

Any line parallel to  $YZ$ -plane is  $x = C$

$\therefore x$ -coordinate is same.

29. The equation  $x = C$  represents a plane parallel to

- (1)  $YZ$ -plane                      (2)  $XY$ -plane                      (3)  $XZ$ -plane                      (4)  $x$ -axis

**Sol.** Answer (1)

$x = C$  is a line parallel to  $YZ$ -plane.

30. The plane uniquely determined by  $x$ -axis and  $y$ -axis is known as

- (1)  $XY$ -plane                      (2)  $YZ$ -plane                      (3)  $XZ$ -plane                      (4) None of these

**Sol.** Answer (1)

$XY$ -plane is uniquely determined by  $x$  and  $y$ -coordinates.

31. Three mutually perpendicular plane divide the space into

- (1) 6 parts (2) 8 parts (3) 4 parts (4) None of these

**Sol.** Answer (2)

Three mutually perpendicular plane divide the space into 8 parts. Each known as octant.

32. The intersection of XY-plane and YZ-plane is known as

- (1) x-axis (2) y-axis (3) z-axis (4) Origin

**Sol.** Answer (2)

Planes intersect in a line, lines of intersection of XY-plane and YZ-plane is y-axis.

33. If the mid-points of the sides of triangle are (1, 2, -3), (3, 0, 1) and (2, -2, 5), then the centroid is

- (1) (0, 0, 0) (2) (1, 2, 3) (3) (2, 0, 1) (4) (2, 3, 0)

**Sol.** Answer (3)

Centroid of the triangle formed by the mid-points of the sides of a triangle is equal to centroid of the given triangle

$$\therefore \text{Centroid is given by } \left( \frac{1+3+2}{3}, \frac{2+0-2}{3}, \frac{-3+1+5}{3} \right) = (2, 0, 1)$$

34. The shortest distance of the point (a, b, c) from y-axis is

- (1) a (2)  $\sqrt{a^2 + b^2 + c^2}$  (3)  $\sqrt{a^2 + c^2}$  (4)  $\sqrt{b^2 + c^2}$

**Sol.** Answer (3)

Shortest distance of the point (a, b, c) from y-axis is length of foot of perpendicular the given point to y-axis.

$$\begin{aligned} \therefore \text{Length of perpendicular} &= \sqrt{(x\text{-coordinate})^2 + (z\text{-coordinate})^2} \\ &= \sqrt{a^2 + c^2} \end{aligned}$$

35. Values of a for which the distance between the points (3, -5, 4) and (a, -8, 4) is 5 is

- (1) -1 or 7 (2) 2 or 7 (3) 3 or 1 (4) 2 or 1

**Sol.** Answer (1)

Let A(3, -5, 4) and B(a, -8, 4) be given two points, then

$$AB = 5 \Rightarrow (AB)^2 = 25$$

$$\Rightarrow (a - 3)^2 + (-8 + 5)^2 + (4 - 4)^2 = 25$$

$$\Rightarrow a^2 + 9 - 6a + 9 = 25$$

$$\Rightarrow a^2 - 6a - 7 = 0$$

$$\Rightarrow (a - 7)(a + 1) = 0$$

$$\therefore a = -1 \text{ and } 7$$

36. The graph of the equation  $x^2 + y^2 = 0$  in the three dimensional space is

- (1) x-axis (2) y-axis (3) z-axis (4) XY-plane

**Sol.** Answer (3)

Any point on z-axis is  $x = 0, y = 0$

$$\Rightarrow x^2 = 0 \text{ and } y^2 = 0$$

$$\Rightarrow x^2 + y^2 = 0$$

37. The coordinates of the point equidistant from the points  $A(0, 0, 0)$ ,  $B(4, 0, 0)$ ,  $C(0, 6, 0)$  and  $D(0, 0, 8)$  is

- (1)  $(3, 5, 1)$  (2)  $(2, 6, 8)$  (3)  $(-2, 3, 4)$  (4)  $(2, 3, 4)$

**Sol.** Answer (4)

Let the point  $P(x, y, z)$  be equal from the given points  $A(0, 0, 0)$ ,  $B(4, 0, 0)$ ,  $C(0, 6, 0)$  and  $D(0, 0, 8)$ , then

$$PA = PB = PC = PD$$

$$\text{Now, } PA = PB \Rightarrow (PA)^2 = (PB)^2$$

$$\Rightarrow x^2 + y^2 + z^2 = (x - 4)^2 + y^2 + z^2$$

$$\Rightarrow (x - 4)^2 = x^2$$

$$\Rightarrow x = 2$$

$$\text{Also, } PA = PC \Rightarrow (PA)^2 = (PC)^2$$

$$\Rightarrow x^2 + y^2 + z^2 = x^2 + (y - 6)^2 + z^2$$

$$\Rightarrow y^2 = (y - 6)^2$$

$$\Rightarrow y = 3$$

$$\text{Again, } PA = PD \Rightarrow (PA)^2 = (PD)^2$$

$$\Rightarrow x^2 + y^2 + z^2 = x^2 + y^2 + (z - 8)^2$$

$$\Rightarrow z^2 = (z - 8)^2$$

$$\Rightarrow z = 4$$

Hence, the required coordinates are  $(2, 3, 4)$ .

38. The plane-XY divides the join of  $(1, -1, 5)$  and  $(2, 3, 4)$  in the ratio  $k : 1$ , then  $k$  is

- (1)  $\frac{5}{4}$  (2)  $-\frac{5}{4}$  (3)  $\frac{3}{2}$  (4)  $\frac{4}{3}$

**Sol.** Answer (2)

XY-plane divides the join of  $(1, -1, 5)$  and  $(2, 3, 4)$  in the ratio  $k : 1$ , then the coordinates of the point are

$$\left( \frac{2k+1}{k+1}, \frac{3k-1}{k+1}, \frac{4k+5}{k+1} \right)$$

Since, z-coordinate on XY-plane is zero

$$\Rightarrow \frac{4k+5}{k+1} = 0$$

$$\Rightarrow k = -\frac{5}{4}$$

39. A parallelopiped is formed by planes drawn through the points  $(1, 2, 3)$  and  $(6, 8, 18)$  parallel to the coordinate planes then which of the following is not the length of an edge of the rectangular parallelopiped?

- (1) 5 units (2) 10 units (3) 15 units (4) 6 units

**Sol.** Answer (2)

Length of the edges are given by

$$(x_2 - x_1), (y_2 - y_1) \text{ and } (z_2 - z_1)$$

$$x_2 - x_1 = 6 - 1 = 5$$

$$y_2 - y_1 = 8 - 2 = 6$$

$$z_2 - z_1 = 18 - 3 = 15$$

40. The coordinates of the point where the line joining  $(3, 4, 1)$  and  $(5, 1, 6)$  crosses the  $XY$ -plane are

- (1)  $\left(\frac{13}{5}, \frac{23}{5}, 0\right)$       (2)  $(0, 0, 0)$       (3)  $\left(-\frac{13}{5}, -\frac{23}{5}, 0\right)$       (4)  $\left(-\frac{13}{5}, \frac{23}{5}, 0\right)$

**Sol.** Answer (1)

Let the point on  $XY$ -plane divides the join of  $(3, 4, 1)$  and  $(5, 1, 6)$  in  $k : 1$ .

Therefore, the coordinates of the point are

$$\left(\frac{5k+3}{k+1}, \frac{k+4}{k+1}, \frac{6k+1}{k+1}\right) \quad \dots(i)$$

$z$ -coordinate on  $XY$ -plane is zero

$$\Rightarrow \frac{6k+1}{k+1} = 0$$

$$\Rightarrow k = \frac{-1}{6}$$

On putting the value of  $k = \frac{-1}{6}$  in (i), we get the coordinates of the point as

$$\left(\frac{13}{5}, \frac{23}{5}, 0\right)$$

41. Two vertices of a triangle are  $(3, 4, 2)$  and  $(1, 3, 3)$ . If the medians of the triangle intersect at  $(0, 3, -1)$ , then the coordinates of the third vertex of the triangle are

- (1)  $(3, 1, -8)$       (2)  $(4, 2, 8)$       (3)  $(0, 0, 0)$       (4)  $(-4, 2, -8)$

**Sol.** Answer (4)

The point of intersection of median is known as centroid. Let the third vertex is  $(x, y, z)$ , then

$$\left(\frac{3+1+x}{3}, \frac{4+3+y}{3}, \frac{2+3+z}{3}\right) = (0, 3, -1)$$

$$\Rightarrow \left(\frac{x+4}{3}, \frac{y+7}{3}, \frac{z+5}{3}\right) = (0, 3, -1)$$

$$\Rightarrow \frac{x+4}{3} = 0, \frac{y+7}{3} = 3 \text{ and } \frac{z+5}{3} = -1$$

$$\Rightarrow x = -4, y = 2 \text{ and } z = -8$$

Hence, the required point is  $(-4, 2, -8)$ .

42. If the points  $(a, b, 3)$ ,  $(2, 0, -1)$  and  $(1, -1, -3)$  are collinear, then the value of  $a$  is

- (1) 2      (2) 4      (3) 3      (4) 1

**Sol.** Answer (2)

Let  $A(a, b, 3)$ ,  $B(2, 0, -1)$  and  $C(1, -1, -3)$  be given points. Also, let  $A$  divides the  $BC$  in  $k : 1$ .

$$\text{Therefore, } \left(\frac{k+2}{k+1}, \frac{-k}{k+1}, \frac{-3k-1}{k+1}\right) = (a, b, 3)$$

$$\Rightarrow \frac{-3k-1}{k+1} = 3$$

$$\Rightarrow -3k - 1 = 3k + 3$$

$$\Rightarrow 6k = -4$$

$$\Rightarrow k = \frac{-2}{3}$$

$$\text{Now, } \frac{k+2}{k+1} = a \text{ and } \frac{-k}{k+1} = b$$

$$\Rightarrow a = 4 \text{ and } b = -2$$

43. The ratio in which the plane  $2x + 3y + 5z = 1$  divides the line segment joining the points  $(1, 0, 0)$  and  $(1, 3, -5)$  is

(1)  $7 : 8$

(2)  $13 : 12$

(3)  $15 : 1$

(4)  $1 : 15$

**Sol.** Answer (4)

Let the join of the given points  $(1, 0, 0)$  and  $(1, 3, -5)$  is divided by the plane  $2x + 3y + 5z = 1$  in  $k : 1$ .

Therefore, the coordinates of the given point are

$$\left( \frac{k+1}{k+1}, \frac{3k}{k+1}, \frac{-5k}{k+1} \right)$$

Since, the point lies on  $2x + 3y + 5z = 1$

$$\Rightarrow 2 + \frac{9k}{k+1} - \frac{25k}{k+1} = 1$$

$$\Rightarrow \frac{-16k}{k+1} = -1$$

$$\Rightarrow -16k = -k - 1$$

$$\Rightarrow k = \frac{1}{15}$$

Hence, the required ratio is  $1 : 15$  internally.

44. The coordinates of a point which divides the join of points  $(3, 3, 7)$  and  $(8, 3, 1)$  internally in the ratio  $2 : 1$  is

(1)  $\left( \frac{19}{3}, 3, 3 \right)$

(2)  $(1, -2, 3)$

(3)  $(3, 3, 0)$

(4)  $\left( \frac{19}{3}, -3, -3 \right)$

**Sol.** Answer (1)

Let  $P(x, y, z)$  divides the join of  $(3, 3, 7)$  and  $(8, 3, 1)$  in  $2 : 1$

$$\Rightarrow (x, y, z) = \left( \frac{16+3}{3}, \frac{6+3}{3}, \frac{2+7}{3} \right)$$

$$\Rightarrow (x, y, z) = \left( \frac{19}{3}, 3, 3 \right)$$

Hence, the required coordinates are  $\left( \frac{19}{3}, 3, 3 \right)$ .



45. If the distance between the points  $(a, 0, 1)$  and  $(0, 1, 2)$  is  $\sqrt{27}$  units, then the value of  $a$  is

(1) 5

(2)  $\pm 5$ (3)  $-5$ 

(4) 6

**Sol.** Answer (2)

$$\sqrt{(a-0)^2 + (0-1)^2 + (1-2)^2} = \sqrt{27}$$

$$\Rightarrow a^2 + 2 = 27$$

$$\Rightarrow a = \pm 5$$

46. The shortest distance of the point  $(1, 3, 5)$  from  $x^2 + y^2 = 0$  is

(1) 5 units

(2)  $\sqrt{10}$  units(3)  $2\sqrt{5}$  units(4)  $\frac{5}{\sqrt{2}}$  units

**Sol.** Answer (2)

$x^2 + y^2 = 0$  is the equation of  $z$ -axis.

Shortest distance from  $(1, 3, 5)$  on  $z$ -axis is

$$\sqrt{(x\text{-coordinate})^2 + (y\text{-coordinate})^2} = \sqrt{1^2 + 3^2} = \sqrt{10} \text{ units}$$

47. If the sum of the squares of the distances of the point  $(x, y, z)$  from the points  $(a, 0, 0)$  and  $(-a, 0, 0)$  is  $2c^2$ , then which one of the following is correct?

(1)  $x^2 - a^2 = c^2 - y^2 - z^2$

(2)  $x^2 + a^2 = c^2 + y^2 + z^2$

(3)  $x^2 + a^2 = 2c^2 - y^2 - z^2$

(4)  $x^2 + a^2 = c^2 - y^2 - z^2$

**Sol.** Answer (4)

Let  $P(x, y, z)$ ,  $A(a, 0, 0)$  and  $B(-a, 0, 0)$  be given points such that

$$(PA)^2 + (PB)^2 = 2c^2$$

$$\Rightarrow (x-a)^2 + y^2 + z^2 + (x+a)^2 + y^2 + z^2 = 2c^2$$

$$\Rightarrow 2[x^2 + a^2 + y^2 + z^2] = 2c^2$$

$$\Rightarrow x^2 + a^2 = c^2 - y^2 - z^2$$

48. The points  $A(0, 0, 0)$ ,  $B(2, 0, 0)$ ,  $C(1, \sqrt{3}, 0)$  and  $D\left(1, \frac{1}{\sqrt{3}}, \frac{2\sqrt{2}}{\sqrt{3}}\right)$  are the vertices of a

(1) Rhombus

(2) Square

(3) Rectangular tetrahedron

(4) None of these

**Sol.** Answer (3)

$$AB = \sqrt{2^2} = 2 \text{ units}$$

$$BC = \sqrt{(1-2)^2 + (\sqrt{3}-0)^2} = 2 \text{ units}$$

$$CD = \sqrt{\left(\frac{1}{\sqrt{3}} - \sqrt{3}\right)^2 + \left(\frac{2\sqrt{2}}{\sqrt{3}} - 0\right)^2}$$

$$= 2 \text{ units}$$

$$AD = \sqrt{(1-0)^2 + \left(\frac{1}{\sqrt{3}} - 0\right)^2 + \left(\frac{2\sqrt{2}}{\sqrt{3}} - 0\right)^2}$$

$$= \sqrt{1 + \frac{1}{3} + \frac{8}{3}}$$

$$= 2 \text{ units}$$

and diagonal,

$$AC = \sqrt{(1-0)^2 + (\sqrt{3}-0)^2}$$

$$= 2 \text{ units}$$

$$BD = \sqrt{(2-1)^2 + \left(\frac{1}{\sqrt{3}} - 0\right)^2 + \left(\frac{2\sqrt{2}}{\sqrt{3}} - 0\right)^2}$$

$$= 2 \text{ units}$$

Since, the all four sides and diagonals are equal. Therefore given points are vertices of a rectangular tetrahedron.

49. XY-plane divides the line joining the points (2, 4, 5) and (-4, 3, -2) in the ratio

(1) 3 : 5

(2) 5 : 2

(3) 1 : 3

(4) 3 : 4

**Sol.** Answer (2)

Let the XY-plane divides the join of (2, 4, 5) and (-4, 3, -2) in  $k : 1$ .

Therefore, the coordinates of the point are

$$\left( \frac{-4k+2}{k+1}, \frac{3k+4}{k+1}, \frac{-2k+5}{k+1} \right)$$

z-coordinate on XY-plane is zero

$$\Rightarrow \frac{-2k+5}{k+1} = 0$$

$$\Rightarrow k = \frac{5}{2}$$

Hence, the required is 5 : 2

50. If the points  $(-1, 3, 2)$ ,  $(-4, 2, -2)$  and  $(5, 5, y)$  are collinear, then  $y$  equals

- (1)  $-10$  (2)  $5$  (3)  $-5$  (4)  $10$

**Sol.** Answer (4)

Let  $A(-1, 3, 2)$ ,  $B(-4, 2, -2)$  and  $C(5, 5, y)$  be the given points and  $C$  divides the join of  $AB$  in  $k : 1$ .

$$\Rightarrow \left( \frac{-4k-1}{k+1}, \frac{2k+3}{k+1}, \frac{-2k+2}{k+1} \right) = (5, 5, y)$$

$$\Rightarrow \frac{-4k-1}{k+1} = 5, \frac{2k+3}{k+1} = 5 \text{ and } \frac{-2k+2}{k+1} = y$$

$$\Rightarrow k = \frac{-2}{3} \text{ and } y = 10$$

51. If the vertices of a triangle be  $(1, 1, 0)$ ,  $(1, 2, 1)$  and  $(-2, 2, -1)$ , then the centroid of the triangle is

- (1)  $\left(0, \frac{5}{3}, 0\right)$  (2)  $(5, 0, 3)$  (3)  $(0, 0, 0)$  (4)  $(1, 2, 3)$

**Sol.** Answer (1)

Centroid is given by

$$\left( \frac{1+1+(-2)}{3}, \frac{1+2+2}{3}, \frac{0+1+(-1)}{3} \right) = \left( 0, \frac{5}{3}, 0 \right)$$

52. The distance between the points  $(1, 4, 5)$  and  $(2, 2, 3)$  is

- (1) 5 units (2) 4 units (3) 3 units (4) 2 units

**Sol.** Answer (3)

Let  $A(1, 4, 5)$  and  $B(2, 2, 3)$  are given points

$$\Rightarrow AB = \sqrt{(2-1)^2 + (2-4)^2 + (3-5)^2}$$

$$= \sqrt{1+4+4}$$

$$= 3 \text{ units}$$

53. The minimum distance of the point  $(1, 2, 3)$  from  $x$ -axis is

- (1) 1 unit (2)  $\sqrt{6}$  units (3)  $\sqrt{13}$  units (4)  $\sqrt{14}$  units

**Sol.** Answer (3)

The minimum distance of the point  $(1, 2, 3)$  from  $x$ -axis is

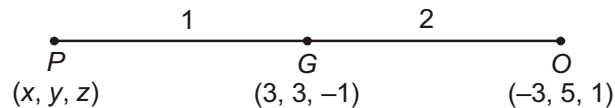
$$\sqrt{(y\text{-coordinate})^2 + (z\text{-coordinate})^2} = \sqrt{2^2 + 3^2}$$

$$= \sqrt{13} \text{ units}$$

54. If the orthocentre and centroid of a triangle are  $(-3, 5, 1)$  and  $(3, 3, -1)$  respectively, then its circumcentre is  
 (1)  $(6, 2, -2)$  (2)  $(1, 4, 0)$  (3)  $(6, 2, 2)$  (4)  $(6, -2, 2)$

**Sol.** Answer (1)

Circumcentre, centroid and orthocentre are collinear. Also centroid is a point of trisection and is closer to the circumcentre. Let the coordinates of circumcentre are  $(x, y, z)$ . Point  $G$  divides line segment  $PO$  in the ratio  $1 : 2$ .



$$\text{So, } \frac{-3+2x}{3} = 3, \frac{5+2y}{3} = 3 \text{ and } \frac{1+2z}{3} = -1$$

$$\Rightarrow x = 6, y = 2 \text{ and } z = -2$$

Hence, the required coordinates are  $(6, 2, -2)$ .

55. The angle between the lines  $2x = 3y = -z$  and  $6x = -y = -4z$  is  
 (1)  $0^\circ$  (2)  $90^\circ$  (3)  $45^\circ$  (4)  $30^\circ$

**Sol.** Answer (2)

$$2x = 3y = -z \text{ and } 6x = -y = -4z$$

$$\cos \theta = \frac{1}{12} - \frac{1}{3} + \frac{1}{4} = 0$$

$$\theta = 90^\circ.$$

56. A line makes the same angle  $\theta$  with each of  $x$  and  $z$  axis. If the angle  $\beta$ , which it makes with the  $y$ -axis, is such that  $\sin^2 \beta = 3 \sin^2 \theta$  then  $\cos^2 \theta$  equals to

- (1)  $\frac{2}{3}$  (2)  $\frac{1}{5}$  (3)  $\frac{3}{5}$  (4)  $\frac{2}{5}$

**Sol.** Answer (3)

$$2 \cos^2 \theta + \cos^2 \beta = 1$$

$$2 \cos^2 \theta = \sin^2 \beta$$

$$= 3 \sin^2 \theta$$

$$3 \sin^2 \theta = 2 (1 - \sin^2 \theta)$$

$$\sin^2 \theta = \frac{2}{5}$$

$$\text{so, } \cos^2 \theta = \frac{3}{5}$$

57. Distance between two parallel planes  $2x + y + 2z = 8$  and  $4x + 2y + 4z + 5 = 0$  is

- (1)  $\frac{3}{2}$  (2)  $\frac{5}{2}$  (3)  $\frac{7}{2}$  (4)  $\frac{9}{2}$

**Sol.** Answer (3)

$$2x + y + 2z = 8$$

$$4x + 2y + 4z = -5 \Rightarrow 2x + y + 2z = -\frac{5}{2}$$

Distance between these two planes

$$= \frac{8 + \frac{5}{2}}{3}$$

$$= \frac{21}{6} = \frac{7}{2}$$

58. The direction cosines of the normal to the plane  $x + 2y - 3z + 4 = 0$  are

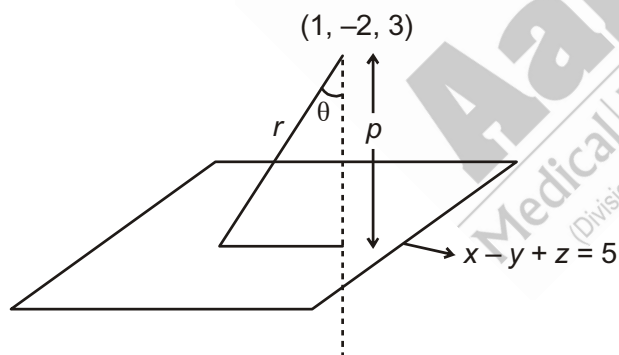
(1)  $\frac{-1}{\sqrt{14}}, \frac{-2}{\sqrt{14}}, \frac{3}{\sqrt{14}}$  (2)  $\frac{1}{\sqrt{14}}, \frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}}$  (3)  $\frac{-1}{\sqrt{14}}, \frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}}$  (4)  $\frac{1}{\sqrt{14}}, \frac{-2}{\sqrt{14}}, \frac{-3}{\sqrt{14}}$

**Sol.** Answer (1)

$$\frac{-1}{\sqrt{14}}, \frac{-2}{\sqrt{14}}, \frac{3}{\sqrt{14}}$$

59. The distance of a point  $(1, -2, 3)$  from the plane  $x - y + z = 5$  and parallel to the line  $\frac{x}{2} = \frac{y}{3} = \frac{z}{-6}$  is

(1) 7 (2) 1 (3) 3 (4) 13

**Sol.** Answer (2)

$$p = \frac{1 + 2 + 3 - 5}{\sqrt{3}} = \frac{1}{\sqrt{3}}$$

$$\text{Distance along given line} = \frac{p}{\cos \theta}$$

$$\cos \theta = \frac{2 - 3 - 6}{\sqrt{3} \sqrt{49}} = \left| \frac{1}{\sqrt{3}} \right|$$

$$\text{Distance along given line} = \frac{1}{\sqrt{3}} \times \frac{\sqrt{3}}{1} = 1$$

60. The coordinates of the point in which the line joining the points  $(2, 5, -7)$  and  $(-3, -1, 8)$  are intersected by the  $y$ - $z$  plane are

(1)  $\left(0, \frac{13}{5}, -1\right)$  (2)  $\left(0, \frac{-13}{5}, -2\right)$  (3)  $\left(0, \frac{-13}{5}, \frac{2}{5}\right)$  (4)  $\left(0, \frac{13}{5}, \frac{2}{5}\right)$

**Sol.** Answer (1)

Equation of  $y$ - $z$  plane  $x = 0$

Let this plane divide the line formed in  $\lambda : 1$  ratio coordinate of that point

$$\left(\frac{3 \times 1 - 2\lambda}{\lambda + 1}, \frac{5 - \lambda}{\lambda + 1}, \frac{-7 + 8\lambda}{\lambda + 1}\right)$$

lies on  $x = 0$

$$\text{so } 3 - 2\lambda = 0 \quad \Rightarrow \lambda = \frac{2}{3}$$

$$\text{Coordinate of point } \left(0, \frac{13}{5}, -1\right)$$

61. Find the angle between the planes  $2x - y + 3z = 6$  and  $x + y + 2z = 7$ .

(1)  $\cos^{-1}\left(\frac{\sqrt{7}}{\sqrt{3}}\right)$  (2)  $\cos^{-1}\left(\frac{1}{2}\sqrt{\frac{7}{3}}\right)$  (3)  $45^\circ$  (4)  $60^\circ$

**Sol.** Answer (2)

$$\cos \theta = \frac{2 \cdot 1 + 6}{\sqrt{14} \sqrt{6}} = \frac{\sqrt{7}}{\sqrt{12}} = \frac{1}{2} \sqrt{\frac{7}{3}}$$

62. If the lines  $\frac{x-0}{1} = \frac{y+1}{2} = \frac{z-1}{-1}$  and  $\frac{x+1}{k} = \frac{y-3}{-2} = \frac{z-2}{1}$  are at right angles, then the value of  $k$  is

(1) 5 (2) 0 (3) 3 (4) -1

**Sol.** Answer (1)

Lines are perpendicular

$$\therefore 1 \times k + 2 \times (-2) + (-1)(1) = 0$$

$$k - 4 - 1 = 0$$

$$k = 5$$

63. The lines  $\frac{x-2}{1} = \frac{y-3}{2} = \frac{z-4}{-2k}$  and  $\frac{x-1}{k} = \frac{y-2}{3} = \frac{z-6}{1}$  are coplanar if

(1)  $k = 0, k = 2$  (2)  $k = \frac{5 + \sqrt{15}}{2}, k = 3$  (3)  $k = \frac{5 \pm \sqrt{15}}{2}$  (4)  $k = 3, k = 5$

**Sol.** Answer (3)

The lines are coplanar if

$$\begin{vmatrix} 1 & 1 & -2 \\ 1 & 2 & -2k \\ k & 3 & 1 \end{vmatrix} = 0$$

$$1(2 + 6k) - 1(1 + 2k^2) - 2(3 - 2k) = 0$$

$$-5 + 10k - 2k^2 = 0$$

$$2k^2 - 10k + 5 = 0$$

$$k = \frac{10 \pm \sqrt{100 - 40}}{4} = \frac{10 \pm 2\sqrt{15}}{4} = \frac{5 \pm \sqrt{15}}{2}$$

64. If the lines  $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z}{5t-1}$  and  $\frac{x+1}{2s+1} = \frac{y}{2} = \frac{z}{4}$  are parallel to each other, then value of  $s, t$  will be

(1)  $6, \frac{5}{7}$

(2)  $\frac{1}{6}, \frac{7}{5}$

(3)  $3, \frac{5}{7}$

(4)  $4, \frac{7}{10}$

**Sol.** Answer (2)

$$\frac{x-1}{2} = \frac{y+1}{3} = \frac{z}{5t-1} \text{ and } \frac{x+1}{2s+1} = \frac{y}{2} = \frac{z}{4}$$

Or

$$\frac{x-1}{\frac{2}{3}} = \frac{y+1}{1} = \frac{z}{\frac{5t-1}{3}} \text{ and } \frac{x+1}{\frac{2s+1}{2}} = \frac{y}{1} = \frac{z}{2}$$

Now comparing direction ratios of two lines as lines are parallel.

$$\frac{2s+1}{2} = \frac{2}{3}$$

$$\frac{5t-1}{3} = 2$$

$$6s + 3 = 4$$

$$5t = 7$$

$$6s = 1 \Rightarrow s = \frac{1}{6}$$

$$t = \frac{7}{5}$$

65. Find value of  $\lambda$  if two lines having direction ratios  $1, -2, 3$  and  $2 - \lambda, 1 + \lambda, -\lambda$  are perpendicular to each other

(1) 1

(2) -1

(3) 0

(4) 2

**Sol.** Answer (3)

As given lines are perpendicular so

$$(2 - \lambda) - 2(1 + \lambda) - 3\lambda = 0$$

$$- \lambda - \lambda - 3\lambda = 0$$

$$5\lambda = 0$$

$$\lambda = 0$$

66. Find the distance of plane through  $(1, 1, 1)$  and perpendicular to the line  $\frac{x-1}{3} = \frac{y-1}{0} = \frac{z-1}{4}$  from the origin is

(1)  $\frac{3}{4}$

(2)  $\frac{4}{3}$

(3)  $\frac{7}{5}$

(4) 1

**Sol.** Answer (3)

Equation of plane will be

$$[\vec{r} - (\hat{i} + \hat{j} + \hat{k})] \cdot (3\hat{i} + 4\hat{j}) = 0$$

$$\vec{r} \cdot \frac{(3\hat{i} + 4\hat{j})}{5} = \frac{+3 + 4}{5}$$

$$\vec{r} \cdot \hat{n} = \left| \frac{+7}{5} \right|$$

67. The equation of the plane through the line of intersection of the planes  $x + y + z - 6 = 0$ ,  $2x + 3y + 4z + 5 = 0$  and the point  $(1, 1, 1)$  is

(1)  $7x - 9y + 8z = 0$

(2)  $7x + y + 8z = 0$

(3)  $2x - 2y - 3z = 14$

(4)  $20x + 23y + 26z - 69 = 0$

**Sol.** Answer (4)

Equation plane will be

$$(x + y + z - 6) + \lambda (2x + 3y + 4z + 5) = 0$$

This passes through  $(1, 1, 1)$  so will satisfy it

$$-3 + \lambda (14) = 0$$

$$\Rightarrow \lambda = \frac{3}{14}$$

Equation plane

$$20x + 23y + 26z - 69 = 0$$

68. The plane passing through the point  $(-2, -2, -3)$  and containing the line joining the points  $(1, 1, 1)$  and  $(1, -1, 2)$

makes an intercept  $a, b, c$  along  $x, y$  and  $z$  axis respectively, then  $\frac{1}{a} + \frac{1}{b} + \frac{1}{c}$  will be

(1) 3

(2) 4

(3) 6

(4) 1

**Sol.** Answer (4)

$$\begin{vmatrix} x+2 & y+2 & z+3 \\ 3 & 3 & 4 \\ 3 & 1 & 5 \end{vmatrix} = 0$$

$$11(x+2) - 3(y+2) - 6(z+3) = 0$$

$$11x - 3y - 6z - 2 = 0$$

$$11x - 3y - 6z = 2$$

$$a = \frac{2}{11}, b = \frac{-2}{3}, c = \frac{-2}{6}$$

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1$$

69. Equation of plane parallel to  $3x + 4y + 5z - 6 = 0$ ,  $6x + 8y + 10z - 16 = 0$  and equidistant from them is

(1)  $3x + 4y + 5z = 7$

(2)  $3x + 4y + 5z = 10$

(3)  $6x + 8y + 10z = 0$

(4)  $6x + 8y + 10z = 3$

**Sol.** Answer (1)

Equation of two given planes

$$3x + 4y + 5z = 6$$

$$3x + 4y + 5z = 8$$

Distance between two parallel planes

$$= \frac{2}{\sqrt{50}}$$

Equation of plane which is equidistant to the two given planes

$$\boxed{3x + 4y + 5z = 7}$$



70. If the plane  $3x - 4y + 5z = 0$  is parallel to  $\frac{2x-1}{4} = \frac{1-y}{-3} = \frac{z-2}{a}$ , then the value of  $a$  is

- (1)  $\frac{6}{4}$  (2)  $\frac{6}{5}$  (3) 1.4 (4)  $\frac{3}{4}$

**Sol.** Answer (2)

$$3 \times 2 - 4 \times 3 + 5a = 0$$

$$-6 + 5a = 0$$

$$a = \frac{6}{5}$$

71. Find the foot of perpendicular from  $(0, 2, 3)$  to the line  $\frac{x+3}{5} = \frac{y-1}{2} = \frac{z+4}{3}$ .

- (1)  $(-2, 3, 4)$  (2)  $(2, -1, 3)$  (3)  $(2, 3, -1)$  (4)  $(3, 2, -1)$

**Sol.** Answer (3)

Any point on the given line

$$x = 5\lambda - 3 \quad x = 2$$

$$y = 2\lambda + 1 \quad y = 3$$

$$z = 3\lambda - 4 \quad z = -1$$

$$5(5\lambda - 3) + 2(2\lambda - 1) + 3(3\lambda - 7) = 0$$

$$38\lambda - 38 = 0 \quad \Rightarrow \lambda = 1$$

Point will be  $(2, 3, -1)$

72. The shortest distance between the lines

$$\frac{x+1}{2} = \frac{y-1}{3} = \frac{z-2}{6} \text{ and } \frac{x-2}{4} = \frac{y+1}{1} = \frac{z-3}{7}$$

- (1)  $\frac{5}{\sqrt{14}}$  (2)  $\frac{3}{\sqrt{17}}$  (3)  $\frac{6}{\sqrt{14}}$  (4)  $\frac{2}{\sqrt{15}}$

**Sol.** Answer (2)

Shortest distance will be

$$= \frac{|(3\hat{i} - 2\hat{j} + \hat{k}) \cdot [(2\hat{i} + 3\hat{j} + 6\hat{k}) \times (4\hat{i} + \hat{j} + 7\hat{k})]|}{|(2\hat{i} + 3\hat{j} + 6\hat{k}) \times (4\hat{i} + \hat{j} + 7\hat{k})|}$$

$$= \frac{|(3\hat{i} - 2\hat{j} + \hat{k}) \cdot [15\hat{i} + 10\hat{j} - 10\hat{k}]|}{|15\hat{i} + 10\hat{j} - 10\hat{k}|}$$

$$= \frac{3}{|3\hat{i} + 2\hat{j} - 2\hat{k}|} = \frac{3}{\sqrt{17}}$$

73. Angle between diagonals of a cube can be

- (1)  $\frac{\pi}{3}$  (2)  $\frac{\pi}{4}$  (3)  $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$  (4)  $\cos^{-1}\left(\frac{1}{3}\right)$

**Sol.** Answer (4)

Angle between diagonals of a cube

$$\theta = \cos^{-1} \frac{a^2}{\sqrt{3}a \times \sqrt{3}a} = \cos^{-1}\left(\frac{1}{3}\right)$$

74. The distance of a plane  $3x + 4y - 5z = 60$  from origin

- (1)  $\frac{3}{\sqrt{50}}$  (2)  $\frac{6}{\sqrt{50}}$  (3)  $6\sqrt{2}$  (4)  $\frac{4}{5\sqrt{2}}$

**Sol.** Answer (3)

Distance of plane from origin

$$= \frac{60}{\sqrt{50}} = \frac{12}{\sqrt{2}} = 6\sqrt{2}$$

75. The equation of line through the point  $(2, -1, 3)$  and which is equally inclined to axis

- (1)  $\frac{x-2}{-\frac{1}{\sqrt{3}}} = \frac{y+1}{\frac{1}{\sqrt{3}}} = \frac{z-3}{\frac{1}{\sqrt{3}}}$  (2)  $\frac{x-2}{-1} = \frac{y+1}{1} = \frac{z-3}{-1}$   
 (3)  $\frac{x-2}{-1} = \frac{y+1}{-1} = \frac{z-3}{1}$  (4)  $\frac{x-2}{\sqrt{3}} = \frac{y+1}{\sqrt{3}} = \frac{z-3}{\sqrt{3}}$

**Sol.** Answer (4)

$$\frac{x-2}{l} = \frac{y+1}{m} = \frac{z-3}{n}$$

$$l^2 + m^2 + n^2 = 1$$

$$3l^2 = 1 \Rightarrow l = \pm \frac{1}{\sqrt{3}}$$

$$\frac{x-2}{\frac{1}{\sqrt{3}}} = \frac{y+1}{\frac{1}{\sqrt{3}}} = \frac{z-3}{\frac{1}{\sqrt{3}}} \Rightarrow \frac{x-2}{\sqrt{3}} = \frac{y+1}{\sqrt{3}} = \frac{z-3}{\sqrt{3}}$$

76. The lines  $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{3}$  and  $\frac{x+1}{2} = \frac{y}{3} = \frac{z}{4}$  are

- (1) Parallel (2) Intersecting (3) Skew (4) At right angles

**Sol.** Answer (3)

$$\begin{vmatrix} 2 & 2 & 3 \\ 1 & 2 & 3 \\ 2 & 3 & 4 \end{vmatrix} = -2 + 4 - 3 = -1$$

skew lines.

77. The equation of plane through the line of intersection of planes  $ax + by + cz + d = 0$ ,  $a'x + b'y + c'z + d' = 0$  and parallel to the line  $y = 0$ ,  $z = 0$  is

- (1)  $(ab' - a'b)x + (bc' - b'c)y + (ad' - a'd)z = 0$   
 (2)  $(ab' - a'b)x + (ad' - a'd)z = 0$   
 (3)  $(ab' - a'b)y + (ac' - a'c)z + (ad' - a'd) = 0$   
 (4)  $(ab' - a'b)x + (bc' - b'c)y + (ad' - a'd) = 0$

**Sol.** Answer (3)

78. Lines  $\vec{r} = \vec{a}_1 + \lambda \vec{b}_1$  and  $\vec{r} = \vec{a}_2 + \mu \vec{b}_2$  will lie in a plane if

(1)  $\vec{a}_1 \times \vec{a}_2 = 0$

(2)  $(\vec{a}_2 - \vec{a}_1) \times \vec{b}_1 \times \vec{b}_2 = 0$

(3)  $(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = 0$

(4)  $\vec{b}_1 \times \vec{b}_2 = 0$

**Sol.** Answer (3)

79. The equation of a plane through the point (2, 3, 1) and (4, -5, 3) and parallel to x-axis

(1)  $x + y + 4z = 7$

(2)  $y + 4z = 7$

(3)  $x - 4z = 7$

(4)  $y + 4x = -7$

**Sol.** Answer (2)

80. The line  $\frac{x-4}{1} = \frac{y-2}{1} = \frac{z-k}{2}$  lies exactly in the plane  $2x - 4y + z = 7$ , then value of  $k$  is

(1) 7

(2) -7

(3) 1

(4) No real value

**Sol.** Answer (1)

Point should lie in the plane

$k = 7$

81. The angle between diagonal of a cube and diagonal of a face of the cube will be

(1)  $\cos^{-1} \sqrt{\frac{2}{3}}$

(2)  $\sin^{-1} \sqrt{\frac{2}{3}}$

(3)  $\sin^{-1} \left( \frac{2}{3} \right)$

(4)  $\cos^{-1} \left( \frac{1}{\sqrt{3}} \right)$

**Sol.** Answer (1)

82. A point at a distance of  $\sqrt{6}$  from the origin which lies on the straight line  $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z+1}{3}$  will be

(1) (1, -1, 2)

(2) (1, 2, -1)

(3)  $\left( \frac{5}{7}, \frac{10}{7}, \frac{-6}{7} \right)$

(4)  $\left( \frac{5}{7}, \frac{2}{7}, \frac{-6}{7} \right)$

**Sol.** Answer (2)

$x = 1 + \lambda, y = 2 + 2\lambda, z = 3\lambda - 1$

Distance from origin

$= \sqrt{(1+\lambda)^2 + 4(1+\lambda)^2 + (3\lambda-1)^2} = \sqrt{6}$

$5(1+\lambda^2+2\lambda) + (9\lambda^2+1-6\lambda) = 6$

$14\lambda^2 + 4\lambda = 0$

$\lambda(14\lambda + 4) = 0 \Rightarrow \lambda = 0, \lambda = \frac{-2}{7}$

Point can be (1, 2, -1)

83. The point where the line through  $A = (3, -2, 7)$  and  $B = (13, 3, -8)$  meets the  $xz$ -plane

- (1)  $(1, 0, 7)$  (2)  $(7, 0, 1)$  (3)  $(0, 1, 7)$  (4)  $(7, 1, 0)$

**Sol.** Answer (2)

Equation of line

$$\frac{x-3}{10} = \frac{y+2}{5} = \frac{z-7}{-15} \text{ or } \frac{x-3}{2} = \frac{y+2}{1} = \frac{z-7}{-3}$$

Point on this line  $x = 2\lambda + 3, y = \lambda - 2, z = -3\lambda + 7$

$xz$ -plane  $y = 0, \lambda = 2$

Point on  $y$ - $z$  plane

$x = 7, y = 0, z = -6 + 7$

$(7, 0, 1)$

84. Find the equation of plane containing the line  $x + y - z = 0 = 2x - y + z$  and passing through the point  $(1, 2, 1)$

- (1)  $3x + 2y + z = 1$  (2)  $x - y + z = 0$  (3)  $x + y + z = 0$  (4)  $2x + 3y + z = 2$

**Sol.** Answer (2)

$$(x + y - z) + \lambda (2x - y + z) = 0$$

$$x(1 + 2\lambda) + y(1 - \lambda) + z(\lambda - 1) = 0$$

Passes through point  $(1, 2, 1)$

$$1 + 2\lambda + 2 - 2\lambda + \lambda - 1 = 0$$

$$\lambda = -2$$

$$-3x + 3y - 3z = 0$$

$$\boxed{x - y + z = 0}$$

85. The direction ratios of the normal to the plane passing through origin and line of intersection of the planes  $x + 2y + 3z = 4$  and  $4x + 3y + 2z = 1$  can be

- (1)  $(1, 2, 3)$  (2)  $(3, 2, 1)$  (3)  $(2, 3, 1)$  (4)  $(3, 1, 2)$

**Sol.** Answer (2)

$$(x + 2y + 3z - 4) + \lambda (4x + 3y + 2z - 1) = 0$$

Passes through origin so

$$-4 - \lambda = 0$$

$$\lambda = -4$$

$$(x + 2y + 3z - 4) - 4(4x + 3y + 2z - 1) = 0$$

$$-15x - 10y - 5z = 0$$

$$3x + 2y + z = 0$$

Direction ratios  $(3, 2, 1)$

86. A unit vector normal to the plane through the points  $\hat{i}, 2\hat{j}$  and  $3\hat{k}$  is

- (1)  $6\hat{i} + 3\hat{j} + 2\hat{k}$  (2)  $\hat{i} + 2\hat{j} + 3\hat{k}$  (3)  $\frac{6\hat{i} + 3\hat{j} + 2\hat{k}}{7}$  (4)  $\frac{\hat{i} + 2\hat{j} + 3\hat{k}}{7}$

**Sol.** Answer (3)

87. The lines  $x = ay + b$ ,  $z = cy + d$  and  $x = a'y + b'$ ,  $z = c'y + d'$  will be perpendicular if and only if

- (1)  $aa' + cc' + 1 = 0$       (2)  $aa' + cc' = 0$       (3)  $aa' + bb' = 0$       (4)  $aa' + bb' + cc' = 0$

**Sol.** Answer (1)

88. The angle between two lines whose direction cosines are given by  $l + m + n = 0$  and  $l^2 + m^2 - n^2 = 0$

- (1)  $\frac{\pi}{6}$       (2)  $\frac{\pi}{3}$       (3)  $\frac{\pi}{2}$       (4)  $\frac{\pi}{4}$

**Sol.** Answer (2)

$$l + m + n = 0 \quad \Rightarrow \quad l = -(m + n)$$

$$l^2 + m^2 - n^2 = 0$$

$$(m + n)^2 + m^2 - n^2 = 0 \quad \Rightarrow \quad 2m^2 + 2mn = 0$$

$$m(m + n) = 0 \quad \Rightarrow \quad m = 0, -n$$

$$m = 0 \quad m = -n$$

$$0l + 1m + 0n = 0$$

$$l + m + n = 0$$

$$\frac{l}{1-0} = \frac{m}{0-0} = \frac{n}{0-1}$$

$$0l + 1m + 1n = 0$$

$$l + m + n = 0$$

$$\frac{l}{1-1} = \frac{m}{1-0} = \frac{n}{0-1}$$

Hence direction cosines of two lines are proportional to

$$\langle 1, 0, -1 \rangle \quad \text{and} \quad \langle 0, 1, -1 \rangle$$

$$\cos \theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$$

**Alternate Method :**

$$l + m + n = 0 \quad \Rightarrow \quad n = -(l + m)$$

$$l^2 + m^2 - n^2 = 0$$

$$\Rightarrow l^2 + m^2 = (l + m)^2$$

$$2lm = 0$$

$$l = 0$$

$$m = -n$$

$$\langle 0, 1, -1 \rangle$$

$$m = 0$$

$$l = -n$$

$$\langle 1, 0, -1 \rangle$$

$$\cos \theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$$

89. The coordinates of the foot of perpendicular drawn from the point  $A(1, 8, 4)$  to the line joining the points  $B(0, -1, 3)$  and  $C(2, -3, -1)$ .

- (1)  $\left(\frac{-5}{3}, \frac{2}{3}, \frac{19}{3}\right)$       (2)  $\left(\frac{5}{3}, \frac{1}{3}, \frac{16}{3}\right)$       (3)  $\left(\frac{2}{3}, \frac{19}{3}, \frac{16}{3}\right)$       (4)  $(1, 2, 3)$

**Sol.** Answer (1)

Equation of line will be

$$\frac{x-0}{2-0} = \frac{y+1}{-3+1} = \frac{z-3}{-1-3}$$

$$\frac{x}{2} = \frac{y+1}{-2} = \frac{z-3}{-4}$$

$$\frac{x}{1} = \frac{y+1}{-1} = \frac{z-3}{-2} = \lambda$$

Any point on the given line

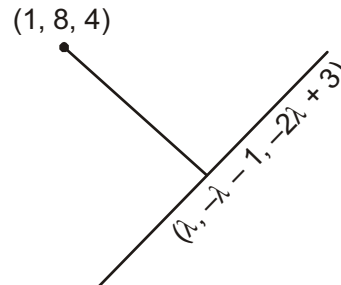
$$(\lambda, -\lambda-1, -2\lambda+3)$$

Direction ratios of a line perpendicular to the given line  $(\lambda-1, -\lambda-9, -2\lambda-1)$ To find  $\lambda$ 

$$(\lambda-1) + (\lambda+9) + 4\lambda + 2 = 0$$

$$6\lambda + 10 = 0 \Rightarrow \lambda = -\frac{5}{3}$$

$$\text{Point will be } \left( -\frac{5}{3}, \frac{2}{3}, \frac{19}{3} \right)$$



90. A line makes equal angles with the coordinate axis. The direction cosines of this line are

$$(1) \left( \frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right) \quad (2) \left( \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right) \quad (3) \left( \frac{1}{\sqrt{3}}, \frac{1}{3}, \frac{1}{3} \right) \quad (4) \left( \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)$$

**Sol.** Answer (2)91. The angle between the two lines whose d.c. are given by equations  $3l + m + 5n = 0$  and  $6mn - 2nl + 5ml = 0$ 

$$(1) \cos^{-1}\left(\frac{1}{3}\right) \quad (2) \cos^{-1}\left(\frac{1}{6}\right) \quad (3) \cos^{-1}\left(\frac{2}{3}\right) \quad (4) \cos^{-1}\left(\frac{1}{4}\right)$$

**Sol.** Answer (2)

$$3l + m + 5n = 0$$

$$6mn - 2nl + 5ml = 0$$

Similar to (37)

$$\theta = \cos^{-1}\left(\frac{1}{6}\right)$$

92. If direction cosines of a line are  $\left( \frac{1}{a}, \frac{1}{a}, \frac{1}{a} \right)$  then

$$(1) a > 0 \quad (2) 0 < a < 1 \quad (3) a = \pm\sqrt{3} \quad (4) a > 2$$

**Sol.** Answer (3)

$$\frac{3}{a^2} = 1 \Rightarrow a = \pm\sqrt{3}$$

93. The planes  $x - 3y + 4z - 1 = 0$  and  $kx - 4y + 3z - 5 = 0$  are perpendicular then value of  $k$  is

- (1) 24 (2) -24 (3) 12 (4) 0

**Sol.** Answer (2)

Planes are perpendicular

$$\therefore k + 12 + 12 = 0$$

$$k = -24$$

94. The distance between the two planes  $2x + 3y + 4z = 9$  and  $4x + 6y + 8z = 16$

- (1)  $\frac{1}{\sqrt{29}}$  (2)  $\frac{6}{\sqrt{58}}$  (3)  $\frac{4}{\sqrt{29}}$  (4)  $\frac{12}{\sqrt{76}}$

**Sol.** Answer (1)

$$d = \frac{1}{\sqrt{29}}$$

95. The equation of a plane passing through the intersection of planes  $x + 2y + 3z - 5 = 0$  and  $3x - 2y - z + 1 = 0$  axes and cutting equal intercepts on the  $ox$  and  $oz$  axes.

- (1)  $5x + 3y + 4z - 3 = 0$  (2)  $5x + 2y + 5z - 9 = 0$   
(3)  $3x + 2y + 4z - 9 = 0$  (4)  $2x - 3y + 4z + 3 = 0$

**Sol.** Answer (2)

$$x(1 + 3\lambda) + y(2 - 2\lambda) + z(3 - \lambda) - 5 + \lambda = 0$$

$$x(3\lambda + 1) + y(2 - 2\lambda) + z(3 - \lambda) = 5 - \lambda$$

$$\frac{5 - \lambda}{3\lambda + 1} = \frac{5 - \lambda}{3 - \lambda} \Rightarrow 3\lambda + 1 = 3 - \lambda$$

$$4\lambda = 2 \Rightarrow \lambda = \frac{1}{2}$$

$$\boxed{5x + 2y + 5z - 9 = 0}$$

96. The locus represented by  $xy + yz = 0$  is

- (1) Pair of perpendicular lines (2) Pair of parallel lines  
(3) Pair of parallel planes (4) Pair of perpendicular planes

**Sol.** Answer (4)

$$y(x + z) = 0$$

Pair of perpendicular planes.

97. The reflection of point  $(\alpha, \beta, \gamma)$  in the  $xy$ -plane is

- (1)  $(\alpha, \beta, 0)$  (2)  $(0, 0, \gamma)$  (3)  $(-\alpha, -\beta, \gamma)$  (4)  $(\alpha, \beta, -\gamma)$

**Sol.** Answer (4)

Reflection of  $(\alpha, \beta, \gamma)$  in  $x$ - $y$  plane =  $(\alpha, \beta, -\gamma)$

98. The sine of the angle between the straight line  $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$  and plane  $2x - 2y + z - 5 = 0$  is

- (1)  $\frac{10}{6\sqrt{5}}$  (2)  $\frac{4}{5\sqrt{2}}$  (3)  $\frac{2\sqrt{3}}{5}$  (4)  $\frac{\sqrt{2}}{10}$

**Sol.** Answer (4)

$$\theta = \sin^{-1} \frac{6-8+5}{3 \times 5\sqrt{2}}$$

$$= \sin^{-1} \left( \frac{1}{5\sqrt{2}} \right)$$

$$= \sin^{-1} \left( \frac{\sqrt{2}}{10} \right)$$

99. If the line  $\frac{x-1}{2} = \frac{y+3}{1} = \frac{z-5}{-1}$  is parallel to the plane  $px + 3y - z + 5 = 0$ , then the value of  $p$

(1) 2

(2) -2

(3)  $\frac{1}{2}$ (4)  $\frac{1}{3}$ **Sol.** Answer (2)

$$2p + 3 + 1 = 0 \Rightarrow p = -2$$

100. The vertices of a tetrahedron are  $O(0, 0, 0)$ ,  $A(1, 2, 1)$ ,  $B(2, 1, 3)$  and  $C(-1, 1, 2)$  the angle between faces  $OAB$  and  $ABC$  is

(1)  $\cos^{-1} \left( \frac{19}{35} \right)$ (2)  $\cos^{-1} \left( \frac{17}{31} \right)$ (3)  $30^\circ$ (4)  $90^\circ$ **Sol.** Answer (1)Normal to plane  $OAB$ 

$$= \overrightarrow{OA} \times \overrightarrow{OB}$$

$$= (\hat{i} + 2\hat{j} + \hat{k}) \times (2\hat{i} + \hat{j} + 3\hat{k})$$

$$= \hat{k} - 3\hat{j} - 2\hat{k} + 6\hat{i} + 2\hat{j} - \hat{i}$$

$$= 5\hat{i} - \hat{j} - 3\hat{k}$$

Normal to plane  $ABC$ 

$$= \overrightarrow{AB} \times \overrightarrow{AC}$$

$$= (\hat{i} - \hat{j} + 2\hat{k}) \times (-2\hat{i} - \hat{j} + \hat{k})$$

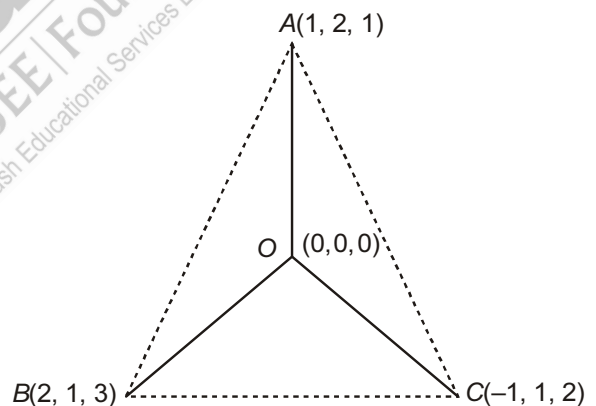
$$= -\hat{k} - \hat{j} - 2\hat{k} - \hat{i} - 4\hat{j} + \hat{i}$$

$$= \hat{i} - 5\hat{j} - 3\hat{k}$$

Angle between the plane  $OAB$  and  $ABC$ 

$$\theta = \cos^{-1} \left( \frac{19}{\sqrt{35} \sqrt{35}} \right)$$

$$= \cos^{-1} \left( \frac{19}{35} \right)$$





101. Equation of the line of the shortest distance between the lines  $\frac{x}{2} = \frac{y}{-3} = \frac{z}{1}$  and  $\frac{x-2}{3} = \frac{y-1}{-5} = \frac{z+2}{2}$  is

$$(1) \quad x - 21 = y + \frac{92}{3} = z - \frac{32}{3}$$

$$(2) \quad x - \frac{62}{3} = y - \frac{92}{3} = z + \frac{31}{3}$$

$$(3) \quad x - 21 = y - \frac{92}{3} = z + \frac{32}{3}$$

$$(4) \quad x - 2 = y + 3 = z - 1$$

**Sol.** Answer (1)

$$\frac{x}{2} = \frac{y}{-3} = \frac{z}{1} \text{ and } \frac{x-2}{-3} = \frac{y-1}{-5} = \frac{z+2}{2}$$

Line of shortest distance

$$x - 21 = y + \frac{92}{3} = z - \frac{32}{3}$$

Or

$$x - \frac{62}{3} = y + 31 = z - \frac{31}{3}$$

102. If distances of  $(-1, 2, 3)$  from  $x, y, z$  axis are  $d_1, d_2, d_3$  respectively and the distances from  $xy, yz, zx$  planes are  $d_4, d_5, d_6$  then the value of  $\sum_{i=1}^6 d_i$  is

$$(1) \quad \sqrt{5} + \sqrt{13} + \sqrt{10} + 6$$

$$(2) \quad \sqrt{2} + \sqrt{3} + 6$$

$$(3) \quad \sqrt{5} + \sqrt{13} + 7$$

$$(4) \quad \sqrt{4} + \sqrt{13} + 7$$

**Sol.** Answer (1)

$$d_1 = \sqrt{4+9} = \sqrt{13}$$

$$\text{Similarly, } d_2 = \sqrt{1+9} = \sqrt{10}, d_3 = \sqrt{1+4} = \sqrt{5}, d_4 = 3, d_5 = 1, d_6 = 2.$$

$$\text{Hence sum} = 6 + \sqrt{10} + \sqrt{5} + \sqrt{13}$$

103. Let  $A \equiv (1, 2, 3)$ ,  $B \equiv (3, 4, 5)$  then

(1) Direction ratios of  $AB$  line cannot be  $(2, 2, 2)$

(2) Direction ratios of  $\overline{AB}$  are  $(-1, 1, -1)$

(3) Direction ratios of  $\overline{AB}$  are  $(1, 1, 1)$

(4) Direction cosine of  $AB$  line may be  $\left(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right)$

**Sol.** Answer (3)

(1) Direction ratios of  $AB = (2, 2, 2)$

(3) Direction ratios of  $\overline{AB} = (1, 1, 1)$

$$\text{And direction cosine of line } AB = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) \text{ or } \left(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right)$$

104. Angle between the lines  $2x = 3y = -z$  and  $6x = -y = -4z$  is

- (1)  $0^\circ$  (2)  $30^\circ$  (3)  $60^\circ$  (4)  $90^\circ$

**Sol.** Answer (4)

$$2x = 3y = -z$$

$$\Rightarrow \frac{x}{3} = \frac{y}{2} = \frac{z}{-6} \quad \dots (i)$$

$$6x = -y = -4z$$

$$\frac{x}{2} = \frac{y}{-12} = \frac{z}{-3} \quad \dots (ii)$$

Let the angle between the lines is  $\theta$

$$\cos\theta = \frac{3 \times 2 + 12 \times (-2) + (-6)(-3)}{\sqrt{9+4+36}\sqrt{4+144+9}} = 0$$

$$\Rightarrow \theta = 90^\circ$$

105. The distance of point  $P(a, b, c)$  from the  $x$ -axis is

- (1)  $\sqrt{b^2 + c^2}$  (2)  $\sqrt{a^2 + c^2}$  (3)  $\sqrt{a^2 + b^2}$  (4)  $a$

**Sol.** Answer (1)

$$\text{The distance of } (a, b, c) \text{ from } x\text{-axis} = \sqrt{b^2 + c^2}$$

106. The equation of a line passing through  $(a, b, c)$  and parallel to  $z$ -axis is

- (1)  $\frac{x-a}{1} = \frac{y-b}{1} = \frac{z-c}{0}$  (2)  $\frac{x-a}{0} = \frac{y-b}{1} = \frac{z-c}{1}$   
 (3)  $\frac{x-a}{1} = \frac{y-b}{1} = \frac{z-c}{1}$  (4)  $\frac{x-a}{0} = \frac{y-b}{0} = \frac{z-c}{1}$

**Sol.** Answer (4)

Direction ratios of the line are  $(0, 0, 1)$

$$\text{Hence equation} = \frac{x-a}{0} = \frac{y-b}{0} = \frac{z-c}{1}$$

107. The equation of the line passing through the point  $(1, 2, -4)$  and perpendicular to lines

$$\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7} \text{ and } \frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5} \text{ is}$$

- (1)  $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+4}{6}$  (2)  $\frac{x-1}{-2} = \frac{y-2}{3} = \frac{z+4}{8}$   
 (3)  $\frac{x-1}{3} = \frac{y-2}{2} = \frac{z+4}{8}$  (4)  $\frac{x+1}{-3} = \frac{y+2}{-2} = \frac{z-4}{8}$

**Sol.** Answer (1)

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -16 & 7 \\ 3 & 8 & -5 \end{vmatrix}$$

$$= 12(2\hat{i} + 3\hat{j} + 6\hat{k})$$

So line

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+4}{6}$$

108. A line with direction cosines proportional to 2, 1, 2 meets each of the lines  $x = y + a = z$  and  $x + a = 2y = 2z$ . The co-ordinates of each of the points of intersection is

(1)  $(2a, 3a, 3a), (2a, a, a)$

(2)  $(3a, 2a, 3a), (a, a, a)$

(3)  $(3a, 2a, 3a), (a, a, 2a)$

(4)  $(3a, 3a, 3a), (a, a, a)$

**Sol.** Answer (2)

$$\frac{x}{1} = \frac{y+a}{1} = \frac{z}{1} = t_1 \quad \dots (i)$$

$$\Rightarrow x = t_1, y = t_1 - a, z = t_1$$

$$\frac{x+a}{1} = \frac{y}{1} = \frac{z}{1} = t_2 \quad \dots (ii)$$

$$x = t_2 - a, y = \frac{t_2}{2}, z = \frac{t_2}{2}$$

Let the line with direction ratio (2, 1, 2) meets (i) and (ii) at  $A(t_1, t_1 - a, t_1)$  and  $B\left(t_2 - a, \frac{t_2}{2}, \frac{t_2}{2}\right)$  respectively.

Direction ratios of  $AB$  are proportional to (2, 1, 1)

$$\Rightarrow \frac{t_2 - a - t_1}{2} = \frac{\frac{t_2}{2} - t_1 + a}{1} = \frac{\frac{t_2}{2} - t_1}{2}$$

$$\Rightarrow t_2 - a - t_1 = t_2 - 2t_1 + 2a$$

$$\text{or } t_1 = 3a$$

$$\Rightarrow A \equiv (3a, 2a, 3a)$$

$$\text{Again by } \frac{\frac{t_2}{2} - t_1 + a}{1} = \frac{\frac{t_2}{2} - t_1}{2}$$

$$t_2 - 2t_1 + 2a = \frac{t_2}{2} - t_1$$

$$\Rightarrow \frac{t_2}{2} = t_1 - 2a$$

$$t_2 = 2(t_1 - 2a) = 2(3a - 2a)$$

$$t_2 = 2a$$

$$\Rightarrow B \equiv (a, a, a)$$

109. Let  $\vec{a} = \hat{i} + \hat{j}$  and  $\vec{b} = 2\hat{i} - \hat{k}$ , then the point of intersection of the lines  $\vec{r} \times \vec{a} = \vec{b} \times \vec{a}$  and  $\vec{r} \times \vec{b} = \vec{a} \times \vec{b}$  is

- (1)  $(-1, 1, 1)$  (2)  $(3, -1, 1)$  (3)  $(3, 1, -1)$  (4)  $(1, -1, -1)$

**Sol.** Answer (3)

$$\vec{r} \times \vec{a} = \vec{b} \times \vec{a}$$

$$\Rightarrow (\vec{r} - \vec{b}) \times \vec{a} = 0$$

$\vec{r} - \vec{b}$  is parallel to  $\vec{a}$

$$\Rightarrow \vec{r} - \vec{b} = t_1 \vec{a}$$

$$\Rightarrow \vec{r} = \vec{b} + \vec{a}t_1 \quad \dots (i)$$

Similarly other line like

$$\vec{r} = \vec{a} + \vec{b}t_2 \quad \dots (ii)$$

By (i) and (ii)

$$= \vec{b} + \vec{a}t_1 = \vec{a} + \vec{b}t_2$$

$$\Rightarrow \text{Clearly } t_1 = t_2 = 1$$

Hence point of intersection in  $\vec{a} + \vec{b} = (\hat{i} + \hat{j}) + (2\hat{i} - \hat{k})$

$$= 3\hat{i} + \hat{j} - \hat{k}$$

$$= (3, 1, -1)$$

110. The lines  $\vec{r} = (\hat{i} + \hat{j} + \hat{k})\alpha + 3\hat{k}$  and  $\vec{r} = (\hat{i} - 2\hat{j} + \hat{k})\beta + 3\hat{k}$

- (1) Intersects at right angle (2) Are skew  
(3) Are parallel (4) Are not perpendicular

**Sol.** Answer (1)

$$\vec{b}_1 = \hat{i} + \hat{j} + \hat{k}$$

$$\vec{b}_2 = \hat{i} - 2\hat{j} + \hat{k}$$

$$\vec{b}_1 \cdot \vec{b}_2 = (\hat{i} + \hat{j} + \hat{k}) \cdot (\hat{i} - 2\hat{j} + \hat{k}) = 1 - 2 + 1 = 0$$

Hence lines are perpendicular and have a common point  $3\hat{k}$ . Hence the lines also intersect.

111. If  $P$  be the point  $(2, 6, 3)$  then the equation of the plane through  $P$ , at right angles to  $OP$ , where 'O' is the origin is

- (1)  $2x + 6y + 3z = 7$  (2)  $2x - 6y + 3z = 7$  (3)  $2x + 6y - 3z = 49$  (4)  $2x + 6y + 3z = 49$

**Sol.** Answer (4)

Equation of the plane passing through the point  $(x_1, y_1, z_1)$  and having DR's of normal vector as  $a, b, c$  is

$$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$$

Hence  $(x_1, y_1, z_1) \equiv (2, 6, 3)$

$$a\hat{i} + b\hat{j} + c\hat{k} = \vec{OP}$$

$$= 2\hat{i} + 6\hat{j} + 3\hat{k}$$

Equation of the plane is

$$2(x-2) + 6(y-6) + 3(z-3) = 0$$

$$2x + 6y + 3z = 4 + 36 + 9 = 49$$

112.  $P$  is a point  $(a, b, c)$ . Let  $A, B, C$  be images of  $P$  in  $y$ - $z$ ,  $z$ - $x$  and  $x$ - $y$  planes respectively, then the equation of the plane  $ABC$  is

$$(1) \quad bcx + acy + abz = 3abc$$

$$(2) \quad bcx + acy + abz = 2abc$$

$$(3) \quad bcx + acy + abz = abc$$

$$(4) \quad \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 0$$

**Sol.** Answer (3)

$$A \equiv (-a, b, c)$$

$$B \equiv (a, -b, c)$$

$$C \equiv (a, b, -c)$$

Equation of plane through  $A, B$  and  $C$  is

$$\begin{vmatrix} x+a & y-b & z-c \\ 2a & -2b & 0 \\ 2a & 0 & -2c \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} x+a & y-b & z-c \\ a & -b & 0 \\ a & 0 & -c \end{vmatrix} = 0$$

$$(x+a)(bc) - (y-b)(-ac-0) + (z-c)(ab) = 0$$

$$xbc + abc + yac - abc + zab - abc = 0$$

$$xbc + yac + zba = abc$$

113. The equation of the plane containing the line  $\frac{x+1}{-3} = \frac{y-3}{2} = \frac{z+2}{1}$  and the point  $(0, 7, -7)$  is

$$(1) \quad x + y + z = 1$$

$$(2) \quad x + y + z = 2$$

$$(3) \quad x + y + z = 0$$

$$(4) \quad x + y + z = 3$$

**Sol.** Answer (3)

Equation of the plane

$$\begin{vmatrix} x+1 & y-3 & z+2 \\ -3 & 2 & 1 \\ -1 & -4 & 5 \end{vmatrix} = 0$$

$$(x+1)(10+4) - (y-3)(-15+1) + (z+2)(12+2) = 0$$

$$14x + 14 + 14y - 42 + 14z + 28 = 0$$

$$14x + 14y + 14z = 0$$

$$x + y + z = 0$$

114. The equation of the plane which passes through the points  $(2, 1, -1)$  and  $(-1, 3, 4)$  and perpendicular to the plane  $x - 2y + 4z = 0$  is

$$(1) \quad 18x + 17y + 4z = 49$$

$$(2) \quad 18x - 17y + 4z = 49$$

$$(3) \quad 18x + 17y - 4z + 49 = 0$$

$$(4) \quad 18x + 17y + 4z + 49 = 0$$

**Sol.** Answer (1)

Let  $\vec{A} = (2, 1, -1)$

$\vec{B} = (-1, 3, 4)$

$\vec{AB} = -3\hat{i} + 2\hat{j} + 5\hat{k}$  and  $\vec{n}_1 = \hat{i} - 2\hat{j} + 4\hat{k}$

Let the normal vector for the plane is  $\vec{n}$

$\vec{n}$  is perpendicular both  $\vec{AB}$  and  $\vec{n}_1$

Hence  $\vec{n}$  can be written as

$\vec{n} = \vec{n}_1 \times \vec{AB}$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 4 \\ -3 & 2 & 5 \end{vmatrix} = \hat{i}(-18) - \hat{j}(5 + 12) + \hat{k}(2 - 6)$$

$= -18\hat{i} - 17\hat{j} - 4\hat{k}$

$\vec{n} = -18\hat{i} - 17\hat{j} - 4\hat{k}$

Equation of the plane is

$-18(x - 2) - 17(y - 1) - 4(z + 1) = 0$

$18x - 36 + 17y - 17 + 4z + 4 = 0$

$18x + 17y + 4z = 49$

115. In the plane  $2x - y + z = -3$ , the image of the point  $(1, 3, 4)$  is

- (1)  $(1, 2, 3)$  (2)  $(3, 1, 4)$  (3)  $(3, -1, 4)$  (4)  $(-3, 5, 2)$

**Sol.** Answer (4)

The image is  $(x, y, z)$

$\frac{x-1}{2} = \frac{y-3}{-1} = \frac{z-4}{1}$

$= \frac{-2(2 \times 1 - 3 + 4 + 3)}{4 + 1 + 1}$

$= \frac{-2(6)}{6} = -2$

$\frac{x-1}{2} = \frac{y-3}{-1} = \frac{z-4}{1} = -2$

$x = -4 + 1 = -3$

$y - 3 = 2$

$y = 5$

$\frac{z-4}{1} = -2$

$\Rightarrow z = 2$

Hence the points is  $(-3, 5, 2)$

116. Length of the perpendicular from origin to the plane passing through the three non-collinear points  $\vec{a}, \vec{b}, \vec{c}$  is

(1)  $\frac{[\vec{a} \vec{b} \vec{c}]}{|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|}$

(2)  $\frac{3[\vec{a} \vec{b} \vec{c}]}{|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|}$

(3)  $[\vec{a} \vec{b} \vec{c}]$

(4)  $\frac{[\vec{a} \vec{b} \vec{c}]^2}{|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|}$

**Sol.** Answer (1)

The equation of the plane passing through three non-collinear points  $\vec{a}, \vec{b}, \vec{c}$  is

$$(\vec{r} - \vec{a}) \cdot \vec{n} = 0$$

$$\text{where } \vec{n} = (\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a})$$

Length of the perpendicular from origin

$$\begin{aligned} &= \frac{|\vec{o} - \vec{a}| \cdot \vec{n}}{|\vec{n}|} = \frac{|\vec{a} \cdot \vec{n}|}{|\vec{n}|} \\ &= \frac{|\vec{a} \cdot (\vec{b} \times \vec{c})|}{|\vec{n}|} \\ &= \frac{|[\vec{a} \vec{b} \vec{c}]|}{|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|} \end{aligned}$$

117. A plane  $x - 3y + 5z = d$  passes through the point  $(1, 2, 4)$ . Intercepts on the axes are

- (1) 15, -5, 3                      (2) 1, -5, 3                      (3) -15, 5, -3                      (4) 1, -6, 20

**Sol.** Answer (1)

$$x - 3y + 5z = d$$

$$\Rightarrow 1 - 3 \times 2 + 5 \times 4 = d$$

$$\Rightarrow 1 - 6 + 20 = d \quad \Rightarrow \quad d = 15$$

Hence the plane is  $x - 3y + 5z = 15$  ... (i)

$$\Rightarrow \frac{x}{15} + \frac{y}{-5} + \frac{z}{3} = 1$$

Intercepts on the axis are 15, -5 and 3 respectively.

118. The plane  $\frac{x}{2} + \frac{y}{3} + \frac{z}{4} = 1$  cuts the axes in  $A, B$  and  $C$ , then the area of the  $\triangle ABC$  is equal to

- (1)  $\sqrt{31}$                       (2)  $\sqrt{61}$                       (3)  $\sqrt{41}$                       (4)  $\sqrt{51}$

**Sol.** Answer (2)

Point  $A$  lies on  $x$ -axis hence coordinates of  $a = (2, 0, 0)$

$$B = (0, 3, 0), C = (0, 0, 4)$$

$$\Rightarrow \vec{AB} = -2\hat{i} + 3\hat{j}$$

$$\vec{BC} = -3\hat{j} + 4\hat{k}$$

$$\text{Area of triangle } ABC = \frac{1}{2} |\vec{AB} \times \vec{BC}|$$

$$\text{Now } \vec{AB} \times \vec{BC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 3 & 0 \\ 0 & -3 & 4 \end{vmatrix}$$

$$= \hat{i}(12 - 0) - \hat{j}(-8) + \hat{k}(6)$$

$$= 12\hat{i} + 8\hat{j} + 6\hat{k}$$

$$\frac{1}{2} \overline{AB} \times \overline{BC} = 12\hat{i} + 8\hat{j} + 6\hat{k}$$

$$\text{Area} = \frac{1}{2} |\overline{AB} \times \overline{BC}| = \sqrt{36 + 16 + 9} = \sqrt{61}$$

119. The position vectors of points A and B are  $\hat{i} - \hat{j} + 3\hat{k}$  and  $3\hat{i} + 3\hat{j} - 3\hat{k}$  respectively. The equation of a plane is  $\vec{r} \cdot (5\hat{i} + 2\hat{j} - 7\hat{k}) = 0$ . The points A and B

- (1) Lie on the plane (2) Are on the same side of the plane  
(3) Are on the opposite sides of the plane (4) Nothing can be said

**Sol.** Answer (3)

$$\vec{r}_1 \cdot (5\hat{i} + 2\hat{j} - 7\hat{k}) = 5 - 2 - 21 = -ve$$

$$\vec{r}_2 \cdot (5\hat{i} + 2\hat{j} - 7\hat{k}) = 15 + 6 + 21 = +ve$$

So points lie on opposite sides of the plane.

120. The vector equation of the plane through the point (2, 1, -1) and passing through the line of intersection of the plane  $\vec{r} \cdot (\hat{i} + 3\hat{j} - \hat{k}) = 0$  and  $\vec{r} \cdot (\hat{j} + 2\hat{k}) = 0$  is

- (1)  $\vec{r} \cdot (\hat{i} + 9\hat{j} + 11\hat{k}) = 0$  (2)  $\vec{r} \cdot (\hat{i} + 9\hat{j} + 11\hat{k}) = 6$  (3)  $\vec{r} \cdot (\hat{i} - 3\hat{j} - 13\hat{k}) = 0$  (4)  $\vec{r} \cdot (2\hat{i} - 3\hat{j} - 4\hat{k}) = 0$

**Sol.** Answer (1)

The required equation of the plane is

$$\vec{r} \cdot (\hat{i} + 3\hat{j} - \hat{k}) + \lambda (\vec{r} \cdot (\hat{j} + 2\hat{k})) = 0, \text{ where } \lambda \in R$$

But  $\vec{r} = 2\hat{i} + \hat{j} - \hat{k}$ , satisfies the equation

$$\Rightarrow (2\hat{i} + \hat{j} - \hat{k}) \cdot (\hat{i} + 3\hat{j} - \hat{k}) + \lambda (2\hat{i} + \hat{j} - \hat{k}) \cdot (\hat{j} + 2\hat{k}) = 0$$

$$\Rightarrow (2 + 3 + 1) + \lambda(1 - 2) = 0$$

$$6 - \lambda = 0$$

$$\lambda = 6$$

Hence the equation of the plane is

$$\vec{r} \cdot (\hat{i} + 3\hat{j} - \hat{k}) + 6\vec{r} \cdot (\hat{j} + 2\hat{k})$$

$$\Rightarrow \vec{r} \cdot (\hat{i} + 9\hat{j} + 11\hat{k}) = 0$$

121. The equation of the line passing through the intersection of planes  $x + 2y - 3z - 4 = 0$  and  $3x - 8y + z + 2 = 0$  is

$$(1) \frac{x-2}{22} = \frac{y+1}{10} = \frac{z-0}{14}$$

$$(2) \frac{x-2}{-22} = \frac{y-1}{-10} = \frac{z-0}{-14}$$

$$(3) \frac{x+2}{22} = \frac{y+1}{10} = \frac{z-0}{14}$$

$$(4) \frac{x-2}{-22} = \frac{y+1}{-10} = \frac{z-0}{-14}$$



**Sol.** Answer (2)

$$\vec{n}_1 = \hat{i} + 2\hat{j} - 3\hat{k}$$

$$\vec{n}_2 = 3\hat{i} - 8\hat{j} + \hat{k}$$

Vector parallel to the line is  $\vec{n}$ , we can be taken  $\vec{n} = \vec{n}_1 \times \vec{n}_2$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -3 \\ 3 & -8 & 1 \end{vmatrix} = \hat{i}(2 - 24) - \hat{j}(1 + 9) + \hat{k}(-8 - 6)$$

$$= -22\hat{i} - 10\hat{j} - 14\hat{k}$$

Hence DR's of the line are  $-22, -10, -14$

or  $22, 10, 14$

To find the common point of both plane we put  $z = 0$ , in both equation.

$$x + 2y = 4 \quad \dots (i)$$

$$3x - 8y = -2 \quad \dots (ii)$$

By (i)  $\times$  (4) + (ii)

$$\Rightarrow x = 2$$

$$y = 1$$

$$z = 0$$

Hence the equation of the line is  $\frac{x-2}{22} = \frac{y-1}{10} = \frac{z-0}{14}$

122. The line  $x + 2y - z - 3 = 0 = x + 3y - z - 4$  is parallel to

(1) x-y plane

(2) y-z plane

(3) z-x plane

(4) z-axis

**Sol.** Answer (3)

The vector parallel to the plane is

$$\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -1 \\ 1 & 3 & -1 \end{vmatrix}$$

$$= \hat{i}(-2 + 3) - \hat{j}(-1 + 1) + \hat{k}(3 - 1)$$

$$= \hat{i} + \hat{k}$$

This is parallel to z-x plane.

123. The shortest distance between the lines  $x = y = z$  and the line  $2x + y + z - 1 = 0 = 3x + y + 2z - 2$  is

(1)  $\frac{1}{\sqrt{2}}$

(2)  $\sqrt{2}$

(3)  $\frac{3}{\sqrt{2}}$

(4)  $\frac{\sqrt{3}}{2}$

**Sol.** Answer (1)

$$x = y = z \quad \dots (i)$$

$$2x + y + z = 1 \quad \dots (ii)$$

$$3x + y + 2z = 2 \quad \dots (iii)$$

Normal vector to plane (ii) is  $\vec{n}_2 = 2\hat{i} + \hat{j} - \hat{k}$

Similarly normal vector to the plane (iii) is  $\vec{n}_3 = 3\hat{i} + \hat{j} + 2\hat{k}$

Vector parallel to the line of intersection is  $\vec{n}_2 \times \vec{n}_3$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 1 \\ 3 & 1 & 2 \end{vmatrix}$$

$$= \hat{i}(2-1) - \hat{j}(4-3) + \hat{k}(2-3)$$

$$= (\hat{i} - \hat{j} - \hat{k}), \text{ hence DR's of the line are } (1, -1, -1)$$

To find point on the line of intersection, we put  $z = 0$  in (ii) and (iii)

$$\Rightarrow 2x + y = 1 \quad \dots (iv)$$

$$3x + y = 2 \quad \dots (v)$$

Solving (iv) and (v)

$$x = 1, y = -1, z = 0$$

Equation of line is

$$\frac{x-1}{1} = \frac{y+1}{-1} = \frac{z-0}{-1} \quad \dots (vi)$$

Let one point on the line (i) is  $\vec{a}_1 = (0\hat{i} + 0\hat{j} + 0\hat{k})$

Similarly the point on the line is  $\vec{a}_2 = (\hat{i} - \hat{j} + 0\hat{k})$

Vector parallel to line (i) is  $\vec{b}_1 = \hat{i} + \hat{j} + \hat{k}$

Similarly the vector parallel to the line (vi) is  $\vec{b}_2 = (1, -1, -1)$

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 1 & -1 & -1 \end{vmatrix} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 0 & 0 \\ 1 & -1 & -1 \end{vmatrix}$$

$$\Rightarrow \hat{i}(0) - \hat{j}(-2) + \hat{k}(-2) = 2\hat{j} - 2\hat{k}$$

$$\begin{aligned} \text{Shortest distance} &= \left| \frac{(\hat{i} - \hat{j}) \cdot (2\hat{j} - 2\hat{k})}{2\sqrt{2}} \right| \\ &= \left| \frac{2}{2\sqrt{2}} \right| = \frac{1}{\sqrt{2}} \end{aligned}$$

124. The lines  $x + y + z - 3 = 0 = 2x - y + 5z - 6$  and  $x - y - z + 1 = 0 = 2x + 3y + 7z - k$  are coplanar then  $k$  equals

(1) 10

(2) 11

(3) 12

(4) 13

**Sol.** Answer (3)

$$x + y + z - 3 = 0 \quad \dots (i)$$

$$2x - y + 5z - 6 = 0 \quad \dots (ii)$$

If  $z = 0$

$$x + y = 3$$

$$2x - y = 6$$

$$\Rightarrow x = 3, y = 0$$

Point on the line is  $(3, 0, 0) = \vec{a}_1$

$$\vec{n}_1 = \hat{i} + \hat{j} + \hat{k}$$

$$\vec{n}_2 = 2\hat{i} - \hat{j} + 5\hat{k}$$

$$\vec{b}_1 = \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 2 & -1 & 5 \end{vmatrix}$$

$$= \hat{i}(5 + 1) - \hat{j}(5 - 2) + \hat{k}(-1 - 2)$$

$$= \vec{b}_1 = 6\hat{i} - 3\hat{j} - 3\hat{k}$$

$$x - y - z = -1$$

... (i)

$$2x + 3y + 7z = k$$

... (ii)

At  $z = 0$ ,

We get,  $x - y = -1$

$$2x + 3y = k$$

$$\text{On solving } x = \frac{k-3}{5}, y = \frac{k+2}{5}$$

Hence point on the second line is  $\left(\frac{k-3}{5}, \frac{k+2}{5}, 0\right) = \vec{a}_2$

The vector parallel to this line is

$$\vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & -1 \\ 2 & 3 & 7 \end{vmatrix}$$

$$= \hat{i}(-7 + 3) - \hat{j}(7 + 2) + \hat{k}(3 + 2)$$

$$= -4\hat{i} - 9\hat{j} + 5\hat{k}$$

If both lines are coplanar then the vector

$\vec{a}_2 - \vec{a}_1, \vec{b}_1, \vec{b}_2$  will be coplanar

$$\Rightarrow \begin{vmatrix} \frac{k-3}{5} - 3 & \frac{k+2}{5} & 0 \\ 6 & -3 & -3 \\ -4 & -9 & 5 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} k-18 & k+2 & 0 \\ 2 & -1 & -1 \\ -4 & -9 & 5 \end{vmatrix} = 0$$

$$\Rightarrow (k-18)(-5-9) - (k+2)(10-4) = 0$$

$$\Rightarrow (k-18)(-5-9) - (k+2)6 = 0$$

$$\Rightarrow (k-18)(-14) - (k+2)6 = 0$$

$$\Rightarrow -7k + 126 - 3k - 6 = 0$$

$$\Rightarrow -10k = -120$$

$$\Rightarrow k = 12$$

125. The line  $\vec{r} = \vec{a} + \lambda\vec{b}$  will not meet the plane  $\vec{r} \cdot \vec{n} = q$ , if

$$(1) \vec{b} \cdot \vec{n} = 0, \vec{a} \cdot \vec{n} = q \quad (2) \vec{b} \cdot \vec{n} \neq 0, \vec{a} \cdot \vec{n} \neq q \quad (3) \vec{b} \cdot \vec{n} = 0, \vec{a} \cdot \vec{n} \neq q \quad (4) \vec{b} \cdot \vec{n} \neq 0, \vec{a} \cdot \vec{n} = q$$

**Sol.** Answer (3)

If  $\vec{r} = \vec{a} + \lambda\vec{b}$  will not meet the plane

$$\vec{r} \cdot \vec{n} = q, \text{ then}$$

Line must be parallel to the plane but not lies in the plane

$$\Rightarrow \vec{b} \cdot \vec{n} = 0 \text{ and } \vec{a} \cdot \vec{n} \neq q$$

126. The plane  $\vec{r} \cdot \vec{n} = q$  will contain the line  $\vec{r} = \vec{a} + \lambda\vec{b}$  if

$$(1) \vec{b} \cdot \vec{n} \neq 0, \vec{a} \cdot \vec{n} = q \quad (2) \vec{b} \cdot \vec{n} \neq 0, \vec{a} \cdot \vec{n} \neq q \quad (3) \vec{b} \cdot \vec{n} = 0, \vec{a} \cdot \vec{n} \neq q \quad (4) \vec{b} \cdot \vec{n} = 0, \vec{a} \cdot \vec{n} = q$$

**Sol.** Answer (4)

If a line lies in the plane then one point of the line must satisfy the equation of plane and the line must parallel to the plane.

$$\Rightarrow \vec{b} \cdot \vec{n} = 0 \text{ and } \vec{a} \cdot \vec{n} = q$$

127. Perpendicular distance of the point (1, 2, 3) from the line  $\frac{x-6}{3} = \frac{y-7}{2} = \frac{z-7}{-2}$  is

$$(1) 5$$

$$(2) 6$$

$$(3) 7$$

$$(4) 8$$

**Sol.** Answer (3)

$$\frac{x-6}{3} = \frac{y-7}{2} = \frac{z-7}{-2} = t$$

General point on the line is  $B(3t+6, 2t+7, -2t+7)$

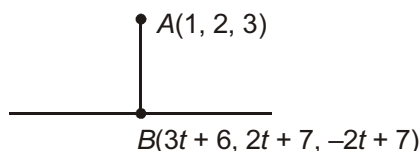
Direction ratios of AB

$$= (3t+6-1, 2t+7-2, -2t+7-3)$$

$$= (3t+5, 2t+5, -2t+4) \equiv (l_2, m_2, n_2)$$

Let direction ratios of the given line are

$$(l_1, m_1, n_1) \equiv (3, 2, -2)$$



As  $AB$  is perpendicular to the given line hence

$$l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$$

$$3(3t + 5) + 2(2t + 5) - 2(-2t + 4) = 0$$

$$9t + 15 + 4t + 10 + 4t - 8 = 0$$

$$17t + 17 = 0$$

$$\Rightarrow t = -1$$

$$\text{Hence } B \equiv (3, 5, 9)$$

$$AB = \sqrt{(3-1)^2 + (5-2)^2 + (9-3)^2} = 7$$

128. Two system of rectangular axes have the same origin. If the plane cuts the axes at distances  $a, b, c$  to the first system and  $a', b', c'$  to the second system of rectangular axes from the origin then

$$(1) \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{(a')^2} + \frac{1}{(b')^2} + \frac{1}{(c')^2}$$

$$(2) \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} + \frac{1}{(a')^2} + \frac{1}{(b')^2} + \frac{1}{(c')^2} = 0$$

$$(3) \frac{1}{a} = \frac{1}{a'}, \frac{1}{b} = \frac{1}{b'}, \frac{1}{c} = \frac{1}{c'}$$

$$(4) \frac{1}{a} = -\frac{1}{a'}, \frac{1}{b} = -\frac{1}{b'}, \frac{1}{c} = -\frac{1}{c'}$$

**Sol.** Answer (1)

Before rotation the equation is  $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

After rotation the equation is  $\frac{x}{a'} + \frac{y}{b'} + \frac{z}{c'} = 1$

As origin and plane are fixed in both cases, hence distance between origin and the plane is constant

$$\Rightarrow \frac{1}{\sqrt{\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2}}} = \frac{1}{\sqrt{\frac{x^2}{a'^2} + \frac{y^2}{b'^2} + \frac{z^2}{c'^2}}}$$

$$\Rightarrow \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{a'^2} + \frac{1}{b'^2} + \frac{1}{c'^2}$$

129. A variable plane at a constant distance ' $p$ ' from origin cuts axes at  $A, B, C$  the locus of centroid of  $\triangle ABC$  is

$$(1) x^{-2} + y^{-2} + z^{-2} = p^{-2}$$

$$(2) x^{-2} + y^{-2} + z^{-2} = 9p^{-2}$$

$$(3) x^{-2} + y^{-2} + z^{-2} = p^2$$

$$(4) x^{-2} + y^{-2} + z^{-2} = 9p^2$$

**Sol.** Answer (2)

Let the plane is  $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

Then  $A \equiv (a, 0, 0)$

$B \equiv (0, b, 0)$

$C \equiv (0, 0, c)$

Let centroid of triangle is  $(\alpha, \beta, \gamma)$

$$\Rightarrow \alpha = \frac{a+0+0}{3} \Rightarrow a = 3\alpha \quad \dots (i)$$

$$\beta = \frac{0+b+0}{3} \Rightarrow b = 3\beta \quad \dots (ii)$$

$$\gamma = \frac{0+0+\gamma}{3} \Rightarrow c = 3\gamma \quad \dots (iii)$$

Now the distance of  $(0, 0, 0)$  from the plane in  $p$

$$\Rightarrow \frac{1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}} = p^2$$

Squaring both sides

$$\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{p^2}$$

By (i), (ii), (iii)

$$\frac{1}{(3\alpha)^2} + \frac{1}{(3\beta)^2} + \frac{1}{(3\gamma)^2} = \frac{1}{p^2}$$

$$\Rightarrow \alpha^{-2} + \beta^{-2} + \gamma^{-2} = 9p^{-2}$$

Hence locus of  $(\alpha, \beta, \gamma)$  is

$$x^{-2} + y^{-2} + z^{-2} = 9p^{-2}$$

130. The planes  $2x + 5y + 3z = 0$ ,  $x - y + 4z = 2$  and  $7y - 5z + 4 = 0$

- (1) Meet in point (2) Meet in a line  
(3) Are equidistant from origin (4) None of these

**Sol.** Answer (2)

$$\Delta = \begin{vmatrix} 2 & 5 & 3 \\ 1 & -1 & 4 \\ 0 & 7 & -5 \end{vmatrix}$$

$$= 2(5 - 28) - 5(-5) + 3(7)$$

$$= -46 + 25 + 21 = 0$$

$$\text{Also } \Delta_1 = 0 = \Delta_2 = \Delta_3$$

So planes meet in a line.

131. Let  $\vec{n}$  be a unit vector perpendicular to the plane containing the point whose position vectors are  $\vec{a}$ ,  $\vec{b}$  and  $\vec{c}$  and, if

$$|[(\vec{r} - \vec{a})(\vec{b} - \vec{a})\vec{n}]| = \lambda |\vec{r} \times (\vec{b} - \vec{a}) - \vec{a} \times \vec{b}| \text{ then } \lambda \text{ is equal to}$$

- (1) 1 (2) 2 (3) 3 (4) 4

**Sol.** Answer (1)

$$\text{Clearly } \vec{n} = (\vec{a} - \vec{b}) \times (\vec{b} - \vec{c})$$

$$\vec{n} = \vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}$$

$$[(\vec{r} - \vec{a})(\vec{b} - \vec{a})\vec{n}] = \lambda [\vec{r} \times (\vec{b} - \vec{a}) - \vec{a} \times \vec{b}]$$

$$\begin{aligned}
 \text{LHS } [\vec{r} - \vec{a} \quad \vec{b} - \vec{a} \quad \vec{n}] &= [\vec{n} \quad \vec{r} - \vec{a} \quad \vec{b} - \vec{a}] \\
 &= \vec{n} \cdot [(\vec{r} - \vec{a}) \times (\vec{b} - \vec{a})] \\
 &= \vec{n} \cdot [\vec{r} \times (\vec{b} - \vec{a}) - \vec{a} \times (\vec{b} - \vec{a})] = \vec{n} \cdot [\vec{r} \times (\vec{b} - \vec{a}) - \vec{a} \times \vec{b}] \\
 \Rightarrow |\vec{r} - \vec{a} \quad \vec{b} - \vec{a} \quad \vec{n}| &= |\vec{r} \times (\vec{b} - \vec{a}) - \vec{a} \times \vec{b}| \text{ because } |\vec{n}| = 1 \\
 &= \lambda |\vec{r} \times (\vec{b} - \vec{a}) - \vec{a} \times \vec{b}| \Rightarrow \lambda = 1
 \end{aligned}$$

132. The co-ordinates of the point  $P$  on the line  $\vec{r} = (\hat{i} + \hat{j} + \hat{k}) + \lambda(-\hat{i} + \hat{j} - \hat{k})$  which is nearest to the origin is

- (1)  $\left(\frac{2}{3}, \frac{4}{3}, \frac{2}{3}\right)$  (2)  $\left(-\frac{2}{3}, -\frac{4}{3}, \frac{2}{3}\right)$  (3)  $\left(\frac{2}{3}, \frac{4}{3}, -\frac{2}{3}\right)$  (4)  $\left(\frac{2}{3}, \frac{2}{3}, \frac{4}{3}\right)$

**Sol.** Answer (1)

Let the point nearest to origin is  $P \quad (\hat{i} + \hat{j} + \hat{k}) + \lambda(-\hat{i} + \hat{j} - \hat{k})$

If  $O$  is origin, then  $\vec{OP} = \hat{i} + \hat{j} + \hat{k} + \lambda(-\hat{i} + \hat{j} - \hat{k})$

$$= \hat{i}(1-\lambda) + \hat{j}(1+\lambda) + \hat{k}(1-\lambda)$$

$\vec{OP}$  is perpendicular to  $-\hat{i} + \hat{j} - \hat{k}$

$$\Rightarrow [\hat{i}(1-\lambda) + \hat{j}(1+\lambda) + \hat{k}(1-\lambda)] \cdot (-\hat{i} + \hat{j} - \hat{k}) = 0$$

$$\Rightarrow -1 + \lambda + 1 + \lambda - 1 + \lambda = 0$$

$$\Rightarrow 3\lambda = 1$$

$$\Rightarrow \lambda = \frac{1}{3}$$

$$\vec{P} = \left(\frac{2}{3}\hat{i} + \frac{4}{3}\hat{j} + \frac{2}{3}\hat{k}\right)$$

$$\Rightarrow \text{Coordinates} = \left(\frac{2}{3}, \frac{4}{3}, \frac{2}{3}\right)$$

133. The projection of the line segment joining the points  $(1, 2, 3)$  and  $(4, 5, 6)$  on the plane  $2x + y + z = 1$  is

- (1) 1 (2)  $\sqrt{3}$  (3) 5 (4) 6

**Sol.** Answer (2)

$$\vec{n} = 2\hat{i} + \hat{j} + \hat{k},$$

$$\vec{b} = (4\hat{i} + 5\hat{j} + 6\hat{k}) - (\hat{i} + 2\hat{j} + 3\hat{k})$$

$$= 3\hat{i} + 3\hat{j} + 3\hat{k}$$

Let  $l = |\vec{b}| \cos\theta$  where  $\theta$  is the angle between  $\vec{n}$  and  $\vec{b}$

$$= \frac{\vec{n} \cdot \vec{b}}{|\vec{n}|} = \frac{6 + 3 + 3}{\sqrt{6}} = \frac{12}{\sqrt{6}} = 2\sqrt{6}$$

$$\text{Length of line segment} = \sqrt{b^2 - (2\sqrt{6})^2} = \sqrt{27 - 24} = \sqrt{3}$$

134. Let  $P(3, 2, 6)$  be a point in space and  $Q$  be a point on the line  $\vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \mu(-3\hat{i} + \hat{j} + 5\hat{k})$ . Then the value of  $\mu$  for which the vector  $\overrightarrow{PQ}$  is parallel to the plane  $x - 4y + 3z = 1$  is [IIT-JEE 2009]

- (1)  $\frac{1}{4}$  (2)  $-\frac{1}{4}$  (3)  $\frac{1}{8}$  (4)  $-\frac{1}{8}$

**Sol.** Answer (1)

$$\text{Let } \overrightarrow{OQ} = (\hat{i} - \hat{j} + 2\hat{k}) + \mu(-3\hat{i} + \hat{j} + 5\hat{k})$$

$$\overrightarrow{OP} = (3\hat{i} + 2\hat{j} + 6\hat{k})$$

$$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP}$$

$$= ((\hat{i} - \hat{j} + 2\hat{k}) + \mu(-3\hat{i} + \hat{j} + 5\hat{k})) - (3\hat{i} + 2\hat{j} + 6\hat{k})$$

$$= (-2\hat{i} - 3\hat{j} - 4\hat{k}) + \mu(-3\hat{i} + \hat{j} + 5\hat{k})$$

$$\overrightarrow{PQ} = \hat{i}(-2 - 3\mu) + \hat{j}(-3 + \mu) + \hat{k}(-4 + 5\mu)$$

$$\text{Vector perpendicular to plane is } \vec{n} = \hat{i} - 4\hat{j} + 3\hat{k}$$

As  $\overrightarrow{PQ}$  is parallel to the plane

$$\Rightarrow \overrightarrow{PQ} \cdot \vec{n} = 0$$

$$\Rightarrow 1(-2 - 3\mu) - 4(-3 + \mu) + 3(-4 + 5\mu) = 0$$

$$\Rightarrow -2 - 3\mu + 12 - 4\mu - 12 + 15\mu = 0$$

$$8\mu - 2 = 0$$

$$\mu = \frac{1}{4}$$

135. A line with positive direction cosines passes through the point  $P(2, -1, 2)$  and makes equal angles with the coordinate axes. The line meets the plane  $2x + y + z = 9$  at point  $Q$ . The length of the line segment  $PQ$  equals [IIT-JEE 2009]

- (1) 1 (2)  $\sqrt{2}$  (3)  $\sqrt{3}$  (4) 2

**Sol.** Answer (3)

The equation of given line is  $\frac{x-2}{1} = \frac{y+1}{1} = \frac{z-2}{1}$  (as direction cosines are equal and positive)

Any points on the line may be taken as  $(r+2, r-1, r+2)$  which lies on  $2x + y + z = 9$

$$\therefore 2(r+2) + r-1 + r+2 = 9$$

$$4r + 5 = 9$$

$$r = 1$$

Point is  $(3, 0, 3)$

$$PQ = \sqrt{(3-2)^2 + (0+1)^2 + (3-2)^2}$$

$$= \sqrt{1+1+1} = \sqrt{3}$$



136. Equation of the plane containing the straight line  $\frac{x}{2} = \frac{y}{3} = \frac{z}{4}$  and perpendicular to the plane containing the straight

lines  $\frac{x}{3} = \frac{y}{4} = \frac{z}{2}$  and  $\frac{x}{4} = \frac{y}{2} = \frac{z}{3}$  is

[IIT-JEE 2010]

- (1)  $x + 2y - 2z = 0$       (2)  $3x + 2y - 2z = 0$       (3)  $x - 2y + z = 0$       (4)  $5x + 2y - 4z = 0$

**Sol.** Answer (3)

The equation of the plane containing the line  $\frac{x}{2} = \frac{y}{3} = \frac{z}{4}$  may be written as

$$ax + by + cz = 0 \text{ while } 2a + 3b + 4c = 0 \quad \dots (i)$$

The equation of the plane containing straight lines  $\frac{x}{3} = \frac{y}{4} = \frac{z}{2}$  and  $\frac{x}{4} = \frac{y}{2} = \frac{z}{3}$  can be put in the form

where  $a_1x + b_1y + c_1z = 0$

$$3a_1 + 4b_1 + 2c_1 = 0$$

$$4a_1 + 2b_1 + 3c_1 = 0$$

$$\Rightarrow \frac{a_1}{8} = \frac{b_1}{-1} = \frac{c_1}{-10}$$

Then the equation of the plane containing given two lines is

$$8x - y - 10z = 0 \quad \dots (ii)$$

Since the plane (i) is perpendicular to the plane (ii),

$$\text{hence } 8a - b - 10c = 0$$

$$\text{Also } 2a + 3b + 4c = 0$$

$$\therefore \frac{a}{-4+30} = \frac{b}{-20-32} = \frac{c}{24+2}$$

$$\Rightarrow \frac{a}{26} = \frac{b}{-52} = \frac{c}{26}$$

$$\Rightarrow \frac{a}{1} = \frac{b}{-2} = \frac{c}{1}$$

Substituting for  $a, b, c$  in equation (i), the required equation of the plane is

$$x - 2y + z = 0$$

137. If the distance of the point  $P(1, -2, 1)$  from the plane  $x + 2y - 2z = \alpha$ , where  $\alpha > 0$ , is 5, then the foot of the perpendicular from  $P$  to the plane is

[IIT-JEE 2010]

- (1)  $\left(\frac{8}{3}, \frac{4}{3}, -\frac{7}{3}\right)$       (2)  $\left(\frac{4}{3}, -\frac{4}{3}, \frac{1}{3}\right)$       (3)  $\left(\frac{1}{3}, \frac{2}{3}, \frac{10}{3}\right)$       (4)  $\left(\frac{2}{3}, -\frac{1}{3}, \frac{5}{3}\right)$

**Sol.** Answer (1)

Distance of point  $P$  from plane = 5

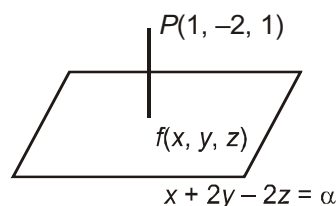
$$\therefore 5 = \left| \frac{1 - 4 - 2 - \alpha}{3} \right|$$

$$\alpha = 10$$

Foot of perpendicular

$$\frac{x-1}{1} = \frac{y+2}{2} = \frac{z-1}{-2} = \frac{5}{3}$$

$$x = \frac{8}{3}, y = \frac{4}{3}, z = -\frac{7}{3}$$



Thus the foot of the perpendicular is  $f\left(\frac{8}{3}, \frac{4}{3}, -\frac{7}{3}\right)$

138. The point  $P$  is the intersection of the straight line joining the points  $Q(2, 3, 5)$  and  $R(1, -1, 4)$  with the plane  $5x - 4y - z = 1$ . If  $S$  is the foot of the perpendicular drawn from the point  $T(2, 1, 4)$  to  $QR$ , then the length of the line segment  $PS$  is **[IIT-JEE 2012]**

- (1)  $\frac{1}{\sqrt{2}}$  (2)  $\sqrt{2}$  (3) 2 (4)  $2\sqrt{2}$

**Sol.** Answer (1)

The equation of the line  $QR$  is

$$\frac{x-2}{-1} = \frac{y-3}{-4} = \frac{z-5}{-1} = k$$

so any arbitrary point is  $(-k+2, -4k+3, -k+5)$

Now this will satisfy the equation of the plane

$$5(-k+2) - 4(-4k+3) - (-k+5) = 1$$

$$\Rightarrow k = \frac{2}{3}$$

So, the point is  $\left(\frac{4}{3}, \frac{1}{3}, \frac{13}{3}\right)$

Now, let point  $S$  be  $(-\lambda+2, -4\lambda+3, -\lambda+5)$

So, D.R's of  $ST$  is  $(-\lambda, -4\lambda+2, -\lambda+1)$

$$\Rightarrow (-\lambda)(-1) - 4(-4\lambda+2) - 1(-\lambda+1) = 0$$

$$\Rightarrow \lambda = \frac{1}{2}$$

The point is  $\left(\frac{3}{2}, 1, \frac{9}{2}\right)$

So,  $PS$  is  $\frac{1}{\sqrt{2}}$

139. The equation of a plane passing through the line of intersection of the planes  $x + 2y + 3z = 2$  and  $x - y + z = 3$  and at a distance  $\frac{2}{\sqrt{3}}$  from the point  $(3, 1, -1)$  is **[IIT-JEE 2012]**

- (1)  $5x - 11y + z = 17$  (2)  $\sqrt{2}x + y = 3\sqrt{2} - 1$  (3)  $x + y + z = \sqrt{3}$  (4)  $x - \sqrt{2}y = 1 - \sqrt{2}$

**Sol.** Answer (1)

Required equation of plane is given by

$$P_1 + \lambda P_2 = 0$$

$$\Rightarrow (x + 2y + 3z - 2) + \lambda(x - y + z - 3) = 0$$

$$\Rightarrow (1 + \lambda)x + (2 - \lambda)y + (3 + \lambda)z - (2 + 3\lambda) = 0 \quad \dots(i)$$

$$\text{As, } \left| \frac{3(1 + \lambda) + (2 - \lambda) - (3 + \lambda) - (2 + 3\lambda)}{\sqrt{(1 + \lambda)^2 + (2 - \lambda)^2 + (3 + \lambda)^2}} \right| = \frac{2}{\sqrt{3}}$$

$$\Rightarrow \lambda = -\frac{7}{2}$$

So, using (i), equation of plane becomes

$$\Rightarrow \boxed{5x - 11y + z = 17}$$

140. Perpendiculars are drawn from points on the line  $\frac{x+2}{2} = \frac{y+1}{-1} = \frac{z}{3}$  to the plane  $x + y + z = 3$ . The feet of perpendiculars lie on the line [JEE(Advanced) 2013]

(1)  $\frac{x}{5} = \frac{y-1}{8} = \frac{z-2}{-13}$     (2)  $\frac{x}{2} = \frac{y-1}{3} = \frac{z-2}{-5}$     (3)  $\frac{x}{4} = \frac{y-1}{3} = \frac{z-2}{-7}$     (4)  $\frac{x}{2} = \frac{y-1}{-7} = \frac{z-2}{5}$

**Sol.** Answer (4)

$$\frac{x+2}{2} = \frac{y+1}{-1} = \frac{z}{3} = \lambda, \quad x + y + z = 3$$

Any point on this line is  $(2\lambda - 2, -\lambda - 1, 3\lambda)$

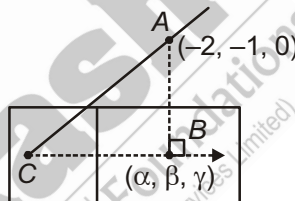
This point will satisfy the plane

$$(2\lambda - 2) + (-\lambda - 1) + (3\lambda) = 3$$

$$\Rightarrow 4\lambda - 3 = 3$$

$$\Rightarrow \lambda = \frac{3}{2}$$

So, point of intersection of plane is  $\left(1, \frac{-5}{2}, \frac{9}{2}\right) \equiv C$ .



Now, point on line is  $(-2, -1, 0)$  and direction ratio of  $AB$  (from figure) is  $\frac{\alpha + 2}{1} = \frac{\beta + 1}{1} = \frac{\gamma}{1} = k$ .

Any general point on line  $AB$  is  $(k - 2, k - 1, k)$ .

This will satisfy equation so,  $(k - 2) + (k - 1) + k = 3$ .

$$\Rightarrow k = 2$$

Therefore,  $(\alpha, \beta, \gamma) \equiv (0, 1, 2)$ .

So, equation of line passing through  $BC$  is

$$\frac{x}{1} = \frac{y-1}{-7} = \frac{z-2}{5}$$

$$\Rightarrow \frac{x}{2} = \frac{y-1}{-7} = \frac{z-2}{5}$$

141. From a point  $P(\lambda, \lambda, \lambda)$ , perpendiculars  $PQ$  and  $PR$  are drawn respectively on the lines  $y = x, z = 1$  and  $y = -x, z = -1$ . If  $P$  is such that  $\angle QPR$  is a right angle, then the possible value(s) of  $\lambda$  is(are)

[JEE(Advanced)-2014]

(1)  $\sqrt{2}$     (2) 1    (3) -1    (4)  $-\sqrt{2}$

**Sol.** Answer (3)

$$\frac{x}{1} = \frac{y}{1} = \frac{z-1}{0} = K \quad \dots(i)$$

$$\Rightarrow Q : (K, K, 1)$$

$$\frac{x}{-1} = \frac{y}{1} = \frac{z+1}{0} = m \quad \dots(ii)$$

$$\Rightarrow R : (-m, m, -1)$$

PQ is perpendicular to line (i)

$$\therefore (\lambda - K) + (\lambda - K) + 0 = 0 \quad \Rightarrow K = \lambda$$

PR is perpendicular to line (ii)

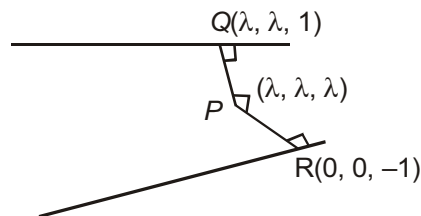
$$\therefore -1(-m + \lambda) + (m - \lambda) + 0 = 0$$

$$\Rightarrow m = 0$$

Also line PQ is perpendicular to PR

$$\Rightarrow 0 + 0 + (\lambda - 1)(\lambda + 1) = 0$$

$$\Rightarrow \lambda = \pm 1$$

 $\lambda = 1$  will be rejected as P will coincide with Q.

142. Let P be the image of the point (3, 1, 7) with respect to the plane  $x - y + z = 3$ . Then the equation of the plane passing through P and containing the straight line  $\frac{x}{1} = \frac{y}{2} = \frac{z}{1}$  is [JEE(Advanced)-2016]

$$(1) \quad x + y - 3z = 0$$

$$(2) \quad 3x + z = 0$$

$$(3) \quad x - 4y + 7z = 0$$

$$(4) \quad 2x - y = 0$$

**Sol.** Answer (3)

$$\frac{x-3}{1} = \frac{y-1}{-1} = \frac{z-7}{1} = \frac{-2(3-1+7-3)}{3} = -4$$

$$\Rightarrow x = -1, y = 5, z = 3$$

$$\therefore P(-1, 5, 3)$$

$$\therefore \text{Equation of plane is } \begin{vmatrix} x & y & z \\ 1 & 2 & 1 \\ -1 & 5 & 3 \end{vmatrix} = 0$$

$$\Rightarrow x - 4y + 7z = 0$$

143. The equation of the plane passing through the point (1, 1, 1) and perpendicular to the planes  $2x + y - 2z = 5$  and  $3x - 6y - 2z = 7$ , is [JEE(Advanced)-2017]

$$(1) \quad -14x + 2y + 15z = 3$$

$$(2) \quad 14x - 2y + 15z = 27$$

$$(3) \quad 14x + 2y - 15z = 1$$

$$(4) \quad 14x + 2y + 15z = 31$$

**Sol.** Answer (4)

$$\text{Required equation of plane is } \begin{vmatrix} x-1 & y-1 & z-1 \\ 2 & 1 & -2 \\ 3 & -6 & -2 \end{vmatrix} = 0$$

$$\Rightarrow -14(x-1) - 2(y-1) + (-15)(z-1) = 0$$

$$\Rightarrow \boxed{14x + 2y + 15z = 31}$$

## SECTION - B

## Objective Type Questions (More than One Options are correct)

1. If the distance between the points  $P(a, -8, 4)$  and  $Q(-3, -5, 4)$  is 5, then possible values of  $a$  is/are  
 (1) 1 (2) -2 (3) 3 (4) -7

**Sol.** Answer (1, 4)

$$\sqrt{(a+3)^2 + (-8+5)^2 + (4-4)^2} = 5 \Rightarrow (a+3)^2 + 9 = 25$$

$$\Rightarrow (a+3)^2 = 16 \Rightarrow a+3 = 4, -4$$

$$a = 1, -7$$

2. If  $A = (2, -3, 7)$ ,  $B = (-1, 4, -5)$  and  $P$  is a point on the line  $AB$  such that  $AP : BP = 3 : 2$ , then  $P$  has the coordinates

(1)  $\left(\frac{4}{5}, \frac{-1}{5}, \frac{11}{5}\right)$  (2)  $\left(\frac{1}{5}, \frac{6}{5}, \frac{-1}{5}\right)$  (3)  $\left(\frac{7}{5}, \frac{-18}{5}, \frac{29}{5}\right)$  (4)  $(-7, 18, -29)$

**Sol.** Answer (2, 4)

Coordinates of  $P$  are  $\left(\frac{3(-1)+2(2)}{3+2}, \frac{3(4)+2(-3)}{3+2}, \frac{3(-5)+2(7)}{3+2}\right) = \left(\frac{1}{5}, \frac{6}{5}, \frac{-1}{5}\right)$  for internal division, and

$$\left(\frac{3(-1)-2(2)}{3-2}, \frac{3(4)-2(-3)}{3-2}, \frac{3(-5)-2(7)}{3-2}\right) = (-7, 18, -29)$$
 for external division.

3. If the direction ratios of a line are  $1 + \lambda$ ,  $1 - \lambda$ , 2, and it makes an angle of  $60^\circ$  with the  $y$ -axis, then  $\lambda$  is  
 (1)  $1 + \sqrt{3}$  (2)  $2 + \sqrt{5}$  (3)  $1 - \sqrt{3}$  (4)  $2 - \sqrt{5}$

**Sol.** Answer (2, 4)

$$a_1 = 1 + \lambda, b_1 = 1 - \lambda, c_1 = 2$$

$$a_2 = 0, b_2 = 1, c_2 = 0$$

$$\cos \theta = \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{\sum a_1^2} \sqrt{\sum a_2^2}}$$

$$\Rightarrow \frac{1}{2} = \frac{0(1+\lambda) + 1(1-\lambda) + 0(2)}{\sqrt{(1+\lambda)^2 + (1-\lambda)^2 + 2^2} \sqrt{0^2 + 1^2 + 0^2}} \Rightarrow \frac{1}{2} = \frac{1-\lambda}{\sqrt{2\lambda^2 + 6}}$$

$$\Rightarrow 2\lambda^2 + 6 = 4(1-\lambda)^2 \Rightarrow 2\lambda^2 - 8\lambda - 2 = 0 \Rightarrow \lambda^2 - 4\lambda - 1 = 0 \Rightarrow \lambda = 2 \pm \sqrt{5}$$

4. The direction cosines of a line whose direction ratios are 3, 4, 12 is/are

(1)  $\frac{3}{13}, \frac{4}{13}, \frac{12}{13}$  (2)  $\frac{-3}{13}, \frac{-4}{13}, \frac{-12}{13}$  (3)  $\frac{1}{13}, \frac{2}{13}, \frac{3}{13}$  (4)  $\frac{-1}{13}, \frac{-2}{13}, \frac{-3}{13}$

**Sol.** Answer (1, 2)

$$a = 3, b = 4, c = 12$$

$$\sqrt{a^2 + b^2 + c^2} = \sqrt{3^2 + 4^2 + 12^2} = 13 \therefore l, m, n \text{ are } \pm \frac{3}{13}, \pm \frac{4}{13}, \pm \frac{12}{13}$$

5. A line segment has length 63 and direction ratios are 3, -2, 6. The components of the line vector are  
 (1) -27, 18, 54 (2) 27, -18, -54 (3) 27, -18, 54 (4) -27, 18, -54

**Sol.** Answer (3, 4)

Direction Ratio's are 3, -2, 6

$\therefore$  Direction cosine's are  $\frac{3}{7}, \frac{-2}{7}, \frac{6}{7}$

$\therefore$  Components are  $\pm \left( \frac{3}{7} \times 63, \frac{-2}{7} \times 63, \frac{6}{7} \times 63 \right) = \pm (27, -18, 54)$

6. A point Q at a distance 3 from the point P (1, 1, 1) lying on the line joining the points A (0, -1, 3) and P, has the coordinates

- (1) (2, 3, -1) (2) (4, 7, -5) (3) (0, -1, 3) (4) (-2, -5, 7)

**Sol.** Answer (1, 3)

Direction ratio's of AP are 1, 2, -2

$\therefore$  Equation of AP is

$$\frac{x-1}{1} = \frac{y-1}{2} = \frac{z-1}{-2} = r \text{ (say)}$$

To put  $r$  as the distance between points P and Q, direction ratios must be converted to direction cosines.

$$\therefore \frac{x-1}{\left(\frac{1}{3}\right)} = \frac{y-1}{\left(\frac{2}{3}\right)} = \frac{z-1}{\left(\frac{-2}{3}\right)} = r$$

Put  $r = 3$ , and  $-3$ , the coordinates of Q are (2, 3, -1) and (0, -1, 3).

7. The direction cosines of two lines are given by the equations  $3m + n + 5l = 0$ ,  $6nl - 2lm + 5mn = 0$ , then the direction cosines are

- (1)  $\frac{1}{\sqrt{6}}, \frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}$  (2)  $\frac{-1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{-1}{\sqrt{6}}$  (3)  $\frac{-1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}$  (4)  $\frac{1}{\sqrt{6}}, \frac{-1}{\sqrt{6}}, \frac{-2}{\sqrt{6}}$

**Sol.** Answer (1, 2, 3, 4)

Eliminating 'n' between the given equations

$$(6l + 5m)(3m + 5l) + 2lm = 0$$

$$\Rightarrow 30l^2 + 15m^2 + 45lm = 0 \Rightarrow 2\left(\frac{l}{m}\right)^2 + 3\left(\frac{l}{m}\right) + 1 = 0$$

$$\frac{l_1}{m_1} = -1, \frac{l_2}{m_2} = -\frac{1}{2}$$

Similarly, eliminating 'm' between the given equations  $6nl - \left(\frac{n+5l}{3}\right)(5n-2l) = 0$

$$\Rightarrow 10l^2 - 5nl - 5n^2 = 0 \Rightarrow 2\left(\frac{l}{n}\right)^2 - \left(\frac{l}{n}\right) - 1 = 0$$

$$\Rightarrow \frac{l_1}{n_1} = \frac{-1}{2}, \frac{l_2}{n_2} = +1 \quad \therefore \frac{l_1}{-1} = \frac{m_1}{1} = \frac{n_1}{2} = \pm \frac{\sqrt{l_1^2 + m_1^2 + n_1^2}}{\sqrt{(-1)^2 + 1^2 + 2^2}} = \pm \frac{1}{\sqrt{16}}$$

$$\frac{l_2}{-1} = \frac{m_2}{2} = \frac{n_2}{-1} = \pm \frac{\sqrt{l_2^2 + m_2^2 + n_2^2}}{\sqrt{(-1)^2 + 2^2 + (-1)^2}} = \pm \frac{1}{\sqrt{6}}$$

∴ The direction-cosines of the lines are

$$\frac{1}{\sqrt{6}}, \frac{-1}{\sqrt{6}}, \frac{-2}{\sqrt{6}}; \frac{-1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}; \frac{-1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{-1}{\sqrt{6}}; \frac{1}{\sqrt{6}}, \frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}$$

8. The ratio in which the sphere  $x^2 + y^2 + z^2 = 504$  divides the line joining the points (12, -4, 8) and (27, -9, 18) is

- (1) 2 : 3 internally      (2) 2 : 3 externally      (3) 3 : 4 internally      (4) 3 : 4 externally

**Sol.** Answer (1, 2)

Let a point on sphere divides joining of two points in the ratio  $\lambda : 1$

∴ Its co-ordinates are  $\frac{27\lambda + 12}{\lambda + 1}, \frac{-9\lambda - 4}{\lambda + 1}, \frac{18\lambda + 8}{\lambda + 1}$

It lies on the sphere, therefore

$$(27\lambda + 12)^2 + (-9\lambda - 4)^2 + (18\lambda + 8)^2 = 504(\lambda + 1)^2$$

$$729\lambda^2 + 144 + 648\lambda + 81\lambda^2 + 16 + 72\lambda + 324\lambda^2 + 64 + 288\lambda = 504\lambda^2 + 1008\lambda + 504$$

$$\Rightarrow 630\lambda^2 = 280$$

$$\lambda^2 = \frac{4}{9} \Rightarrow \lambda = \pm \frac{2}{3}$$

∴ (1) & (2) are correct

9. The direction cosines of a line equally inclined with the co-ordinate axes are

- (1)  $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$       (2)  $-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$       (3)  $\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$       (4)  $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}$

**Sol.** Answer (1, 2, 3, 4)

$$|\cos\alpha| = |\cos\beta| = |\cos\gamma| = k$$

$$\text{But } \cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$$

$$\Rightarrow 3k^2 = 1 \Rightarrow k = \pm \frac{1}{\sqrt{3}}$$

$$\text{Direction cosines are } \left( \pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}} \right)$$

10. Let  $PN$  be the perpendicular from the point  $P(1, 2, 3)$  to  $xy$ -plane. If  $OP$  makes an angle  $\alpha$  with positive direction of the  $z$ -axis and  $ON$  makes an angle  $\beta$  with the positive direction of  $x$ -axis, where  $O$  is the origin ( $\alpha$  and  $\beta$  are acute angles), then

- (1)  $\sin\alpha \sin\beta = \frac{2}{\sqrt{14}}$       (2)  $\cos\alpha \cos\beta = \frac{2}{\sqrt{14}}$       (3)  $\tan\alpha = \frac{\sqrt{5}}{3}$       (4)  $\tan\beta = 2$

**Sol.** Answer (1, 3, 4)

Let  $O \equiv (0, 0, 0)$

$$\text{Direction of } OP \equiv \left( \frac{1}{\sqrt{14}}, \frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}} \right)$$

$$\text{or } \left( -\frac{1}{\sqrt{14}}, -\frac{2}{\sqrt{14}}, -\frac{3}{\sqrt{14}} \right)$$

$$\text{Similarly direction of } ON = \left( \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}, 0 \right)$$

$$\text{or } \left( -\frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}}, 0 \right)$$

$$\text{Hence } \cos \alpha = \frac{3}{\sqrt{14}} \quad \text{or} \quad -\frac{3}{\sqrt{14}}$$

$$\Rightarrow \sin \alpha = \frac{\sqrt{5}}{\sqrt{14}}, -\frac{\sqrt{5}}{\sqrt{14}} \quad \text{and} \quad \cos \beta = \frac{1}{\sqrt{5}}, -\frac{1}{\sqrt{5}}$$

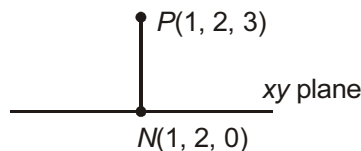
$$\Rightarrow \sin \beta = \frac{2}{\sqrt{5}}, -\frac{2}{\sqrt{5}}$$

$$\Rightarrow \sin \alpha \sin \beta = \frac{2}{\sqrt{14}},$$

$$\cos \alpha \cos \beta = \frac{3}{\sqrt{70}} \quad \text{or} \quad -\frac{3}{\sqrt{70}}$$

$$\tan \alpha = \frac{\sqrt{5}}{3} \quad \text{or} \quad -\frac{\sqrt{5}}{3}$$

$$\tan \beta = 2, -2$$



11. The lines  $\vec{r} = 2\hat{i} - 3\hat{j} + 7\hat{k} + \lambda(2\hat{i} + p\hat{j} + 5\hat{k})$  and  $\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \mu(3\hat{i} + p\hat{j} + p\hat{k})$  are perpendicular then  $p$  is equal to

(1)  $-2$

(2)  $2$

(3)  $5$

(4)  $-3$

**Sol.** Answer (1, 4)

For perpendicular lines

$$(2\hat{i} + p\hat{j} + 5\hat{k}) \cdot (3\hat{i} + p\hat{j} + p\hat{k}) = 0$$

$$\Rightarrow 6 + p^2 + 5p = 0$$

$$\Rightarrow p^2 + 5p + 6 = 0$$

$$(p + 2)(p + 3) = 0$$

$$p = -2, p = 3$$



12. The plane passing through (0, 0, 0) and containing the lines whose direction cosines proportional to 1, -2, 2 and 2, 3, -1 passes through the point

(1) (1, -2, 2)                      (2) (2, 3, -1)                      (3) (3, 1, 1)                      (4) (4, 0, 7)

**Sol.** Answer (1, 2, 3)

The equation of the plane is

$$\begin{vmatrix} x-0 & y-0 & z-0 \\ 1 & -2 & 2 \\ 2 & 3 & -1 \end{vmatrix} = 0$$

$$x(2-6) - y(-1-4) + z(3+4) = 0$$

$$-4x + 5y + 7z = 0$$

$$4x - 5y - 7z = 0 \quad \dots (i)$$

The points (1, -2, 2), (2, 3, -1) and (3, 1, 1) satisfies the equation (i), hence lie on the plane.

13. The plane passing through the point (-2, -2, 2) and containing the line joining the points (1, 1, 1) and (1, -1, 2) makes intercepts of lengths a, b, c on the axes of x, y and z respectively then

(1)  $a = 3b$                       (2)  $b = 2c$                       (3)  $a + b + c = 12$                       (4)  $a + 2b + 2c = 0$

**Sol.** Answer (1, 2, 3)

Equation of the plane is

$$\begin{vmatrix} x+2 & y+2 & z-2 \\ 3 & 3 & -1 \\ 3 & 1 & 0 \end{vmatrix} = 0$$

$$\Rightarrow (x+2)(0+1) - (y+2)(0+3) + (z-2)(3-9) = 0$$

$$\Rightarrow (x+2) - 3(y+2) + (-6)(z-2) = 0$$

$$\Rightarrow x+2 - 3y-6 - 6z+12 = 0$$

$$\Rightarrow x - 3y - 6z = -8$$

$$\Rightarrow \frac{x}{-8} + \frac{y}{8} + \frac{z}{6} = 1$$

$$\therefore a = |-8| = 8, b = \frac{8}{3}, c = \frac{8}{6} = \frac{4}{3}$$

$$\Rightarrow a = 3b, b = 2c$$

$$\Rightarrow a + b + c = 8 + \frac{8}{3} + \frac{4}{3} = 12$$

14. An equation of the line passing through  $3\hat{i} - 5\hat{j} + 7\hat{k}$  and perpendicular to the plane  $3x - 4y + 5z = 8$  is

(1)  $\frac{x-3}{3} = \frac{y+5}{-4} = \frac{z-7}{5}$

(2)  $\frac{x-3}{3} = \frac{y+4}{-5} = \frac{z-5}{7}$

(3)  $\vec{r} = (3\hat{i} - 5\hat{j} + 7\hat{k}) + \lambda(3\hat{i} - 4\hat{j} + 5\hat{k})$

(4)  $\vec{r} = (3\hat{i} - 4\hat{j} - 5\hat{k}) + \mu(3\hat{i} + 5\hat{j} + 7\hat{k})$

**Sol.** Answer (1, 3)

The vector parallel to the line is perpendicular to the plane

Hence the equation of the line is  $\frac{x-3}{3} = \frac{y+5}{-4} = \frac{z-7}{5}$  or  $\vec{r} = (3\hat{i} - 5\hat{j} + 7\hat{k}) + \lambda(3\hat{i} - 4\hat{j} + 5\hat{k})$

15. If the straight lines  $\frac{x-1}{2} = \frac{y+1}{k} = \frac{z}{2}$  and  $\frac{x+1}{5} = \frac{y+1}{2} = \frac{z}{k}$  are coplanar, then the plane(s) containing these two lines is (are) [IIT-JEE 2012]

(1)  $y + 2z = -1$                       (2)  $y + z = -1$                       (3)  $y - z = -1$                       (4)  $y - 2z = -1$

**Sol.** Answer (2, 3)

Given planes are co-planar  $\Rightarrow \begin{vmatrix} 2 & 0 & 0 \\ 2 & k & 2 \\ 5 & 2 & k \end{vmatrix} = 0$

$\Rightarrow 2(k^2 - 4) = 0$

$\Rightarrow k = \pm 2$

Equation of plane containing, these two lines are

$\begin{vmatrix} x-1 & y+1 & z \\ 2 & 2 & 2 \\ 5 & 2 & 2 \end{vmatrix} = 0$  and  $\begin{vmatrix} x-1 & y+1 & z \\ 2 & -2 & 2 \\ 5 & 2 & -2 \end{vmatrix} = 0$

$\Rightarrow y - z + 1 = 0$  and  $y + 1 + z = 0$

16. Two lines  $L_1 : x = 5, \frac{y}{3-\alpha} = \frac{z}{-2}$  and  $L_2 : x = \alpha, \frac{y}{-1} = \frac{z}{2-\alpha}$  are coplanar. Then  $\alpha$  can take value(s)

[JEE(Advanced) 2013]

(1) 1                                              (2) 2  
(3) 3                                              (4) 4

**Sol.** Answer (1, 4)

$L_1 : \frac{x-5}{0} = \frac{y}{3-\alpha} = \frac{z}{-2}$

$L_2 : \frac{x-\alpha}{0} = \frac{y}{-1} = \frac{z}{2-\alpha}$

As the two lines are coplanar so  $\begin{vmatrix} 5-\alpha & 0 & 0 \\ 0 & 3-\alpha & -2 \\ 0 & -1 & 2-\alpha \end{vmatrix} = 0$

$(5-\alpha)[(\alpha-2)(\alpha-3)-2] = 0$

$(5-\alpha)[\alpha^2 - 5\alpha + 6 - 2] = 0$

$(5-\alpha)[\alpha^2 - 5\alpha + 4] = 0$

$\alpha = 5, 4, 1$

17. A line  $l$  passing through the origin is perpendicular to the lines

$$l_1 : (3+t)\hat{i} + (-1+2t)\hat{j} + (4+2t)\hat{k}, \quad -\infty < t < \infty$$

$$l_2 : (3+2s)\hat{i} + (3+2s)\hat{j} + (2+s)\hat{k}, \quad -\infty < s < \infty$$

Then, the coordinate(s) of the point(s) on  $l_2$  at a distance of  $\sqrt{17}$  from the point of intersection of  $l$  and  $l_1$  is (are)

[JEE(Advanced) 2013]

(1)  $\left(\frac{7}{3}, \frac{7}{3}, \frac{5}{3}\right)$

(2)  $(-1, -1, 0)$

(3)  $(1, 1, 1)$

(4)  $\left(\frac{7}{9}, \frac{7}{9}, \frac{8}{9}\right)$

**Sol.** Answer (2, 4)

$$\frac{x}{a} = \frac{y}{b} = \frac{z}{c} = \lambda \quad (\text{Equation of line } l)$$

$$\text{Equation of a line } l_1, \quad \frac{x-3}{1} = \frac{y+1}{2} = \frac{z-4}{2} = t$$

$$\text{Equation of a line } l_2, \quad \frac{x-3}{2} = \frac{y-3}{2} = \frac{z-2}{1} = s$$

$$\begin{aligned} \text{Direction ratio of a line } l \text{ is given by, } & \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 2 \\ 2 & 2 & 1 \end{vmatrix} \\ & = -2\hat{i} + 3\hat{j} - 2\hat{k} \end{aligned}$$

$$\text{Equation of a line } l \text{ is } \frac{x}{-2} = \frac{y}{3} = \frac{z}{-2} = \lambda$$

Point of intersection of  $l$  and  $l_1$ ,

$$-2\lambda = 3 + t \quad \dots(1)$$

$$3\lambda = 2t - 1 \quad \dots(2)$$

$$\text{Put the value of } t, \quad 3\lambda = 2(-2\lambda - 3) - 1$$

$$3\lambda = -4\lambda - 6 - 1$$

$$7\lambda = -7$$

$$\lambda = -1$$

Point of intersection is  $(2, -3, 2)$

$$\text{So, } \sqrt{(3+2s-2)^2 + (3+2s+3)^2 + (2+s-2)^2} = \sqrt{17}$$

$$\Rightarrow 4s^2 + 4s + 1 + 36 + 24s + 4s^2 + s^2 = 17$$

$$\Rightarrow 9s^2 + 28s + 20 = 0$$

$$\Rightarrow s = -2, -\frac{10}{9}$$

$$\text{i.e., intersection points are } (-1, -1, 0) \text{ and } \left(\frac{7}{9}, \frac{7}{9}, \frac{8}{9}\right)$$

18. In  $\mathbb{R}^3$ , consider the planes  $P_1 : y = 0$  and  $P_2 : x + z = 1$ . Let  $P_3$  be a plane, different from  $P_1$  and  $P_2$ , which passes through the intersection of  $P_1$  and  $P_2$ . If the distance of the point  $(0, 1, 0)$  from  $P_3$  is 1 and the distance of a point  $(\alpha, \beta, \gamma)$  from  $P_3$  is 2, then which of the following relations is (are) true? [JEE(Advanced)-2015]

- (1)  $2\alpha + \beta + 2\gamma + 2 = 0$  (2)  $2\alpha - \beta + 2\gamma + 4 = 0$   
 (3)  $2\alpha + \beta - 2\gamma - 10 = 0$  (4)  $2\alpha - \beta + 2\gamma - 8 = 0$

**Sol.** Answer (2, 4)

$$P_1 = y = 0$$

$$P_2 = x + z = 1$$

$$P_3 = y + \lambda(x + z - 1) = 0$$

$$(\alpha, \beta, \gamma)$$

$$(0, 1, 0)$$



$$y + \lambda(x + z - 1) = 0$$

$$y + \lambda(x + z - 1) = 0$$

$$\therefore \left| \frac{\beta + \lambda(\alpha + \gamma - 1)}{\sqrt{1 + 2\lambda^2}} \right| = \pm 2 \quad \frac{\gamma + \lambda\alpha + \lambda\gamma - \lambda}{\sqrt{1 + 2\lambda^2}} = \pm 1$$

$$\therefore \beta + \lambda(\alpha + \gamma - 1) = \pm 2(1 - \lambda) \quad \frac{1 - \lambda}{\sqrt{1 + 2\lambda^2}} = \pm 1$$

$$\lambda = 0 \text{ (not possible) or } -2$$

$$\text{Putting } \lambda = -2$$

$$\therefore \beta - 2(\alpha + \gamma - 1) = \pm 2(3)$$

$$\beta - 2\alpha - 2\gamma + 2 = 6 \quad \beta - 2(\alpha + \gamma - 1) = -6$$

$$\boxed{2\alpha - \beta + 2\gamma + 4 = 0}$$

$$\boxed{2\alpha - \beta + 2\gamma - 8 = 0}$$

19. In  $\mathbb{R}^3$ , let  $L$  be a straight line passing through the origin. Suppose that all the points on  $L$  are at a constant distance from the two planes  $P_1 : x + 2y - z + 1 = 0$  and  $P_2 : 2x - y + z - 1 = 0$ . Let  $M$  be the locus of the feet of the perpendiculars drawn from the points on  $L$  to the plane  $P_1$ . Which of the following points lie(s) on  $M$ ? [JEE(Advanced)-2015]

(1)  $\left(0, -\frac{5}{6}, -\frac{2}{3}\right)$

(2)  $\left(-\frac{1}{6}, -\frac{1}{3}, \frac{1}{6}\right)$

(3)  $\left(-\frac{5}{6}, 0, \frac{1}{6}\right)$

(4)  $\left(-\frac{1}{3}, 0, \frac{2}{3}\right)$

**Sol.** Answer (1, 2)

Line  $L$  will lie on one of the angle bisector planes and will be parallel to line of intersection of the given planes.

Also locus of foot of perpendicular of line  $L$  in plane  $P_1$  will be line  $M$ , which will be parallel  $L$ .

Foot of perpendicular from  $(0, 0, 0)$  on plane  $P_1$  is  $\left(-\frac{1}{6}, -\frac{1}{3}, \frac{1}{6}\right)$

Hence equation of line  $M$

$$\frac{x + \frac{1}{6}}{1} = \frac{y + \frac{1}{3}}{-3} = \frac{z - \frac{1}{6}}{-5}$$

where  $\hat{i} - 3\hat{j} - 5\hat{k}$  is vector parallel to line of intersection of  $P_1$  and  $P_2$ .

On checking  $\left(0, -\frac{5}{6}, -\frac{2}{3}\right), \left(-\frac{1}{6}, -\frac{1}{3}, \frac{1}{6}\right)$  satisfy the above equation.

20. Let  $P_1 : 2x + y - z = 3$  and  $P_2 : x + 2y + z = 2$  be two planes. Then, which of the following statements(s) is (are) TRUE? [JEE(Advanced)-2018]

- (1) The line of intersection of  $P_1$  and  $P_2$  has direction ratios 1, 2, -1
- (2) The line  $\frac{3x-4}{9} = \frac{1-3y}{9} = \frac{z}{3}$  is perpendicular to the line of intersection of  $P_1$  and  $P_2$
- (3) The acute angle between  $P_1$  and  $P_2$  is  $60^\circ$
- (4) If  $P_3$  is the plane passing through the point  $(4, 2, -2)$  and perpendicular to the line of intersection of  $P_1$  and  $P_2$ , then the distance of the point  $(2, 1, 1)$  from the plane  $P_3$  is  $\frac{2}{\sqrt{3}}$

**Sol.** Answer (3, 4)

Equation of planes :  $P_1 : 2x + y - z = 3$

$P_2 : x + 2y + z = 2$

Let direction ratios of line of intersection of planes

$P_1$  and  $P_2$  are  $\langle a, b, c \rangle$

$$\therefore 2a + b - c = 0$$

$$\text{and } a + 2b + c = 0.$$

$$\therefore \text{Direction ratios} = \langle 1, -1, 1 \rangle$$

$$\text{The given line is : } \frac{x-4/3}{3} = \frac{y-1/3}{-3} = \frac{z}{3}$$

It is parallel to the line of intersection of  $P_1$  and  $P_2$ .

$$\therefore \cos \theta = \frac{|2+2-1|}{\sqrt{6}\sqrt{6}} = \frac{1}{2} \Rightarrow \theta = 60^\circ$$

equation of plane  $P_3$  is

$$1.(x - 4) - (y - 2) + (z + 2) = 0$$

$$\therefore P_3 : x - y + z = 0$$

Distance of plane  $P_3$  from point  $(2, 1, 1)$

$$= \left| \frac{2-1+1}{\sqrt{1+1+1}} \right| = \frac{2}{\sqrt{3}} \text{ units}$$

## SECTION - C

### Linked Comprehension Type Questions

#### Comprehension I

If the direction cosines of two lines are  $(l_1, m_1, n_1)$  and  $(l_2, m_2, n_2)$  and the angle between them is  $\theta$ , then

$$l_1^2 + m_1^2 + n_1^2 = 1 = l_2^2 + m_2^2 + n_2^2 \text{ and } \cos\theta = l_1l_2 + m_1m_2 + n_1n_2$$

Now give the answers of the following questions.

1. If  $l_1 = \frac{1}{\sqrt{3}}, m_1 = \frac{1}{\sqrt{3}}$ , then the value of  $n_1$  is equal to

(1)  $\pm \frac{1}{\sqrt{3}}$

(2)  $+\frac{1}{\sqrt{3}}$

(3)  $-\frac{1}{\sqrt{3}}$

(4) 0

**Sol.** Answer (1)

$$l_1^2 + m_1^2 + n_1^2 = 1$$

$$\Rightarrow \frac{1}{3} + \frac{1}{3} + n_1^2 = 1$$

$$\Rightarrow n_1 = \pm \frac{1}{\sqrt{3}}$$

2. If the angle between the lines is  $60^\circ$  then the value of  $l_1(l_1 + l_2) + m_1(m_1 + m_2) + n_1(n_1 + n_2)$  is

(1) 0

(2)  $\frac{3}{2}$

(3)  $\frac{1}{2}$

(4) 1

**Sol.** Answer (2)

$$\cos 60^\circ = l_1l_2 + m_1m_2 + n_1n_2$$

$$\Rightarrow l_1l_2 + m_1m_2 + n_1n_2 = \frac{1}{2}$$

$$l_1(l_1 + l_2) + m_1(m_1 + m_2) + n_1(n_1 + n_2)$$

$$= (l_1^2 + m_1^2 + n_1^2) + (l_1l_2 + m_1m_2 + n_1n_2)$$

$$= 1 + \frac{1}{2} = \frac{3}{2}$$

3. The angle between the lines whose direction cosines are  $\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}}\right)$  and  $\left(-\frac{1}{2}, -\frac{1}{2}, \frac{1}{\sqrt{2}}\right)$  is

- (1)  $0^\circ$  (2)  $60^\circ$  (3)  $90^\circ$  (4)  $120^\circ$

**Sol.** Answer (3)

$$\begin{aligned}\cos\theta &= \frac{1}{2} \times \left(-\frac{1}{2}\right) + \frac{1}{2} \times \left(-\frac{1}{2}\right) + \left(\frac{1}{\sqrt{2}}\right) \left(\frac{1}{\sqrt{2}}\right) \\ &= -\frac{1}{4} - \frac{1}{4} + \frac{1}{2} = 0\end{aligned}$$

$$\theta = 90^\circ$$

### Comprehension II

Let  $A(1, 1, 1)$ ,  $B(3, 7, 4)$  and  $C(-1, 3, 0)$  be three points in a plane.

1. The co-ordinates of the foot of perpendicular drawn from point  $C$  to the line segment  $AB$  is

- (1)  $\left(\frac{59}{49}, \frac{79}{49}, \frac{64}{49}\right)$  (2)  $\left(\frac{-59}{49}, \frac{79}{49}, \frac{64}{49}\right)$  (3)  $\left(\frac{59}{49}, \frac{-79}{49}, \frac{64}{49}\right)$  (4)  $\left(\frac{59}{49}, \frac{79}{49}, \frac{-64}{49}\right)$

**Sol.** Answer (1)

Let  $D \equiv (x, y, z)$

Then

$$\frac{x-1}{3-1} = \frac{y-1}{7-1} = \frac{z-1}{4-1}$$

$$\text{i.e. } \frac{x-1}{2} = \frac{y-1}{6} = \frac{z-1}{3} \quad \dots(1)$$

Direction ratio's of  $CD$  are  $x+1, y-3, z-0$

Since  $CD \perp AB$

$$\therefore (3-1)(x+1) + (7-1)(y-3) + (4-1)(z-0) = 0$$

$$\Rightarrow 2(x+1) + 6(y-3) + 3z = 0 \quad \dots(2)$$

By equation (1)

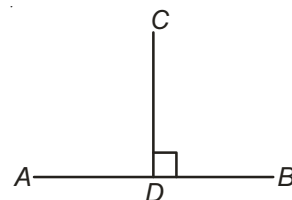
$$x = 2k + 1, y = 6k + 1, z = 3k + 1$$

Substituting in (2)

$$2(2k+2) + 6(6k-2) + 3(3k+1) = 0$$

$$\Rightarrow 49k = 5 \quad \Rightarrow k = \frac{5}{49}$$

$$\therefore x = \frac{59}{49}, y = \frac{79}{49}, z = \frac{64}{49}$$



2. The perpendicular distance from  $C$  to  $AB$  is

- (1)  $\frac{\sqrt{25312}}{49}$  (2)  $\frac{\sqrt{23571}}{49}$  (3)  $\frac{(14321)^{1/2}}{49}$  (4)  $\frac{\sqrt{20384}}{49}$

**Sol.** Answer (4)

Required distance is

$$\sqrt{\left(\frac{59}{49}+1\right)^2 + \left(\frac{79}{49}-3\right)^2 + \left(\frac{64}{49}-0\right)^2} = \frac{\sqrt{20384}}{49}$$

3. Area of the triangle  $ABC$  is (in sq. units)

- (1)  $\frac{\sqrt{23571}}{14}$  (2)  $\frac{\sqrt{20384}}{14}$  (3)  $\frac{\sqrt{25312}}{14}$  (4)  $\frac{\sqrt{14321}}{14}$

**Sol.** Answer (2)

$$\text{Distance } AB = \sqrt{(3-1)^2 + (7-1)^2 + (4-1)^2} = 7$$

$$\therefore \text{Area of } \triangle ABC = \frac{1}{2}(AB)(CD) = \frac{\sqrt{20384}}{14}$$

### Comprehension-III

Let  $A(-2, 2, 3)$  and  $B(13, -3, 13)$  be two points and  $L$  be a line passing through the point  $A$ .

1. Co-ordinates of point  $P$  which divides the join of  $A$  and  $B$  in the ratio of 2 : 3 internally are

- (1)  $\left(\frac{33}{5}, -\frac{2}{5}, 9\right)$  (2)  $(4, 0, 7)$  (3)  $\left(\frac{32}{5}, -\frac{12}{5}, \frac{17}{5}\right)$  (4)  $(20, 0, 35)$

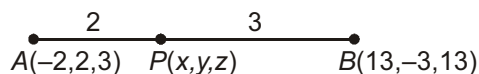
**Sol.** Answer (2)

$$x = \frac{2 \times 13 + 3 \times (-2)}{2 + 3} = \frac{26 - 6}{2 + 3} = 4$$

$$y = \frac{2 \times (-3) + 3 \times 2}{2 + 3} = 0$$

$$z = \frac{2 \times 13 + 3 \times 3}{2 + 3} = \frac{26 + 9}{5} = \frac{35}{5} = 7$$

$$P \equiv (4, 0, 7)$$



2. Equation of line  $L$ , perpendicular to line  $AB$  is

- (1)  $\frac{x+2}{15} = \frac{y-2}{-5} = \frac{z-3}{10}$  (2)  $\frac{x-2}{3} = \frac{y+2}{13} = \frac{z+3}{2}$  (3)  $\frac{x+2}{3} = \frac{y-2}{13} = \frac{z-3}{2}$  (4)  $\frac{x-2}{15} = \frac{y+2}{-5} = \frac{z+3}{10}$



**Sol.** Answer (3)

The equation of line perpendicular to  $AB$  will be that line whose parallel vector is perpendicular to the vector parallel to the given line

$$\overrightarrow{AB} = 15\hat{i} - 5\hat{j} + 10\hat{k}$$

$$\text{Let } \vec{b} = l\hat{i} + m\hat{j} + n\hat{k}$$

$$\overrightarrow{AB} \cdot \vec{b} = 0$$

$$\Rightarrow 15l - 5m + 10n$$

Only option (3) satisfies the given condition.

3. Direction ratios of the normal to the plane passing through the origin and the points  $A$  and  $B$  are

(1) 15, -5, 10

(2) 11, -1, 16

(3) 3, 13, 2

(4) 7, 13, -4

**Sol.** Answer (4)

$$\text{Let } O \equiv (0, 0, 0)$$

$$A = (-2, 2, 3)$$

$$B \equiv (13, -3, 13)$$

$$\overrightarrow{OA} = -2\hat{i} + 2\hat{j} + 3\hat{k}$$

$$\overrightarrow{OB} = 13\hat{i} - 3\hat{j} + 13\hat{k}$$

The vector which is normal to the plane containing  $O, A, B$  is the vector parallel to  $\overrightarrow{OA} \times \overrightarrow{OB}$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 2 & 3 \\ 13 & -3 & 13 \end{vmatrix}$$

$$= \hat{i}(26 + 9) - \hat{j}(-26 - 39) + \hat{k}(6 - 26)$$

$$= 35\hat{i} + 65\hat{j} - 20\hat{k}$$

$$\text{Direction ratios} = (35, 65, -20)$$

$$\text{or Direction ratios are } \equiv (7, 13, -4)$$

**Comprehension-IV**

Let  $\pi_1$  and  $\pi_2$  be two planes and  $L$  be a line such that

$$\pi_1 : x + 2y + 3z = 14 \quad \pi_2 : 2x - y + 3z = 27$$

$$L : \frac{x+1}{2} = \frac{y+1}{3} = \frac{z+1}{4}$$

1. If the line  $L$  meets the plane  $\pi_1$  in  $P(\alpha, \beta, \gamma)$ . Then  $\alpha^2 + \beta^2 + \gamma^2$  is equal to

(1) 3

(2) 14

(3) 28

(4) 29

**Sol.** Answer (2)

$$\pi_1 : x + 2y + 3z = 14, \pi_2 : 2x - y + 3z = 27$$

$$L : \frac{x+1}{1} = \frac{y+2}{2} = \frac{z+3}{3}$$

$$\text{Let } \frac{x+1}{1} = \frac{y+2}{2} = \frac{z+3}{3} = t$$

$$x = t - 1, y = 2t - 2, z = 3t - 3$$

$$\text{Let } P \equiv (t - 1, 2t - 2, 3t - 3)$$

If  $P$  lies in the plane then it satisfies the equation of the plane

$$\Rightarrow (t - 1) + 2(2t - 2) + 3(3t - 3) = 14$$

$$\Rightarrow t = 2$$

$$\text{Hence } P = (1, 2, 3)$$

$$\Rightarrow \alpha = 1, \beta = 2, \gamma = 3$$

$$\alpha^2 + \beta^2 + \gamma^2 = 1 + 4 + 9 = 14$$

2. The line through  $P$  perpendicular to the plane  $\pi_1$  passes through the point

- (1)  $(1, 1, 1)$                       (2)  $(0, 1, 0)$                       (3)  $(0, 0, 0)$                       (4)  $(0, 0, 1)$

**Sol.** Answer (3)

If line is perpendicular to  $\pi_1$ , then its direction ratio are given by the vector normal to  $\pi_1$ .

Hence direction ratios of the line is  $(1, 2, 3)$

Equation of line is

$$\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{3}$$

which is satisfied by  $(0, 0, 0)$

3. If the line in the last question meets the plane  $\pi_2$  in the point  $Q$  then the co-ordinates of mid-point of  $P$  and  $Q$  is

- (1)  $(1, 2, 3)$                       (2)  $(3, 6, 9)$                       (3)  $(2, 3, 4)$                       (4)  $(2, 4, 6)$

**Sol.** Answer (4)

$$\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{3} = t$$

$$x = t + 1$$

$$y = 2t + 2$$

$$z = 3t + 3$$

$$\text{Let } Q \equiv (t + 1, 2t + 2, 3t + 3)$$

This satisfies  $\pi_2$

$$\Rightarrow 2(t + 1) - (2t + 2) + 3(3t + 3) = 27$$

$$\Rightarrow 2t + 2 - 2t - 2 + 9t + 9 = 27$$

$$\Rightarrow 9t + 9 = 27$$

$$\Rightarrow 9t = 18$$

$$\Rightarrow t = 2$$

$$\Rightarrow Q = (3, 6, 9)$$

$$\text{and } P = (1, 2, 3)$$

$$\text{Mid-point of } P \text{ and } Q \text{ is } \left( \frac{3+1}{2}, \frac{2+6}{2}, \frac{3+9}{2} \right) = (2, 4, 6)$$

### Comprehension-V

Let  $L_1$  and  $L_2$  be two lines such that  $L_1: \frac{x+1}{-3} = \frac{y-3}{2} = \frac{z+2}{1}$ ,  $L_2: \frac{x}{1} = \frac{y-7}{-3} = \frac{z+7}{2}$

1. The lines  $L_1$  and  $L_2$  are

(1) Perpendicular

(2) Parallel

(3) Coplanar

(4) Skew lines

**Sol.** Answer (3)

$$L_1 = \frac{x+1}{-3} = \frac{y-3}{2} = \frac{z+2}{1}$$

$$L_2 = \frac{x}{1} = \frac{y-7}{-3} = \frac{z+7}{2}$$

$\Rightarrow \vec{a}, \vec{b}$  are the vectors parallel to the line  $L_1$  and  $L_2$

$$\Rightarrow \vec{a} = -3\hat{i} + 2\hat{j} + \hat{k}$$

$$\vec{b} = \hat{i} - 3\hat{j} + 2\hat{k}$$

$$\vec{a} \cdot \vec{b} = -3 - 6 + 2 = -7 \neq 0$$

Hence lines are not perpendicular

Also  $\vec{a} \neq t\vec{b}$ , hence the lines are not parallel

For co-planarity we find the value of  $\Delta$ , where

$$\Delta = \begin{vmatrix} -1-0 & 3-7 & -2+7 \\ -3 & 2 & 1 \\ 1 & -3 & 2 \end{vmatrix}$$

$$\Rightarrow \Delta = \begin{vmatrix} 1 & -4 & 5 \\ -3 & 2 & 1 \\ 1 & -3 & 2 \end{vmatrix} = -1(4+3) + 4(-6-1) + 5(9-2) = -7 - 28 + 35 = 0$$

Since  $\Delta = 0$ , hence lines are coplanar

2. The point of intersection of  $L_1$  and  $L_2$  is

- (1)  $(-3, 2, 1)$  (2)  $(2, 1, -3)$  (3)  $(1, -3, 2)$  (4)  $(0, 0, 0)$

**Sol.** Answer (2)

$$\text{Let } \frac{x+1}{-3} = \frac{y-3}{2} = \frac{z+2}{1} = t_1$$

$$x = -3t_1 - 1$$

$$y = 2t_1 + 3$$

$$z = t_1 - 2$$

$$\text{and } \frac{x}{1} = \frac{y-7}{-3} = \frac{z+7}{2} = t_2$$

$$x = t_2$$

$$y = -3t_2 + 7$$

$$z = 2t_2 - 7$$

For point of intersection of the lines

$$t_2 = -3t_1 - 1$$

$$t_2 + 3t_1 + 1 = 0 \quad \dots (i)$$

$$-3t_2 + 7 = 2t_1 + 3$$

$$-3t_2 - 2t_1 + 4 = 0$$

$$3t_2 + 2t_1 - 4 = 0 \quad \dots (ii)$$

$$2t_2 - 7 = t_1 - 2$$

$$2t_2 - t_1 = 5 \quad \dots (iii)$$

By (i) and (ii)  $t_1 = -1$ ,  $t_2 = 2$

which satisfies equation (iii)

Hence the lines intersect at the point of intersection is

$$(3t_1 - 1, 2t_1 + 3, t_1 - 2) \equiv (2, 1, -3)$$

3. Equation of a plane containing  $L_1$  and  $L_2$  is

- (1)  $x + y + z = 0$  (2)  $3x - 2y - z = 0$  (3)  $x - 3y + 2z = 0$  (4)  $x + y + z = 42$

**Sol.** Answer (1)

Equation of the plane containing  $L_1$  and  $L_2$  is

$$\begin{vmatrix} x+1 & y-3 & z+2 \\ -3 & 2 & 1 \\ 1 & -3 & 2 \end{vmatrix} = 0$$

$$\Rightarrow (x+1)(4+3) - (y-3)(-6-1) + (z+2)(9-2) = 0$$

$$\Rightarrow 7(x+1) + (y-3)(+7) + (z+2)(7) = 0$$

$$\Rightarrow (x+1) + (y-3) + (z+2) = 0$$

$$\Rightarrow x + y + z = 0$$

## Comprehension-VI

Consider the lines :  $L_1 : \frac{x+1}{3} = \frac{y+2}{1} = \frac{z+1}{2}$  ;  $L_2 : \frac{x-2}{1} = \frac{y+2}{2} = \frac{z-3}{3}$

[IIT-JEE 2008]

1. The unit vector perpendicular to both  $L_1$  and  $L_2$  is

(1)  $\frac{-\hat{i} + 7\hat{j} + 7\hat{k}}{\sqrt{99}}$  (2)  $\frac{-\hat{i} - 7\hat{j} + 5\hat{k}}{5\sqrt{3}}$  (3)  $\frac{-\hat{i} + 7\hat{j} + 5\hat{k}}{5\sqrt{3}}$  (4)  $\frac{7\hat{i} - 7\hat{j} - \hat{k}}{\sqrt{99}}$

**Sol.** Answer (2)

The equations of given lines in vector form may be written as

$$L_1 : \vec{r} = (-\hat{i} - 2\hat{j} - \hat{k}) + \lambda(3\hat{i} + \hat{j} + 2\hat{k}) \text{ and } L_2 : \vec{r} = (2\hat{i} - 2\hat{j} + 3\hat{k}) + \mu(\hat{i} + 2\hat{j} + 3\hat{k})$$

$\therefore$  Vector perpendicular to both  $L_1$  and  $L_2$  is

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & 2 \\ 1 & 2 & 3 \end{vmatrix} = -\hat{i} - 7\hat{j} + 5\hat{k}$$

$$\therefore \text{ Required unit vector} = \frac{(-\hat{i} - 7\hat{j} + 5\hat{k})}{\sqrt{(-1)^2 + (-7)^2 + (5)^2}} = \frac{1}{5\sqrt{3}}(-\hat{i} - 7\hat{j} + 5\hat{k})$$

2. The shortest distance between  $L_1$  and  $L_2$  is

(1) 0 (2)  $\frac{17}{\sqrt{3}}$  (3)  $\frac{41}{5\sqrt{3}}$  (4)  $\frac{17}{5\sqrt{3}}$

**Sol.** Answer (4)

Shortest distance between  $L_1$  and  $L_2$  is

$$\left| \frac{\{(2 - (-1))\hat{i} + (2 - 2)\hat{j} + (3 - (-1))\hat{k}\} \cdot (-\hat{i} - 7\hat{j} + 5\hat{k})}{5\sqrt{3}} \right| = \left| \frac{(3\hat{i} + 4\hat{k}) \cdot (-\hat{i} - 7\hat{j} + 5\hat{k})}{5\sqrt{3}} \right| = \frac{17}{5\sqrt{3}} \text{ units}$$

3. The distance of the point (1, 1, 1) from the plane passing through the point (-1, -2, -1) and whose normal is perpendicular to both the lines  $L_1$  and  $L_2$  is

(1)  $\frac{2}{\sqrt{75}}$  (2)  $\frac{7}{\sqrt{75}}$  (3)  $\frac{13}{\sqrt{75}}$  (4)  $\frac{23}{\sqrt{75}}$

**Sol.** Answer (3)

The equation of the plane passing through the point (-1, -2, -1) and whose normal is perpendicular to both the given lines  $L_1$  and  $L_2$  may be written as

$$(x + 1) + 7(y + 2) - 5(z + 1) = 0$$

$$\text{i.e., } x + 7y - 5z + 10 = 0$$

$$\text{The distance of the point (1, 1, 1) from the plane} = \left| \frac{1 + 7 - 5 + 10}{\sqrt{1 + 49 + 25}} \right| = \frac{13}{\sqrt{75}} \text{ units}$$

## SECTION - D

## Assertion-Reason Type Questions

1. STATEMENT-1 : The triangle with vertices (1, 3, 5), (2, 4, 6) and (0, 5, 7) must be a right angle triangle.

and

STATEMENT-2 : If the dot product of two non-zero vectors is zero then they must be perpendicular.

**Sol.** Answer (1)

Let A, B, C be the points (1, 3, 5), (2, 4, 6) and (0, 5, 7) respectively.

Direction Ratio's of AB, BC, and CA are 1, 1, 1 ; -2, 1, 1 ; 1, -2, -2

respectively.

Clearly,  $-2(1) + 1(1) + 1(1) = 0$

$\therefore AB \perp BC$ .

In vector form  $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$

$$= \hat{i} + \hat{j} + \hat{k} \quad (\overrightarrow{AB} \neq 0)$$

$$\overrightarrow{BC} = \overrightarrow{OC} - \overrightarrow{OB} = -2\hat{i} + \hat{j} + \hat{k} \quad (\overrightarrow{BC} \neq 0)$$

$$\overrightarrow{AB} \cdot \overrightarrow{BC} = -2(1) + 1(1) + 1(1) = 0$$

2. STATEMENT-1 : The points (2, 3, 5), (7, 5, 7) and (-3, 1, 3) are vertices of an equilateral triangle.

and

STATEMENT-2 : The triangle with equal sides is called an equilateral triangle.

**Sol.** Answer (4)

Let points (2, 3, 5), (7, 5, 7) and (-3, 1, 3) be represented by A, B and C respectively.

$$AB = \sqrt{(7-2)^2 + (5-3)^2 + (7-5)^2} = \sqrt{33}$$

$$BC = \sqrt{(-3-7)^2 + (1-5)^2 + (3-7)^2} \neq \sqrt{33}$$

$\therefore \Delta ABC$  is not equilateral

3. STATEMENT-1 : If a line making an angle  $\pi/4$  with x-axis,  $\pi/4$  with y-axis then it must be perpendicular to z-axis.

and

STATEMENT-2 : If direction ratios of two lines are  $l_1, m_1, n_1$  and  $l_2, m_2, n_2$  then the angle between them is given by  $\theta = \cos^{-1} (l_1 l_2 + m_1 m_2 + n_1 n_2)$ .

**Sol.** Answer (3)

The angle given as  $\theta = \cos^{-1} (l_1 l_2 + m_1 m_2 + n_1 n_2)$  is the angle between the lines whose direction-cosines are  $l_1, m_1, n_1$  and  $l_2, m_2, n_2$  (not direction ratio's)

4. STATEMENT-1 : The centroid of a tetrahedron with vertices (0, 0, 0), (4, 0, 0), (0, -8, 0), (0, 0, 12) is (1, -2, 3).

and

STATEMENT-2 : The centroid of a triangle with vertices  $(x_1, y_1, z_1)$ ,  $(x_2, y_2, z_2)$  and  $(x_3, y_3, z_3)$  is

$$\left( \frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3} \right)$$

**Sol.** Answer (2)

Centroid of a tetrahedron with vertices  $(x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3), (x_4, y_4, z_4)$  is

$$\left( \frac{x_1 + x_2 + x_3 + x_4}{4}, \frac{y_1 + y_2 + y_3 + y_4}{4}, \frac{z_1 + z_2 + z_3 + z_4}{4} \right)$$

$\therefore$  The centroid of the tetrahedron with given vertices is

$$\left( \frac{0+4+0+0}{4}, \frac{0+0-8+0}{4}, \frac{0+0+0+12}{4} \right) = (1, -2, 3)$$

Statement 2 is true, but does not explain statement 1.

5. STATEMENT-1 : The points  $A(2, 9, 12), B(1, 8, 8), C(-2, 11, 8)$  and  $D(-1, 12, 12)$  are the vertices of rhombus.  
**and**

STATEMENT-2 :  $AB = BC = CD = DA$  and  $AC \neq BD$ , then the quadrilateral  $ABCD$  is called a rhombus.

**Sol.** Answer (4)

Direction ratios of  $AC = (-2 - 2, 11 - 9, 8 - 12) = (-4, 2, 4) = (l_1, m_1, n_1)$

Direction of  $BD = (-1 - 1, 12 - 8, 12 - 8) = (-2, 4, 4) = (l_2, m_2, n_2)$

$$l_1 l_2 + m_1 m_2 + n_1 n_2 = 8 + 8 - 16 = 0$$

Hence  $AC$  and  $BD$  are perpendicular

$$\text{Again } AB = \sqrt{(1-2)^2 + (8-9)^2 + (8-12)^2} = \sqrt{1+1+16} = \sqrt{18}$$

$$BC = \sqrt{(-2-1)^2 + (11-8)^2 + (8-8)^2} = \sqrt{18}$$

$$CD = \sqrt{(-1+2)^2 + (12-11)^2 + (12-8)^2} = \sqrt{18}$$

$$AD = \sqrt{(-1-2)^2 + (12-9)^2 + (12-12)^2} = \sqrt{18}$$

$$\text{Length of } AC = \sqrt{16+4+16} = 6$$

$$\text{Length of } BD = \sqrt{4+16+16} = 6$$

$$AC = BD$$

Hence  $ABCD$  is not a rhombus

Statement-1 is false but statement-2 is true.

6. STATEMENT-1 : The direction ratios of a line joining the points  $(0, 0, 0)$  and  $(x, y, z)$  must be  $x, y, z$ .  
**and**

STATEMENT-2 : If  $P(x, y, z)$  is a point in space and  $OP = r$ , then direction cosines of  $OP$  are  $\frac{x}{r}, \frac{y}{r}, \frac{z}{r}$ .

**Sol.** Answer (1)

Direction ratios  $= (x - 0, y - 0, z - 0) = (x, y, z)$

Hence statement-1 is true also statement-2 is true and statement-2 is correct explanation of statement-1

7. STATEMENT-1 : The equation of the plane through the intersection of the planes  $x + y + z = 6$  and  $29x + 23y + 17z = 276$ , is  $5x + 4y + 3z = 47$ .

**and**

STATEMENT-2 : Equation of the plane through the line of intersection of the planes  $P_1 = 0$  and  $P_2 = 0$  is  $P_1 + \lambda P_2 = 0, \lambda \neq 0$ .

**Sol.** Answer (1)

The required equation of the plane is

$$(x + y + z - 6) + \lambda(29x + 23y + 17z - 276) = 0$$

If  $\lambda = 1$  then we get

$$30x - 24y + 18z - 282 = 0$$

$$\text{or } 5x + 4y + 3z - 47 = 0$$

Hence statement-1 is true

Statement-2 is also clearly true and statement-2 is correct explanation of statement-1.

8. STATEMENT-1 : The shortest distance between the skew lines  $\frac{x+3}{-4} = \frac{y-6}{3} = \frac{z}{2}$  and  $\frac{x+2}{-4} = \frac{y}{1} = \frac{z-7}{1}$  is 9.

**and**

STATEMENT-2 : Two lines are skew lines if there exists no plane passing through them.

**Sol.** Answer (2)

Here  $\vec{a}_1 = -3\hat{i} + 6\hat{j}$  = One point on one line

$\vec{a}_2 = -2\hat{i} + 0\hat{j} + 7\hat{k}$  = One point on the other line

$$\vec{b}_1 = -4\hat{i} + 3\hat{j} + 2\hat{k}$$

$$\vec{b}_2 = -4\hat{i} + \hat{j} + \hat{k}$$

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -4 & 3 & 2 \\ -4 & 1 & 1 \end{vmatrix}$$

$$= \hat{i}(3-2) - \hat{j}(-4+8) + \hat{k}(-4+12)$$

$$= \hat{i} - 4\hat{j} + 8\hat{k}$$

$$\vec{a}_2 - \vec{a}_1 = \hat{i} - 6\hat{j} + 7\hat{k}$$

Shortest distance between the lines

$$\frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|} = \frac{(\hat{i} - 6\hat{j} + 7\hat{k}) \cdot (\hat{i} - 4\hat{j} + 8\hat{k})}{\sqrt{1+16+64}} = \frac{1+24+56}{9} = \frac{81}{9} = 9$$

Hence statement-1 is true and statement-2 is also true but statement-2 is not the correct explanation of statement-1.

9. Let  $P_1$  and  $P_2$  be two planes containing the lines  $L_1$  and  $L_2$  respectively.

STATEMENT-1 : If  $P_1$  and  $P_2$  are parallel then  $L_1$  and  $L_2$  must be parallel.

**and**

STATEMENT-2 : If  $P_1$  and  $P_2$  are parallel then  $L_1$  and  $L_2$  may not have a common point.

**Sol.** Answer (4)

Statement-1 is false

Because of  $P_1$  and  $P_2$  are parallel, then  $L_1$  and  $L_2$  may or may not be parallel

Statement-2 is clearly true.



10. Consider the planes  $3x - 6y - 2z = 15$  and  $2x + y - 2z = 5$ .

[IIT-JEE 2007]

STATEMENT-1 : The parametric equations of the line of intersection of the given planes are  $x = 3 + 14t$ ,  $y = 1 + 2t$ ,  $z = 15t$ .

and

STATEMENT-2 : The vector  $14\hat{i} + 2\hat{j} + 15\hat{k}$  is parallel to the line of intersection of given planes.

**Sol.** Answer (4)

Statement-2 is true but correct parametric form of  $y$  is  $y = -1 + 2t$ . Thus statement-1 is false.

11. Consider three planes

[IIT-JEE 2008]

$$P_1 : x - y + z = 1$$

$$P_2 : x + y - z = -1$$

$$P_3 : x - 3y + 3z = 2$$

Let  $L_1, L_2, L_3$  be the lines of intersection of the planes  $P_2$  and  $P_3$ ,  $P_3$  and  $P_1$ ,  $P_1$  and  $P_2$ , respectively

STATEMENT-1 : At least two of the lines  $L_1, L_2$  and  $L_3$  are non-parallel.

and

STATEMENT-2 : The three planes do not have a common point.

**Sol.** Answer (4)

The given equations are

$$x - y + z = 1$$

$$x + y - z = -1$$

$$x - 3y + 3z = 2$$

The system of equations can be put in matrix form as  $Ax = B$

$$\text{i.e. } \begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ 1 & -3 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & -2 \\ 0 & -2 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \quad \begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_1 \end{array}$$

$$\sim \begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix}$$

which is inconsistent as

$\Rightarrow$  The three planes do not have a common point.

$\Rightarrow$  Statement-2 is true.

Since planes  $P_1, P_2, P_3$  are pairwise intersecting, their lines of intersection are parallel.

Statement-1, is false.

## SECTION - E

## Matrix-Match Type Questions

1. Match the following :

## Column-I

(A) Centroid of the triangle with vertices

$A(2, 3, 7), B(6, 7, 5), C(1, 2, 3)$

(B) Mid-point of the line joining the points

$A(7, 9, 11) \text{ and } B(-5, 3, -1)$

(C) A point on the line  $\frac{x}{2} = \frac{y}{3} = \frac{z}{5}$ , at a distance 2 from the origin(D) Coordinates of the point dividing the join of  $(5, 5, 0)$  and  $(0, 0, 5)$  in the ratio 2 : 3

## Column-II

(p)  $(1, 6, 5)$

(q)  $(3, 4, 5)$

(r)  $(3, 3, 2)$

(s)  $\left(\frac{4}{\sqrt{38}}, \frac{6}{\sqrt{38}}, \frac{10}{\sqrt{38}}\right)$

Sol. Answer : A(q), B(p), C(s), D(r)

(A)  $\left(\frac{2+6+1}{3}, \frac{3+7+2}{3}, \frac{7+5+3}{3}\right) = (3, 4, 5)$

(B)  $\left(\frac{7-5}{2}, \frac{9+3}{2}, \frac{11-1}{2}\right) = (1, 6, 5)$

(C) Any point on  $\frac{x}{2} = \frac{y}{3} = \frac{z}{5}$  can be taken as  $(2k, 3k, 5k)$ .

$$\therefore \sqrt{(2k-0)^2 + (3k-0)^2 + (5k-0)^2} = 2$$

$$\Rightarrow k = \pm \frac{2}{\sqrt{38}}$$

$$\therefore \text{Required point is } \left(\frac{4}{\sqrt{38}}, \frac{6}{\sqrt{38}}, \frac{10}{\sqrt{38}}\right)$$

(D)  $\left(\frac{2 \times 0 + 3 \times 5}{2+3}, \frac{2 \times 0 + 3 \times 5}{2+3}, \frac{2 \times 5 + 0 \times 3}{2+3}\right) = (3, 3, 2)$

2. If  $\alpha, \beta, \gamma$  are the angles which a line makes with positive direction of the axes, then match the quantities in column-I to their values in column-II.

## Column-I

(A)  $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma$

(B)  $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma$

(C)  $\cos 2\alpha + \cos 2\beta + \cos 2\gamma$

(D)  $2 \cos(\alpha - \beta) \cos(\alpha + \beta) + 2 \cos(\beta - \gamma) \cos(\beta + \gamma) + 2 \cos(\gamma - \alpha) \cos(\gamma + \alpha)$

## Column-II

(p)  $-2$

(q)  $1$

(r)  $-1$

(s)  $2$

**Sol.** Answer : A(s), B(q), C(r), D(p)

$$\text{Since } \cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$$

$$(1 - \sin^2\alpha) + (1 - \sin^2\beta) + (1 - \sin^2\gamma) = 1$$

$$\Rightarrow \sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$$

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 2(\cos^2\alpha + \cos^2\beta + \cos^2\gamma) - 3$$

$$= 2(1) - 3 = -1$$

$$\text{Since, } 2\cos(\alpha + \beta) \cos(\alpha - \beta) = \cos 2\alpha + \cos 2\beta$$

$$\therefore \sum 2\cos(\alpha + \beta) \cos(\alpha - \beta) = 2(\cos^2\alpha + \cos^2\beta + \cos^2\gamma) = -2$$

3. Match column I to column II according to the given condition.

In column I the direction ratios of lines are given. In column II angle between them is given.

**Column I**

**Column II**

(A) (1, 2, 3), (1, 1, -1)

(p)  $90^\circ$

(B) (1, 2, 3), (2, 4, 6) then acute angle is less than

(q)  $60^\circ$

(C) (1, 1, 1), (2, 1, -1) the acute angle is greater than

(r)  $0^\circ$

(D) (1, 0, 0), (0, 1, 0) the angle is greater than

(s)  $30^\circ$

(t)  $70^\circ$

**Sol.** Answer A(p), B(p, q, s, t), C(q, r, s), D(q, r, s, t)

$$(A) 1 \times 1 + 2 \times 1 + 3 \times (-1) = 0$$

$$\text{hence } \theta = 90^\circ$$

$$(B) \frac{1}{2} = \frac{2}{4} = \frac{3}{6}, \text{ hence angle between the lines is zero.}$$

$$(C) \cos \theta = \frac{1 \times 2 + 1 \times 1 + 1 \times (-1)}{\sqrt{1+1+1} \sqrt{4+1+1}} = \frac{2}{\sqrt{3} \sqrt{6}} = \frac{2}{3\sqrt{2}} = \frac{\sqrt{2}}{3}$$

$$\cos \theta \approx \frac{1.411}{3} \approx 0.47 < \frac{1}{2} \Rightarrow \theta > 60^\circ$$

$$(D) 1 \times 0 + 0 \times 1 + 0 \times 0 = 0 \Rightarrow \theta = 90^\circ$$

4. Match the following.

**Column I**

**Column II**

(A) The projection of the line segment joining the point (1, 2, 3)

(p) 8

$$\text{and } (4, 5, 6) \text{ on the line } \frac{x-1}{2} = \frac{y-3}{2} = \frac{z+4}{1} \text{ is}$$

$$(B) \text{ If the lines } \frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{4} \text{ and } \frac{x-3}{1} = \frac{y-k}{2} = \frac{z}{1} \text{ intersect}$$

(q) -5

then the value of  $2k$  is

- (C) Number of vectors which are equally inclined to axes and having modulus  $3\sqrt{3}$  is (r) 5
- (D) The ratio in which the join of (2, 1, 5) and (3, 4, 3) is divided by the plane is  $2x + 2y - 2z - 1 = 0$  is  $\frac{a}{b}$  such that H.C.F.(a, b) = 1 then a is equal to (s) 9
- (t) Composite number

**Sol.** Answer : A(r), B(s, t), C(p, t), D(r)

(A) Projection is  $\frac{(4-1) \cdot 2 + (5-2) \cdot 2 + (6-3) \cdot 1}{\sqrt{4+4+1}} = 5$

(B) Compare the general point

$$2\lambda + 1 = \mu + 3, \quad 3\lambda - 1 = 2\mu + k, \quad 4\lambda + 1 = \mu$$

$$\Rightarrow k = \frac{9}{2}$$

(C) Line which is equally inclined have direction cosines

$$\left( \pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}} \right)$$

Gives the 8 vectors.

(D) The ratio  $\lambda : 1$

$$\Rightarrow 2(3\lambda + 2) + 2(4\lambda + 1) - 2(3\lambda + 5) - \lambda - 1 = 0$$

$$6\lambda + 4 + 8\lambda + 2 - 6\lambda - 10 - \lambda - 1 = 0$$

$$\Rightarrow 7\lambda = 5$$

5. Match the equation of the plane in column I and column II

**Column I**

- (A) Containing the line  $\vec{r} = \vec{a} + t\vec{b}$  and passing through a point  $\vec{c}$
- (B) Containing the line  $\vec{r} = \vec{a} + t\vec{b}$  and perpendicular to the plane  $\vec{r} \cdot \vec{c} = q$
- (C) Through points with position vectors  $\vec{a}, \vec{b}, \vec{c}$
- (D) Through two points whose position vectors  $\vec{a}, \vec{b}$  and parallel to  $\vec{c}$

**Column II**

- (p)  $\vec{r} = (1-t-p)\vec{a} + t\vec{b} + p\vec{c}$
- (q)  $\vec{r} = \vec{a} + t(\vec{c} - \vec{a}) + p\vec{b}$
- (r)  $\vec{r} = \vec{a} + t(\vec{b} - \vec{a}) + p\vec{c}$
- (s)  $\vec{r} = \vec{a} + t\vec{b} + p\vec{c}$

**Sol.** Answer : A(q), B(s), C(p), D(r)

(A) Equation of the plane

$$\vec{r} = \vec{a} + p(\vec{c} - \vec{a}) + t(\vec{b})$$

(B)  $\vec{r} = \vec{a} + p\vec{c}$

(C)  $(\vec{r} - \vec{a}) \cdot ((\vec{b} - \vec{a}) \times (\vec{c} - \vec{a})) = 0$

(D)  $\vec{r} = \vec{a} + t(\vec{b} - \vec{a}) + p\vec{c}$

6. Consider the following linear equations

[IIT-JEE 2007]

$$ax + by + cz = 0$$

$$bx + cy + az = 0$$

$$cx + ay + bz = 0$$

Match the conditions/expression in Column I with statements in Column II.

**Column I**

(A)  $a + b + c \neq 0$  and  $a^2 + b^2 + c^2 = ab + bc + ca$

(B)  $a + b + c = 0$  and  $a^2 + b^2 + c^2 \neq ab + bc + ca$

(C)  $a + b + c \neq 0$  and  $a^2 + b^2 + c^2 \neq ab + bc + ca$

(D)  $a + b + c = 0$  and  $a^2 + b^2 + c^2 = ab + bc + ca$

**Column II**

(p) the equations represent planes meeting only at a single point

(q) the equations represent the line  $x = y = z$

(r) the equations represent identical planes

(s) the equations represent the whole of the three dimensional space

**Sol.** Answer A(r), B(q), C(p), D(s)

7. Match the statements/expressions given in Column I with the values given in Column II.

[IIT-JEE 2009]

**Column I**

(A) The number of solutions of the equation

$$xe^{\sin x} - \cos x = 0 \text{ in the interval } \left(0, \frac{\pi}{2}\right)$$

(B) Value(s) of  $k$  for which the planes  $kx + 4y + z = 0$ ,  $4x + ky + 2z = 0$  and  $2x + 2y + z = 0$  intersect in a straight line

(C) Value(s) of  $k$  for which  $|x - 1| + |x - 2| + |x + 1| + |x + 2| = 4k$  has integer solution(s)

(D) If  $y' = y + 1$  and  $y(0) = 1$ , then value(s) of  $y(\ln 2)$

**Column II**

(p) 1

(q) 2

(r) 3

(s) 4

(t) 5

**Sol.** Answer A(p), B(q, s), C(q, r, s, t), D(r)

(A)  $xe^{\sin x} - (0) x = 0$

$$xe^{\sin x} = \cos x$$

Clearly only one solution exists.

(B) For intersecting in a line given system of equations must have a non-zero solution

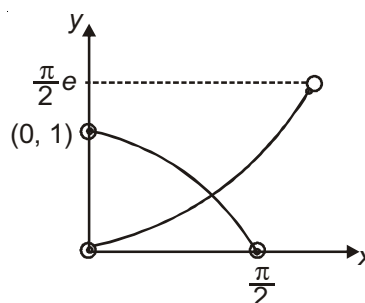
$$\text{i.e., } \begin{vmatrix} k & 4 & 1 \\ 4 & k & 2 \\ 2 & 2 & 1 \end{vmatrix} = 0$$

$$\Rightarrow k(k - 4) - 4(4 - 2) + 2(8 - k) = 0$$

$$\Rightarrow k^2 - 4k - 8 + 16 - 2k = 0$$

$$\Rightarrow k^2 - 6k + 8 = 0$$

$$\Rightarrow k = 4, 2$$



(C)  $|x - 1| + |x - 2| + |x + 1| + |x + 2| = 4k$

Following cases may arise :

Case (i)

$$x \leq -2$$

$$-x + 1 - x + 2 - x - 1 - x - 2 = 4k$$

$$-4x = 4k \Rightarrow x = -k \Rightarrow k = 2, 3, 4, 5, \dots$$

Case (ii)

$$-2 < x \leq -1$$

$$x + 2 - x - 1 - x + 2 - x + 1 = 4k$$

$$-2x + 4 = 4k, \text{ no integral value of } k$$

Case (iii)

$$-1 < x \leq 1$$

$$x + 2 + x + 1 - x + 1 - x + 2 = 4k$$

$$6 = 4k, \text{ no integral value of } k$$

$$1 < x \leq 2$$

$$x - 1 + x + 2 + x + 1 + x + 2 = 4k$$

$$= 2x + 4 = 4k$$

$$x = 2(k - 1) \Rightarrow k = 2, 3$$

Case (iv)

$$x > 2$$

$$x - 1 + x - 2 + x + 1 + x + 2 = 4k$$

$$4x = 4k$$

$$x = k \Rightarrow k = 3, 4, 5, \dots$$

(D)  $\frac{dy}{dx} = y + 1$

$$\Rightarrow \int \frac{dy}{y+1} = \int dx$$

$$\Rightarrow \ln(y + 1) = x + c$$

$$\Rightarrow (y + 1) = ke^x$$

$$\Rightarrow y = k.e^x - 1$$

$$\text{At } x = 0, y = 1$$

$$1 = k - 1 \Rightarrow k = 2$$

$$\therefore y = 2e^x - 1$$

$$y(\ln 2) = 2e^{\ln 2} - 1 = 4 - 1 = 3$$

8. Match the statements in Column-I with the values in Column-II.

[IIT-JEE 2010]

## Column I

## Column II

(A) A line from the origin meets the lines  $\frac{x-2}{1} = \frac{y-1}{-2} = \frac{z+1}{1}$  and

(p) -4

$$\frac{x-\frac{8}{2}}{\frac{3}{2}} = \frac{y+3}{-1} = \frac{z-1}{1} \text{ at } P \text{ and } Q \text{ respectively. If length } PQ = d,$$

then  $d^2$  is(B) The values of  $x$  satisfying  $\tan^{-1}(x+3) - \tan^{-1}(x-3) = \sin^{-1}\left(\frac{3}{5}\right)$  are

(q) 0

(C) Non-zero vectors  $\vec{a}$ ,  $\vec{b}$  and  $\vec{c}$  satisfy  $\vec{a} \cdot \vec{b} = 0$ ,

(r) 4

 $(\vec{b} - \vec{a}) \cdot (\vec{b} + \vec{c}) = 0$  and  $2|\vec{b} + \vec{c}| = |\vec{b} - \vec{a}|$ . If  $\vec{a} = \mu\vec{b} + 4\vec{c}$ , then the possible values of  $\mu$  are
(D) Let  $f$  be the function on  $[-\pi, \pi]$  given by  $f(0) = 9$  and

(s) 5

$$f(x) = \sin\left(\frac{9x}{2}\right) / \sin\left(\frac{x}{2}\right) \text{ for } x \neq 0. \text{ The value of } \frac{2}{\pi} \int_{-\pi}^{\pi} f(x) dx \text{ is}$$

(t) 6

**Sol.** Answer A(t), B(p, r), C(q), D(r)(A) Equation of line passing through origin is  $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$ 

$$\therefore \frac{x}{a} = \frac{y}{b} = \frac{z}{c}$$

$$\Rightarrow a(-1) - b(3) + c(-5) = 0$$

$$-a - 3b - 5c = 0$$

$$a + 3b + 5c = 0 \quad \dots (1)$$

$$\text{Also, } \begin{vmatrix} \frac{8}{3} & -3 & 1 \\ 2 & -1 & 1 \\ a & b & c \end{vmatrix} = 0$$

$$a(-2) - b\left(\frac{2}{3}\right) + c\left(\frac{10}{3}\right) = 0$$

$$2a + \frac{2b}{3} - \frac{10c}{3} = 0$$

$$3a + b - 5c = 0 \quad \dots (2)$$

$$\text{Now, } a + 3b + 5c = 0$$

$$3a + b - 5c = 0$$

$$\frac{a}{-20} = \frac{b}{20} = \frac{c}{-8}$$

$$\frac{a}{5} = \frac{b}{-5} = \frac{c}{4}$$

Equation of line is

$$\frac{x}{5} = \frac{y}{-5} = \frac{z}{+4} \quad \dots (3)$$

$$\frac{x-2}{1} = \frac{y-1}{-2} = \frac{z+1}{1} \quad \dots (4)$$

$$\text{Now, } \frac{x-\frac{8}{3}}{2} = \frac{y+3}{-1} = \frac{z-1}{1} \quad \dots (5)$$

Point on (3) is  $(5\lambda, -5\lambda, +4\lambda)$

Point on (4) is  $(2 + k_1, 1 - 2k_1, -1 + k_1)$

Point on (5) is  $\left(\frac{8}{3} + 2k_2, -3 - k_2, 1 + k_2\right)$

On solving,

$$2 + k_1 + 1 - 2k_1 = 0$$

$$-k_1 + 3 = 0$$

$$k_1 = 3$$

$$P \equiv (5, -5, 2)$$

Again for Q,

$$\frac{8}{3} + 2k_2 - 3 - k_2 = 0$$

$$k_2 - \frac{1}{3} = 0$$

$$k_2 = \frac{1}{3}$$

$$Q \equiv \left(\frac{10}{3}, -\frac{10}{3}, \frac{4}{3}\right)$$



$$PQ = \sqrt{\left(\frac{5}{3}\right)^2 + \left(\frac{5}{3}\right)^2 + \left(\frac{2}{3}\right)^2}$$

$$= \frac{\sqrt{54}}{3}$$

$$PQ^2 = d^2 = \frac{54}{9} = 6$$

$$(B) \tan^{-1}\left(\frac{x+3-x+3}{1+(x^2-9)}\right) = \tan^{-1}\left(\frac{3}{4}\right)$$

$$\frac{6}{x^2-8} = \frac{3}{4}$$

$$3x^2 - 24 = 24$$

$$3x^2 = 48$$

$$x = \pm 4$$

$$(C) (\vec{b} - \vec{a}) \cdot \left(\vec{b} + \frac{\vec{a} - \mu\vec{b}}{4}\right) = 0$$

$$(\vec{b} - \vec{a}) \cdot (4\vec{b} + \vec{a} - \mu\vec{b}) = 0$$

$$(4 - \mu)b^2 - a^2 = 0 \quad \dots (1)$$

$$\text{Also, } 2\left|\vec{b} + \frac{\vec{a} - \mu\vec{b}}{4}\right| = |\vec{b} - \vec{a}|$$

$$2\left|\frac{(4 - \mu)\vec{b} + \vec{a}}{4}\right|$$

$$\frac{(4 - \mu)^2 b^2}{4} + \frac{a^2}{4} = b^2 + a^2$$

$$\frac{3a^2}{4} = \frac{(4 - \mu)^2 - 4}{4} \cdot b^2$$

$$3a^2 = ((4 - \mu)^2 - 4)b^2 \quad \dots (2)$$

From equations (1) and (2),

$$3(4 - \mu) = (4 - \mu)^2 - 4$$

$$(4 - \mu)^2 - 3(4 - \mu) - 4 = 0$$

$$\Rightarrow \mu = 0, 5$$

$\mu = 5$  is not admissible.

(D)  $f(0) = 9$ ,

$$f(x) = \frac{\sin\left(\frac{9x}{2}\right)}{\sin\frac{x}{2}}$$

$$= \left(3 - 4 \sin^2 \frac{x}{2}\right) \left(3 - 4 \sin^2 \frac{3x}{2}\right)$$

$$= 9 - 12 \sin^2 \frac{x}{2} - 12 \sin^2 \frac{3x}{2} + 16 \sin^2 \frac{x}{2} \cdot \sin^2 \frac{3x}{2}$$

$$= 9 - 6(1 - \cos x) - 6(1 - \cos 3x) + 4(1 - \cos x)(1 - \cos 3x)$$

$$= 1 + 6 \cos x + 6 \cos 3x - 4 \cos x - 4 \cos 3x + 4 \cos x \cos 3x$$

$$I = \frac{2}{\pi} \int_{-\pi}^{\pi} \frac{\sin \frac{9x}{2}}{\sin \frac{x}{2}} dx$$

$$= \frac{4}{\pi} \int_0^{\pi} 1 + 2 \cos x + 2 \cos 3x + 2(\cos 4x + \cos x) dx$$

$$= \frac{4}{\pi} \times \pi = 4$$

9. Consider the lines  $L_1 : \frac{x-1}{2} = \frac{y}{-1} = \frac{z+3}{1}$ ,  $L_2 : \frac{x-4}{1} = \frac{y+3}{1} = \frac{z+3}{2}$  and the planes  $P_1 : 7x + y + 2z = 3$ ,  $P_2 : 3x + 5y - 6z = 4$ . Let  $ax + by + cz = d$  be the equation of the plane passing through the point of intersection of lines  $L_1$  and  $L_2$ , and perpendicular to planes  $P_1$  and  $P_2$ . Match Column I with Column II and select the correct answer using the code given below the lists:

[JEE(Advanced) 2013]

**Column I**

(A)  $a =$

(B)  $b =$

(C)  $c =$

(D)  $d =$

**Column II**

(p) 13

(q) -3

(r) 1

(s) -2

**Codes :**

| A     | B | C | D |
|-------|---|---|---|
| (1) r | q | s | p |
| (2) p | r | s | q |
| (3) r | q | p | s |
| (4) q | d | p | r |

**Sol.** Answer (1)

$$L_1 \equiv \frac{x-1}{2} = \frac{y}{-1} = \frac{z+3}{1} = t_1 \text{ and } L_2 \equiv \frac{x-4}{1} = \frac{y+3}{1} = \frac{z+3}{2} = t_2$$

For finding point of intersection, we have

$$1 + 2t_1 = 4 + t_2 \quad \dots(i)$$

$$\text{and } -t_1 = -3 + t_2 \quad \dots(ii)$$

Solving, we get

$$t_1 = 2, t_2 = 1$$

i.e., point of intersection is (5, -2, -1).

Now, equation of plane  $P$  be

$$\begin{vmatrix} x-5 & y+2 & z+1 \\ 7 & 1 & 2 \\ 3 & 5 & -6 \end{vmatrix} = 0$$

$$\Rightarrow (-16)(x-5) + 48(y+2) + 32(z+1) = 0$$

$$\Rightarrow (x-5) - 3(y+2) - 2(z+1) = 0$$

$$\Rightarrow x - 3y - 2z = 13$$

$$\text{i.e., } a = 1, b = -3, c = -2, d = 13$$

**SECTION - F****Integer Answer Type Questions**

1. If the direction ratios of two lines are  $(1, \lambda, 2)$  and  $(\lambda, \lambda + 1, \lambda)$  and the angle between the lines is  $90^\circ$  then the modulus of sum of all values of  $\lambda$  is \_\_\_\_\_.

**Sol.** Answer (4)

$$(1 \times \lambda) + (\lambda(\lambda + 1)) + 2\lambda = 0$$

$$\Rightarrow \lambda + \lambda^2 + \lambda + 2\lambda = 0$$

$$\lambda^2 + 4\lambda = 0$$

$$\lambda = 0, -4$$

$$\text{sum} = 0 - 4 = -4$$

$$\text{Modulus of sum} = 4$$

2. If  $A \equiv (2, 3, 4)$ ,  $B \equiv (3, 4, 5)$ . The direction cosine of a line are  $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$ . Now integral value of the projection of  $AB$  on the given line is \_\_\_\_\_.

**Sol.** Answer (1)

$$\text{Projection} = \frac{1}{\sqrt{3}}(3-2) + \frac{1}{\sqrt{3}}(4-3) + \frac{1}{\sqrt{3}}(5-4) = \sqrt{3}$$

$$= \sqrt{3}$$

$$\text{Integral value} = 1$$

3. The direction ratios of a line are  $(-2, 3, 6)$ . If the line makes an acute angle with positive direction of x-axis then the modulus of integral value of sum of all direction cosines, is \_\_\_\_\_.

**Sol.** Answer (1)

$$\text{Direction cosine} = \left( \pm \left( -\frac{2}{7} \right), \pm \frac{3}{7}, \pm \frac{6}{7} \right)$$

$$= (\cos\alpha, \cos\beta, \cos\gamma)$$

But  $\cos\alpha > 0$

Hence directions cosines are

$$\frac{2}{7}, -\frac{3}{7}, -\frac{6}{7}$$

$$\text{sum} = \frac{2-3-6}{7} = -1$$

Modulus of sum = 1

4. If the centroid of triangles formed by the vertices  $(1, 2, 3)$ ,  $(2, 1, 0)$  and  $(3, 1, 4)$  is  $(\alpha, \beta, \gamma)$  then the value of  $[\alpha] + [\beta] + [\gamma]$ , where  $[ ]$  represents the greatest integer function, is \_\_\_\_\_.

**Sol.** Answer (5)

$$\alpha = \frac{1+2+3}{3} = 2$$

$$\beta = \frac{2+1+1}{3} = \frac{4}{3}$$

$$\gamma = \frac{3+0+4}{3} = \frac{7}{3}$$

$$[\alpha] = 2, [\beta] = 1, [\gamma] = 2$$

$$[\alpha] + [\beta] + [\gamma] = 5$$

5. If the distance of the point  $(1, 2, 3)$  from the plane  $2x - y + z = 6$  measured parallel to line  $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$  is

$$\sqrt{a}\sqrt{b} \text{ where } a \text{ and } b \text{ are two prime numbers then find max. } (a, b).$$

**Sol.** Answer (7)

$$\text{Consider the line } \frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{3}$$

Any point on it  $P(r+1, 2r+2, 3r+3)$

Its on plane, so

$$2(r+1) - 2r - 2 + 3r + 3 = 6$$

So  $r = 1$ , point  $P(2, 4, 6)$

$$\text{So distance} = \sqrt{14}$$

6. If the straight line  $\frac{x-1}{3} = \frac{y-2}{2} = \frac{z-3}{2}$  intersect the curve  $ax^2 + by^2 = 1, z = 0$  find the value of  $49a + 4b$ .

**Sol.** Answer (4)

Any point on the line may be taken as  $(3r + 1, 2r + 2, 2r + 3)$ , which lies on the curve,

$$\text{So } r = -\frac{3}{2} \text{ required point is } \left(-\frac{7}{2}, -1, 0\right)$$

$$\text{Hence } \frac{49}{4}a + b = 1$$

$$49a + 4b = 4$$

7. Let  $M$  and  $N$  are feet of perpendiculars from  $P$  to  $x$ - $z$  and  $x$ - $y$  plane respectively. If  $OP$  makes  $\theta, \alpha, \beta, \gamma$  with plane  $OMN$  and the co-ordinate planes respectively such that  $7 + \cot^2 \theta = \cot^2 \alpha + \cot^2 \beta + \cot^2 \gamma + k$ , then find  $k$ .

**Sol.** Answer (9)

Let  $P(x_1, y_1, z_1)$  then

$M(x_1, 0, z_1)$  and  $N(x_1, y_1, 0)$

Then equation of plane

$$-y_1 z_1(x) + x_1 z_1(y) + x_1 y_1(z) = 0$$

$$\text{So } \sin \theta = \frac{x_1(-y_1 z_1) + y_1(x_1 z_1) + z_1(x_1 y_1)}{\sqrt{x_1^2 + y_1^2 + z_1^2} \sqrt{(y_1^2 z_1^2 + x_1^2 z_1^2 + x_1^2 y_1^2)}}$$

$$\Rightarrow \operatorname{cosec}^2 \theta = \frac{\left(\sum x_1^2\right)\left(\sum x_1^2 y_1^2\right)}{x_1 y_1 z_1}$$

$$\text{Similarly } \operatorname{cosec}^2 \alpha = \frac{\sum x_1^2}{x_1^2}$$

$$\operatorname{cosec}^2 \beta = \frac{\sum x_1^2}{y_1^2}$$

$$\operatorname{cosec}^2 \gamma = \frac{\sum x_1^2}{z_1^2}$$

$$\Rightarrow \sum \operatorname{cosec}^2 \alpha = \left(\sum x_1^2\right) \left(\sum \frac{1}{x_1^2}\right) = \operatorname{cosec}^2 \theta$$

$$\Rightarrow 3 + \sum \cot^2 \alpha = 1 + \cot^2 \theta$$

$$\Rightarrow 9 + \sum \cot^2 \alpha = 7 + \cot^2 \theta$$

So  $k = 9$

8. If the distance between the plane  $Ax - 2y + z = d$  and the plane containing the lines  $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$  and

$$\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5} \text{ is } \sqrt{6}, \text{ then } |d| \text{ is}$$

[IIT-JEE 2010]

**Sol.** Answer (6)

Equation of plane containing the given lines is

$$\begin{vmatrix} x-1 & y-2 & z-3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix} = 0$$

$$\Rightarrow (x-1)(-1) - (y-2)(-2) + (z-3)(-1)$$

$$\Rightarrow -x + 1 + 2y - 4 - z + 3 = 0$$

$$\Rightarrow -x + 2y - z = 0 \quad \dots(i)$$

Given plane is

$$x - 2y + z = d \quad \dots(ii)$$

(i) and (ii) are parallel

Clearly  $A = 1$

$$\text{Now, distance between plane} = \left| \frac{d}{\sqrt{1+4+1}} \right| = \sqrt{6}$$

$$\Rightarrow |d| = 6$$

9. Let  $P$  be a point in the first octant, whose image  $Q$  in the plane  $x + y = 3$  (that is, the line segment  $PQ$  is perpendicular to the plane  $x + y = 3$  and the mid-point of  $PQ$  lies in the plane  $x + y = 3$ ) lies on the  $z$ -axis. Let the distance of  $P$  from the  $x$ -axis be 5. If  $R$  is the image of  $P$  in the  $xy$ -plane, then the length of  $PR$  is \_\_\_\_\_.

[JEE(Advanced) 2018]

**Sol.** Answer (8.00)

$P = (\alpha, \beta, \gamma)$ ,  $Q = (0, 0, K)$  as it lies on  $z$ -axis

midpoint of  $PQ$ , i.e.,  $\left( \frac{\alpha}{2}, \frac{\beta}{2}, \frac{\gamma+K}{2} \right)$  satisfies

$$x + y = 3$$

$$\Rightarrow \alpha + \beta = 6 \quad \dots(i)$$

$$\text{d.r. of } PQ = (\alpha - 0, \beta - 0, \gamma - K) = (p, p, 0)$$

$$\Rightarrow \alpha = \beta \text{ and } \gamma = K \quad \dots(ii)$$

$$\therefore (i) \text{ and } (ii) \Rightarrow \alpha = \beta = 3$$

$$\therefore P = (\alpha, \alpha, K) = (3, 3, K)$$

distance of  $P$  from  $x$ -axis = 5

$$\Rightarrow \beta^2 + \gamma^2 = 25 \Rightarrow \alpha^2 + K^2 = 25 \Rightarrow 3^2 + K^2 = 25$$

$$\Rightarrow |K| = 4$$

$$\text{Length of } PR = 2|K| = 8$$

## SECTION - G

## Multiple True-False Type Questions

1. STATEMENT-1 : The distance of (3, 4, 5) from x-axis 5.

STATEMENT-2 : The distance of (x, y, z) from xy plane is z.

STATEMENT-3 : The projection of a line segment on the axes are 1, 2, 3 then the length of segment is  $\sqrt{14}$ .

(1) T F T

(2) T T T

(3) F F F

(4) F F T

**Sol.** Answer (1)

STATEMENT-1 distance =  $\sqrt{3^2 + 4^2} = 5$

STATEMENT-2 distance = |z|

STATEMENT-3 length =  $\sqrt{1^2 + 2^2 + 3^2} = \sqrt{14}$

2. STATEMENT-1 : The angle between the lines whose direction ratios are (1, 2, 3) and (-12, 3, 2) is  $90^\circ$ .

STATEMENT-2 : The direction cosines of a line whose direction ratios are (3, 4, 12) are  $\left(\frac{3}{13}, \frac{4}{13}, \frac{12}{13}\right)$ .

STATEMENT-3 : If  $\cos\alpha, \cos\beta, \cos\gamma$  are the direction cosines of a line then  $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$ .

(1) T F T

(2) T T T

(3) F F F

(4) F F T

**Sol.** Answer (1)

STATEMENT-1

$$\cos\theta = \frac{1 \times (-12) + (2 \times 3) + (3 \times 2)}{\sqrt{1^2 + 2^2 + 3^2} \sqrt{(-12)^2 + 3^2 + 2^2}} = \frac{-12 + 6 + 6}{\sqrt{14} \sqrt{169}} = 0, \theta = 90^\circ$$

STATEMENT-2

$$\text{Direction cosine} = \pm \frac{3}{13}, \pm \frac{4}{13}, \pm \frac{12}{13}$$

STATEMENT-3

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$$

$$1 - \sin^2\alpha + 1 - \sin^2\beta + 1 - \sin^2\gamma = 1$$

$$\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$$

3. STATEMENT-1 : Equation of the plane through (2, 3, 3) and (1, -3, -4) and parallel to  $\frac{x-1}{3} = \frac{y-3}{4} = \frac{z+1}{5}$  is  $x + y - 7z = 16$ .

STATEMENT-2 : The shortest distance between two non-intersecting lines  $\vec{r} = \vec{a} + \lambda\vec{b}$  and  $\vec{r} = \vec{c} + \mu\vec{d}$  is

$$\frac{|[(\vec{c} - \vec{a}) \cdot \vec{b} \times \vec{d}]|}{|\vec{b} \times \vec{d}|}$$

STATEMENT-3 : The vector equation of a plane through a point having position vector  $\vec{a}$  and parallel to vector  $\vec{b}$  and  $\vec{c}$  is,  $\vec{r} = \vec{a} + \lambda\vec{b} + \mu\vec{c}$ .

(1) F F F

(2) F F T

(3) T F T

(4) F T F

**Sol.** Answer (2)

STATEMENT-1

Let plane  $a(x - 2) + b(y - 3) + c(z - 3) = 0$ Point  $(1, -3, -4)$  is on it.So  $a(1 - 2) + b(-3 - 3) + c(-4 - 3) = 0$ or  $a + 6b + 7c = 0 \quad \dots(i)$ 

Also normal of the plane will be perpendicular to the given line

 $\therefore 3a + 4b + 5b = 0 \quad \dots(ii)$ From (i) and (ii)  $\Rightarrow \frac{a}{1} = \frac{b}{1} = \frac{c}{-7}$ 

So equation of plane

 $(x - 2) + (y - 3) - 7(z - 3) = 0$  $x + y - 7z + 16 = 0$ 

STATEMENT-2

False because formula is true only for nonparallel lines.

STATEMENT-3

 $\therefore$  Equation is  $r = \vec{a} + \lambda \vec{b} + \mu \vec{c}$ 

4. STATEMENT-1 : If  $l, m, n$  are direction ratios of a straight line then minimum value of  $l^2 + m^2 + n^2$  will be 1.

STATEMENT-2 : If the line joining the origin to the point  $(0, -1, 9)$  makes angles  $\alpha, \beta$  and  $\gamma$  with the positive direction of the axes then the value of  $\cos 2\alpha + \cos 2\beta + \cos 2\gamma$  is  $-1$ .

STATEMENT-3 : The angle between two lines that have the direction ratios  $(1, 2, 3)$  and  $(3, -2, 1)$  is  $\cos^{-1}\left(\frac{1}{7}\right)$ .

(1) T T T

(2) F T F

(3) F F T

(4) F T T

**Sol.** Answer (4)

STATEMENT-1

If  $\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0$  are direction cosines, then direction ratios may be  $\frac{1}{2\sqrt{2}}, -\frac{1}{2\sqrt{2}}, 0$

STATEMENT-2

$$\sum \cos 2\alpha = 2\left(\sum \cos^2 \alpha\right) - 3 = 2 - 3 = -1$$

STATEMENT-3

$$\text{Use } \cos \theta = \frac{\sum a_1 a_2}{\sqrt{\sum a_1^2} \sqrt{\sum a_2^2}}$$



## SECTION - H

## Aakash Challengers Questions

1. Prove that two lines whose direction cosines are given by the relations  $pl + qm + rn = 0$  and  $a^2l^2 + b^2m^2 + c^2n^2 = 0$  are perpendicular or parallel according as  $p^2(b + c) + q^2(c + a) + r^2(a + b) = 0$  or

$$\frac{p^2}{a} + \frac{q^2}{b} + \frac{r^2}{c} = 0.$$

**Sol.** Let  $l_1, m_1, n_1$  and  $l_2, m_2, n_2$  be the direction cosines of the two lines satisfying the relations

$$pl + qm + rn = 0 \text{ and } a^2l^2 + b^2m^2 + c^2n^2 = 0$$

Eliminating  $n$ , we get

$$a^2l^2 + b^2m^2 + c \left( -\frac{pl + qm}{r} \right)^2 = 0$$

$$\Rightarrow al^2r^2 + bm^2r^2 + cp^2l^2 + cq^2m^2 + 2cpqlm = 0$$

$$\Rightarrow (ar^2 + cp^2) \left( \frac{l}{m} \right)^2 + 2cpq \left( \frac{l}{m} \right) + (br^2 + cq^2) = 0$$

$$\Rightarrow \frac{l_1l_2}{m_1m_2} = \frac{br^2 + cq^2}{ar^2 + cp^2}$$

$$\Rightarrow \frac{l_1l_2}{br^2 + cq^2} = \frac{m_1m_2}{ar^2 + cp^2}$$

from symmetry of the result we can write

$$\frac{l_1l_2}{br^2 + cq^2} = \frac{m_1m_2}{ar^2 + cp^2} = \frac{n_1n_2}{aq^2 + bp^2}$$

The two straight lines are perpendicular if

$$l_1l_2 + m_1m_2 + n_1n_2 = 0$$

$$\Rightarrow br^2 + cq^2 + ar^2 + cp^2 + aq^2 + bp^2 = 0$$

$$\Rightarrow p^2(b + c) + q^2(c + a) + r^2(a + b) = 0$$

Also, for the two lines to be parallel, their direction cosines are equal and hence the discriminant of the above equation is equal to zero.

$$\Rightarrow 4c^2p^2q^2 - 4(ar^2 + cp^2)(br^2 + cq^2) = 0$$

$$\Rightarrow c^2p^2q^2 - abr^4 - acq^2r^2 - bcp^2r^2 - c^2p^2q^2 = 0$$

$$\Rightarrow abr^2 + acq^2 + bcp^2 = 0$$

$$\Rightarrow \frac{p^2}{a} + \frac{q^2}{b} + \frac{r^2}{c} = 0.$$

2. Find the point in which the join of  $(-9, 4, 5)$  and  $(11, 0, -1)$  is meet by the perpendicular to it from the origin.

**Sol.** Let required point be  $P(x, y, z)$

d.r. of  $AB$   $(11 + 9, 0 - 4, -1 - 5)$

$$(20, -4, -6)$$

Or  $(10, -2, -3)$

Also d.r. of line  $AB$  can be taken as

$$(x - 11, y - 0, z + 1)$$

$$\Rightarrow \frac{x-11}{10} = \frac{y}{-2} = \frac{z+1}{-3} = k \text{ (say)}$$

$$x = 10k + 11$$

$$y = -2k$$

$$z = -3k - 1$$

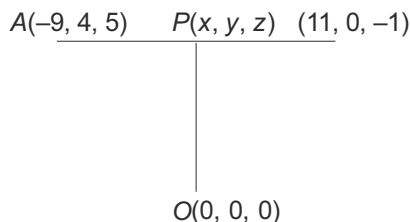
D.R. of  $OP$  are  $(10k + 11, -2k, -3k - 1)$

$\therefore AB$  is perpendicular to  $OP$

$$\therefore (10k + 11) \times 10 + (-2k)(-2) + (-3k - 1)(-3) = 0$$

$$\Rightarrow 113k + 113 = 0 \Rightarrow k = -1$$

$$\therefore \text{Co-ordinate of } P \text{ are } (10 \times -1 + 11, -2 \times -1, -3 \times (-1 - 1)) = (1, 2, 2)$$



3. A rectangular parallelopiped is formed by drawing planes through the points  $(1, 2, 5)$  and  $(-1, -1, -1)$  and parallel to the co-ordinate planes. The length of diagonal of the parallelopiped is .....

**Sol.** Clearly  $(1, 2, 5)$  and  $(-1, -1, -1)$  will be the end points of the diagonal

$$\therefore \text{Length of diagonal} = \sqrt{(1+1)^2 + (2+1)^2 + (5+1)^2} = 7$$

4. If  $d$  is the shortest distance between the lines  $\vec{r} = (3\hat{i} + 5\hat{j} + 7\hat{k}) + \lambda(\hat{i} + 2\hat{j} + \hat{k})$  and

$$\vec{r} = (-\hat{i} - \hat{j} - \hat{k}) + \mu(7\hat{i} - 6\hat{j} + \hat{k}) \text{ then } 125d^2 \text{ is equal to } \underline{\hspace{2cm}}.$$

**Sol.** Here  $\vec{a}_1 = 3\hat{i} + 5\hat{j} + 7\hat{k}$ ,  $\vec{b}_1 = \hat{i} + 2\hat{j} + \hat{k}$

$$\vec{a}_2 = -\hat{i} - \hat{j} - \hat{k}, \vec{b}_2 = 7\hat{i} - 6\hat{j} + \hat{k}$$

$$\vec{a}_2 - \vec{a}_1 = -4\hat{i} - 6\hat{j} - 8\hat{k}$$

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 1 \\ 7 & -6 & 1 \end{vmatrix} = \hat{i}(2+6) - \hat{j}(1-7) + \hat{k}(-6-14) = 8\hat{i} + 6\hat{j} - 20\hat{k}$$

$$\begin{aligned} d &= \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|} \\ &= \frac{|(-4\hat{i} - 6\hat{j} - 8\hat{k}) \cdot (8\hat{i} + 6\hat{j} - 20\hat{k})|}{\sqrt{64 + 36 + 400}} \\ &= \frac{|-32 - 36 + 160|}{\sqrt{500}} \end{aligned}$$

$$d = \frac{92}{\sqrt{500}}$$

$$\Rightarrow 125d^2 = \frac{92 \times 92 \times 125}{500} = 23 \times 92 = 2116$$

5. If the equation of the plane passing through the point (4, 5, 1), (3, 9, 4), (-4, 4, 4) is  $ax + by + cz + d = 0$  then  $a^4 - b^3 + c^2 + d$  is equal to \_\_\_\_\_.

**Sol.** Equation of plane is

$$\begin{vmatrix} x-4 & y-5 & z-1 \\ 3-4 & 9-5 & 4-1 \\ -4-4 & 4-5 & 4-1 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} x-4 & y-5 & z-1 \\ -1 & 4 & 3 \\ -8 & -1 & 3 \end{vmatrix} = 0$$

$$\Rightarrow (x-4)(12+3) - (y-5)(-3+24) + (z-1)(1+32) = 0$$

$$\Rightarrow (x-4)(15) - (y-5)(21) + (z-1)(33) = 0$$

$$\Rightarrow 15x - 60 - 21y + 105 + 33z - 33 = 0$$

$$\Rightarrow 15x - 21y + 33z + 12 = 0$$

$$\Rightarrow 5x - 7y + 11z + 4 = 0$$

$$\Rightarrow 5x - 7y + 11z = -4$$

$$\Rightarrow a = 5, b = -7, c = 11, d = -4$$

$$\begin{aligned} a^4 - b^3 + c^2 + d &= 625 - (-343) + 121 - (-4) \\ &= 625 + 343 + 125 \\ &= 1093 \end{aligned}$$

6. If  $P_1$  and  $P_2$  are the perpendicular distances of the points with position vectors  $\vec{a} = 3\hat{i} - 5\hat{j} + 8\hat{k}$  and  $\vec{b} = 2\hat{i} - 41\hat{j} + 21\hat{k}$  from the plane  $\vec{r} \cdot (2\hat{i} + 2\hat{j} - \hat{k}) = 12$ , then  $P_1^2 + P_2^2$  is equal to \_\_\_\_\_.

$$\text{Sol. } P_1 = \left| \frac{(3\hat{i} - 5\hat{j} + 8\hat{k}) \cdot (2\hat{i} + 2\hat{j} - \hat{k}) - 12}{\sqrt{4+4+1}} \right|$$

$$= \left| \frac{6 - 10 - 8 - 12}{3} \right| = \left| \frac{-24}{3} \right| = 8$$

$$P_2 = \left| \frac{(2\hat{i} - 41\hat{j} + 21\hat{k}) \cdot (2\hat{i} + 2\hat{j} - \hat{k}) - 12}{\sqrt{4+4+1}} \right|$$

$$P_2 = \left| \frac{4 - 82 - 21 - 12}{3} \right|$$

$$= \left| \frac{-78 - 33}{3} \right| = 37$$

$$P_1^2 + P_2^2 = 8^2 + (37)^2 = 64 + 1369 = 1433$$

7. If  $d$  is the distance between the point  $(-1, -5, -10)$  and the point of intersection of the line  $\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12}$  with the plane  $x - y + z = 5$ , then  $3d^3$  is equal to \_\_\_\_\_.

**Sol.**  $\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12} = t$

$$x = 3t + 2$$

$$y = 4t - 1$$

$$z = 12t + 2$$

For point of intersection the above point will satisfy the equation of the plane

$$\Rightarrow x - y + z = 5$$

$$\Rightarrow (3t + 2) - (4t - 1) + (12t + 2) = 5$$

$$\Rightarrow 11t = 0 \Rightarrow t = 0$$

Hence point =  $(2, -1, 2)$

$$d = \sqrt{(2+1)^2 + (-1+5)^2 + (2+10)^2}$$

$$= \sqrt{9 + 16 + 144}$$

$$= 13$$

$$\Rightarrow 3d^3 = 3 \times (13)^3 = 6591$$

8. A plane passes through  $(1, 2, 2)$  and is perpendicular to two planes  $2x - 2y + z = 0$  and  $x - y + 2z = 4$ . Square of the distance of the plane from the point  $(52, 53, 57)$  is equal to \_\_\_\_\_.

**Sol.**  $\vec{n}_1 = 2\hat{i} - 2\hat{j} + \hat{k}$

$$\vec{n}_2 = \hat{i} - \hat{j} + 2\hat{k}$$

$$\vec{n} = \vec{n}_1 \times \vec{n}_2$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -2 & 1 \\ 1 & -1 & 2 \end{vmatrix}$$

$$= \hat{i}(-4 + 1) - \hat{j}(4 - 1) + \hat{k}(-2 + 2)$$

$$\vec{n} = -3\hat{i} - 3\hat{j}$$

Equation of the plane is  $(\vec{r} - \vec{a}) \cdot \vec{n} = 0$

where  $\vec{a} = (\hat{i} + 2\hat{j} + 2\hat{k})$

$$\text{or } (\vec{r} - (\hat{i} + 2\hat{j} + 2\hat{k})) \cdot (-3\hat{i} - 3\hat{j}) = 0$$

$$\text{or } (\vec{r} - (\hat{i} + 2\hat{j} + 2\hat{k})) \cdot (3\hat{i} + 3\hat{j}) = 0$$

$$d = \left| \frac{((52\hat{i} + 53\hat{j} + 57\hat{k}) - (\hat{i} + 2\hat{j} + 2\hat{k})) \cdot (3\hat{i} + 3\hat{j})}{\sqrt{9+9}} \right|$$

$$\Rightarrow d = \left| \frac{(51\hat{i} + 51\hat{j} + 55\hat{k}) \cdot (3\hat{i} + 3\hat{j})}{3\sqrt{2}} \right|$$

$$= \left| \frac{51+51}{\sqrt{2}} \right| = 51\sqrt{2}$$

$$\Rightarrow d^2 = 51 \times 51 \times 2 = 51 \times 102 = 5202$$

9. If  $l_1, m_1, n_1; l_2, m_2, n_2; l_3, m_3, n_3$  are direction cosines of three mutually perpendicular lines  $OA, OB, OC$  then show that the line  $OP$  whose direction cosines are proportional to  $l_1 + l_2 + l_3, m_1 + m_2 + m_3, n_1 + n_2 + n_3$ , make equal angles with them.

**Sol.** According to the question,  $OA, OB$  and  $OC$  are mutually perpendicular to each other,

$$\text{hence } l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$$

$$l_2 l_3 + m_2 m_3 + n_2 n_3 = 0$$

$$l_3 l_1 + m_3 m_1 + n_3 n_1 = 0$$

Let  $l, m, n$  be the direction cosines of  $OP$ , then

$$\frac{l}{l_1 + l_2 + l_3} = \frac{m}{m_1 + m_2 + m_3} = \frac{n}{n_1 + n_2 + n_3} = \frac{\sqrt{l^2 + m^2 + n^2}}{\sqrt{l_1^2 + m_1^2 + n_1^2 + l_2^2 + m_2^2 + n_2^2 + l_3^2 + m_3^2 + n_3^2}} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow l = \frac{l_1 + l_2 + l_3}{\sqrt{3}}, m = \frac{m_1 + m_2 + m_3}{\sqrt{3}}, n = \frac{n_1 + n_2 + n_3}{\sqrt{3}}$$

Let  $\theta$ , be the angle between  $OA$  and  $OP$ , then

$$\cos \theta = ll_1 + mm_1 + nn_1$$

$$= \frac{l_1 + l_2 + l_3}{\sqrt{3}} l_1 + \frac{m_1 + m_2 + m_3}{\sqrt{3}} m_1 + \frac{n_1 + n_2 + n_3}{\sqrt{3}} n_1$$

$$= \frac{1}{\sqrt{3}}$$

Similarly,  $\phi$  and  $\psi$  be the angle between  $OB$  and  $OP$ , between  $OC$  and  $OP$  respectively, then

$$\cos \phi = ll_2 + mm_2 + nn_2 = \frac{1}{\sqrt{3}}$$

$$\cos \psi = ll_3 + mm_3 + nn_3 = \frac{1}{\sqrt{3}}$$

$$\therefore \cos \theta = \cos \phi = \cos \psi$$

$\Rightarrow OP$  makes equal angle with  $OA, OB$  and  $OC$ .

10. The plane  $ax + by = 0$  is rotated through an angle  $\alpha$  about its line of intersection with the plane  $z = 0$ . Show that the equation to the plane in new position is  $ax + by \pm z \sqrt{a^2 + b^2} \tan \alpha = 0$ .

**Sol.** The equation of the plane passing through the line of intersection of the plane  $ax + by = 0$ , and  $z = 0$  may be written as  $ax + by + kz = 0$

The direction cosines of the normal to the plane are  $\frac{a}{\sqrt{a^2 + b^2 + k^2}}, \frac{b}{\sqrt{a^2 + b^2 + k^2}}, \frac{k}{\sqrt{a^2 + b^2 + k^2}}$  and the direction cosines of the plane  $ax + by = 0$  are  $\frac{a}{\sqrt{a^2 + b^2}}, \frac{b}{\sqrt{a^2 + b^2}}, 0$ .

Since the angle between the planes  $ax + by = 0$  and  $z = 0$  is  $\alpha$ , hence

$$\cos \alpha = \frac{a \cdot a + b \cdot b + k \cdot 0}{\sqrt{a^2 + b^2 + k^2} \cdot \sqrt{a^2 + b^2}} = \frac{a^2 + b^2}{\sqrt{a^2 + b^2 + k^2} \sqrt{a^2 + b^2}}$$

$$\Rightarrow (a^2 + b^2 + k^2) \cos^2 \alpha = a^2 + b^2$$

$$\Rightarrow k^2 = \frac{(a^2 + b^2) \sin^2 \alpha}{\cos^2 \alpha} = (a^2 + b^2) \tan^2 \alpha$$

$$\Rightarrow k = \pm (\sqrt{a^2 + b^2}) \tan \alpha$$

Hence the required equation of the plane is  $ax + by \pm (\sqrt{a^2 + b^2} \tan \alpha) z = 0$ .

11. Find the equations of the projection of the line  $\frac{x+1}{-2} = \frac{y-1}{3} = \frac{z+2}{4}$  on the plane  $2x + y + 4z = 1$ .

**Sol.** The required line is the intersection of the given plane and the plane through the given lines perpendicular to the given plane.

Equation of plane through given line and perpendicular to given plane is

$$a(x+1) + b(y-1) + c(z+2) = 0 \text{ where}$$

$$-2a + 3b + 4c = 0 \text{ and } 2a + b + 4c = 0$$

$$\Rightarrow 2a = b = -2c \Rightarrow a = \frac{b}{2} = \frac{c}{-1}$$

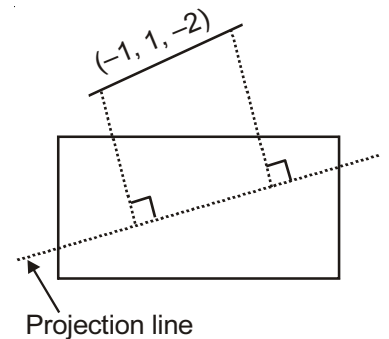
$$\therefore \text{Equation of plane is } x + 2y - z = 3$$

$$\text{Required line is } x + 2y - z = 3 \text{ and } 2x + y + 4z = 1$$

Let  $a, b, c$  be direction ratio of line

$$\Rightarrow a + 2b - c = 0, 2a + b + 4c = 0$$

$$\Rightarrow a = 3, b = -2, c = -1$$



Put  $z = 0$ ,  $x + 2y = 3$ ,  $2x + y = 1$

$$x = \frac{-1}{3}, y = \frac{5}{3}$$

Required equation of line is  $\frac{x + \frac{1}{3}}{3} = \frac{y - \frac{5}{3}}{-2} = \frac{z}{-1}$

12. Find the shortest distance between the diagonal of a cuboid. The lengths of whose coterminal edges are  $a$ ,  $b$ ,  $c$  and edges not meeting.

**Sol.** Let  $a$ ,  $b$ ,  $c$  be length of sides  $OA$ ,  $OB$  and  $OC$  respectively.

$$\overrightarrow{OA} = a\hat{i}, \overrightarrow{OB} = b\hat{j}, \overrightarrow{OC} = c\hat{k},$$

$$\overrightarrow{OP} = a\hat{i} + b\hat{j} + c\hat{k}$$

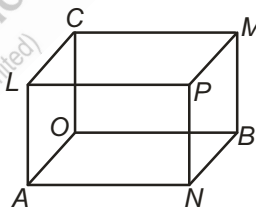
The edges which do not meet the diagonal  $OP$  are  $AL$ ,  $AN$ ,  $BN$  and their parallels are  $BM$ ,  $CM$ ,  $CL$ .

The shortest distance between two non-intersecting lines in space (two skew lines) passing through the points

whose position vectors are  $\vec{a}$  and  $\vec{b}$  and are parallel to the vectors  $\vec{c}$  and  $\vec{d}$  respectively is given by  $\frac{[\vec{a} - \vec{b} \ \vec{c} \ \vec{d}]}{|\vec{c} \times \vec{d}|}$ .

The shortest distance between  $OP$  and  $BN$  is given by

$$p = \frac{[-b\hat{j}, \hat{i} \ a\hat{i} + b\hat{j} + c\hat{k}]}{1\hat{i} \times a(\hat{i} + b\hat{j} + c\hat{k})} = \frac{1}{\sqrt{b^2 + c^2}} \begin{vmatrix} 0 & -b & 0 \\ 1 & 0 & 0 \\ a & b & c \end{vmatrix} = \frac{bc}{\sqrt{b^2 + c^2}}$$



Similarly, shortest distance between  $OP$  and  $AN$  is  $\frac{ca}{\sqrt{c^2 + a^2}}$  and  $OP$  and  $AL$  is  $\frac{ab}{\sqrt{a^2 + b^2}}$ .

13. Prove that the planes  $x - z - 1 = 0$ ,  $x + y - 2z - 3 = 0$  and  $x - 2y + z - 3 = 0$  form a triangular prism.

**Sol.**  $x - z - 1 = 0$  ... (i)

$x + y - 2z - 3 = 0$  ... (ii)

Let  $l$ ,  $m$ ,  $n$  be the direction ratio or line represented by given two equations.

$$\therefore \frac{l}{1} = \frac{m}{1} = \frac{n}{1}$$

(1, 2, 0) be point on line.

Line represented by (i) & (ii) is

$$\frac{x-1}{1} = \frac{y-2}{1} = \frac{z}{1}$$

Clearly, this line is parallel to  $x - 2y + z - 3 = 0$

The three planes form a triangular prism.

14. The plane  $x - y - z = 2$  is rotated through an angle  $\frac{\pi}{2}$  about its line of intersection with the plane  $x + 2y + z = 2$ .

Find its equation in the new position.

**Sol.**  $x - y - z = 2$  ... (i)

$x + 2y + z = 2$  ... (ii)

Any plane through the line of intersection (i) and (ii) is  $x - y - z - 2 + \lambda(x + 2y + z - 2) = 0$

$\Rightarrow (1 + \lambda)x + (2\lambda - 1)y + (\lambda - 1)z - (2\lambda + 2) = 0$  ... (iii)

Direction ratios of normal to (iii) are  $[1 + \lambda, 2\lambda - 1, \lambda - 1]$

Direction ratios of normal to (i) are  $[1, -1, -1]$  as (i) and (iii) are perpendicular,

$$1 + \lambda - 2\lambda + 1 - \lambda + 1 = 0 \Rightarrow \lambda = \frac{3}{2}$$

Substituting in (iii),

The required plane is  $5x + 4y + z - 10 = 0$

